



Dolby Conference Phone 5.0

Administrator's Guide

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Notices

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Contents

1 Introduction.....	7
1.1 About this documentation.....	8
1.2 New in this version.....	8
1.3 Related documentation.....	8
1.4 Configuration guides.....	8
1.5 Problem reports.....	9
1.6 Documentation feedback.....	10
2 Solution architecture.....	11
2.1 Dolby Conference Phone software application.....	12
2.2 Network configurations.....	12
2.3 Operation modes of Dolby Conference Phone	13
2.3.1 Integrating with your conferencing service provider.....	13
2.3.2 Supported IP PBX call control platforms.....	14
3 Preparation and requirements.....	15
3.1 Conference room requirements.....	16
3.2 Power requirements.....	16
3.3 Network requirements.....	16
3.4 Supported DHCP parameters.....	17
3.5 Network ports used.....	18
3.6 Bandwidth planning and codec selection.....	18
3.7 Primary and secondary VLANs.....	19
4 Configuring the provisioning method.....	24
4.1 Automatic provisioning method.....	25
4.1.1 Preparation: DNS server administrative access.....	25
4.1.2 Out-of-the-box discovery of the provisioning server.....	26
4.1.3 Rediscovering the provisioning server.....	26
4.2 Plug-and-play provisioning method.....	26
4.2.1 How the plug and play provisioning method works.....	27
4.2.2 Server addresses.....	27
4.3 Manual provisioning method.....	28
5 Configuring the phone with Dolby Conferencing Console	29
5.1 Reasons for using Dolby Conferencing Console	30
5.1.1 The purpose of a provisioning server.....	30
5.1.2 Overview of Dolby Conferencing Console.....	30
5.1.3 How the phone connects to Dolby Conferencing Console	31
5.2 Supported provisioning protocols with Dolby Conferencing Console	31
5.3 Updating phone software using Dolby Conferencing Console	31
6 Deploying the phone	33

6.1 Setting up the phone using the plug and play provisioning method.....	35
6.2 Setting up the phone using the provisioning method.....	37
6.3 Setting up the phone using the manual provisioning method.....	38
6.3.1 Connecting the phone to Dolby Conferencing Console or the provisioning server using the manual provisioning method.....	39
6.4 Deploying the phone in a secure network.....	41
6.4.1 802.1X.....	41
6.4.2 Configuring 802.1X during out-of-the-box setup.....	41
6.4.3 Configuring 802.1X after out-of-the-box setup.....	42
6.4.4 HTTPS provisioning.....	43
6.5 Changing network configuration after out-of-the-box setup.....	44
6.5.1 Configuration and rediscovery of the provisioning server.....	44
6.5.2 Network configuration.....	44
6.5.3 Configuring IP Network options.....	44
6.5.4 Configuring LLDP.....	45
6.5.5 Configuring the secondary VLAN.....	46
6.5.6 Configuring the time server.....	47
6.5.7 Using IEEE 802.1p/Q.....	47
6.5.8 Configuring PAC and WPAD.....	48
6.5.9 Proxy authentication.....	48
7 Configuring phone features.....	50
7.1 Configuration parameters.....	51
7.2 Setting and viewing configuration parameters.....	52
7.3 Configuring certificates.....	53
7.3.1 Device certificate.....	53
7.3.2 Viewing the device certificate issuer.....	54
7.3.3 Adding certificate authority certificates.....	55
7.3.4 Viewing certificate authority certificates status.....	55
7.4 Configuring preferences.....	56
7.4.1 Using the recent calls list.....	56
7.4.2 Enabling user feedback.....	56
7.4.3 Configuring call forwarding.....	57
7.4.4 Enabling automatic answer.....	57
7.4.5 Enabling do not disturb.....	58
7.4.6 Setting the language.....	58
7.4.7 Setting the time and date display.....	59
7.5 Provisioning options.....	61
7.5.1 Scheduling a software update.....	61
7.5.2 Setting recurring provisioning.....	61
7.5.3 Disabling the reboot prompt.....	62
7.6 Phone access control	62
7.6.1 Changing the administrative password.....	62
7.7 Directory options.....	63
7.7.1 Corporate directory.....	63
7.7.2 Local directory.....	63
7.7.3 Configuring the corporate directory.....	64

7.7.4 Creating a local contact directory from the provisioning server.....	65
7.7.5 Clearing the local contact directory.....	65
7.8 Configuring IP PBX.....	65
7.8.1 Prerequisites for configuring IP PBX.....	66
7.8.2 Support for SIP high availability.....	66
7.8.3 Configuring IP PBX using the web interface.....	67
7.8.4 Configuring IP PBX using the Dolby Conferencing Console.....	68
7.8.5 Adjusting codec preferences.....	70
7.8.6 Configuring voice activation detection.....	71
7.8.7 Configuring IP PBX for SIP URI dialing.....	71
7.9 Optimizing Dolby Conference Phone for small rooms.....	72
7.10 Disabling Bluetooth advertisement.....	72
8 Administering the phone.....	74
8.1 Error messages.....	75
8.2 Viewing the status menu.....	78
8.3 Accessing the Dolby Conference Phone web interface.....	79
8.4 Dolby Conference Phone web interface main page.....	80
8.5 Setting the Dolby Conference Phone operational mode.....	81
8.6 Viewing and managing event logs.....	82
8.6.1 Event logging.....	82
8.6.2 Log levels.....	82
8.6.3 Uploading log files to a provisioning server.....	83
8.6.4 Audio logging.....	84
8.6.5 syslog logging.....	85
8.6.6 Viewing log files.....	85
8.7 Exporting the Dolby Conference Phone configuration.....	86
8.8 Rebooting the Dolby Conference Phone	87
8.9 Call and audio statistics.....	87
8.10 Running Dolby Conference Phone diagnostics.....	88
8.11 Administering with the diagnostic shell.....	89
8.11.1 Diagnostic shell.....	89
8.11.2 Opening the diagnostic shell.....	90
8.12 Bluetooth LE initialization and advertisement.....	91
8.13 Confirming Bluetooth LE initialization and advertisement.....	91
8.14 Collection of usage data.....	92
8.14.1 Enabling Privacy block.....	92
8.15 Phone memory resources.....	93
8.16 Resetting to factory defaults.....	93
8.17 LLDP and supported type-length values.....	94
8.17.1 Supported type-length values.....	95
8.18 Trusted certificates and issuers.....	96
8.18.1 TLS cipher suites.....	98
9 Troubleshooting.....	100
9.1 Report to Dolby.....	101
9.2 Phone issues and solutions.....	102

9.2.1 Power and start-up issues.....	102
9.2.2 Hardware issues.....	102
9.2.3 Network access issues.....	103
9.2.4 Provisioning issues.....	105
9.2.5 Authentication issues.....	107
9.2.6 IP PBX calling issues.....	108
9.2.7 Audio quality issues.....	111
9.2.8 Software update issues.....	112
9.2.9 Corporate directory issues.....	112
9.2.10 Fatal error issues.....	112
9.2.11 Dolby Satellite Microphone issues.....	113
9.3 Plug and play provisioning method issues and solutions.....	113
9.3.1 Dolby Voice Console error messages.....	114
9.3.2 Provisioning service error messages.....	118
9.3.3 Dolby Conference Phone setup app error messages.....	119
9.4 Decommissioning and disposing of the phone.....	122
10 Appendix A: Configuring the provisioning server.....	123
10.1 Reasons for using a provisioning server.....	124
10.2 Building the provisioning server.....	124
10.2.1 Requirements for provisioning servers.....	124
10.2.2 Supported provisioning protocols with provisioning servers.....	124
10.3 Security for provisioning servers.....	124
10.4 Setting up an FTP server.....	125
10.5 HTTPS server setup.....	125
10.6 Master configuration file.....	125
10.6.1 Setting up the CA certificate files for the phone to download.....	127
10.7 Template files.....	127
10.8 Configuration file (JSON).....	128
10.9 Recommended file and directory layout.....	128
10.10 Redundant provisioning servers.....	129
10.11 Log directory.....	129
10.12 Updating preinstalled software using a provisioning server.....	129
10.12.1 Available software packages.....	129
10.12.2 Upgrading and downgrading phone software.....	130
10.12.3 Upgrading phones to the latest Dolby Conference Phone software.....	130
10.12.4 Downgrading phones to an earlier version of Dolby Conference Phone software.....	131
10.13 Sample configurations.....	131
10.13.1 Directory structure.....	131
10.13.2 Master configuration file.....	132
10.13.3 Simple IP PBX-only configuration.....	132
Glossary.....	134

Introduction

The Dolby Conference Phone offers an intuitive touch-screen interface that lets you easily join and manage meetings. The phone is a dual-mode device that can be used for Dolby Voice meetings and for traditional telephony.

- [About this documentation](#)
- [New in this version](#)
- [Related documentation](#)
- [Configuration guides](#)
- [Problem reports](#)
- [Documentation feedback](#)

1.1 About this documentation

Use this documentation to provision and deploy the Dolby Conference Phone.

We assume that users of this documentation are IT administrators or equivalent and are familiar with:

- Basics of computer networking and Linux administration
- IP private branch exchange (PBX) call controls used by your organization
- Conferencing service provider functionality used by your organization

This documentation provides information about:

- Dolby Conference Phone architecture
- Preparation and requirements for setting up the phone
- Configuring and managing phones with the Dolby Conferencing Console or a provisioning server
- Configuring the phone using the phone user interface and phone web interface
- Troubleshooting common issues with the phone

1.2 New in this version

This release of Dolby Conference Phone includes new and updated features.

- Removed expiring CA certificates

Related information

[Trusted certificates and issuers](#) on page 96

1.3 Related documentation

Review all of the Dolby Conference Phone documents to ensure that the phone is set up correctly and that you have the best conferencing experience possible.

These documents are available:

- *Dolby Conference Phone Quick Start Guide*

This guide describes the contents of the phone package, how to assemble the phone, and how to connect the phone to the network. The quick start guide is included in the phone package. It is also available from the Dolby Conference Phone support pages.

- *Dolby Conference Phone Quick User's Guide*

This short document provides a high-level overview of common tasks and how to use the phone as an IP phone, and some basic troubleshooting tips.

- *Dolby Conference Phone Third-Party Software Guide*

This guide describes the open source software used in the Dolby Conference Phone.

- *Dolby Conference Phone Quick Start Guide*

This guide describes the contents of the microphone package and how to connect them to the Dolby Conference Phone. The quick start guide is included in the microphone package. It is also available from the Dolby Conference Phone support pages.

- *Dolby Voice Product Compatibility Guide*

This guide describes the compatibility relationships between the various Dolby Voice products.

1.4 Configuration guides

For information about integrating the Dolby Conference Phone with your conferencing service provider and a supported IP PBX, see the configuration guide for your call control platform.

The following configuration guides are available:

- *Dolby Conference Phone Configuration Guide for Avaya Aura Platform 6*
- *BlueJeans Rooms with Dolby Conference Phone Advanced Configuration Guide*
- *Dolby Conference Phone Configuration Guide for BT MeetMe with Dolby Voice*
- *Dolby Conference Phone Configuration Guide for Cisco Unified Communications Manager*
- *Dolby Conference Phone Configuration Guide for iMeet*
- *Dolby Conference Phone Configuration Guide for Microsoft Skype for Business*
- *Dolby Conference Phone Configuration Guide for MiVoice Business*
- *Dolby Conference Phone Configuration Guide for RingCentral*
- *Dolby Conference Phone Configuration Guide for Unify OpenScape*
- *Dolby Conference Phone Configuration Guide for West Conferencing*

Related information

[Server addresses](#) on page 27

1.5 Problem reports

When you report problems to your conferencing service provider, be prepared to provide basic information about the phone, configuration files and logs, and a full description of the issue.

- Basic information about the phone:
 - Phone serial number
 - Software version on the phone
- Configuration files and logs:
 - Phone configuration. If the Dolby Conferencing Console is not used as the provisioning server, provide the relevant configuration files from the provisioning server for the phone, or export your entire phone configuration.
 - If you can reproduce the problem, detail how.
 - Any relevant log files showing the problem occurring, or anything that might be relevant in the log. Did anything unusual occur? Did the system generate any high-severity log messages? If so, attach an extract. If you have not enabled logging, enable it for the phone experiencing the issue, reproduce the issue, and provide the logs.
- Description of the issue:
 - When did the problem occur? How often does it occur? Is there any pattern or trend to the occurrence?
 - What was the scope of the problem? How many users did it affect? Was there any pattern or trend to the affected users?
 - Any symptoms observed.
 - Any error messages displayed on the phone user interface or web interface.
 - Any obvious triggers. (What was the user doing before the error message displayed?)
 - If the issue was reproducible. (Does it happen regularly or after certain tasks are performed?)
 - How widespread the issue appears to be. (Does it affect only a single user or the whole organization?)
 - Any other information you have that might assist in understanding the issue.
 - What operating system and version are being used by the client? What browser and version are being used by the client?
 - What kind of Internet connection is the client using (ADSL/cable/wireless/mobile)?
 - What other observations have you made? Is there anything else you think might assist us in identifying the root cause of the problem?

Related information

[Configuring the provisioning method](#) on page 24

[Uploading log files to a provisioning server](#) on page 83

[Exporting the Dolby Conference Phone configuration](#) on page 86

[Master configuration file](#) on page 132

1.6 Documentation feedback

If you have comments or feedback about this documentation, send us an email at dolbyvoicedocs@dolby.com.

Solution architecture

The Dolby Conference Phone supports advanced functionality, including local conferencing, multiple call appearances, Hypertext Transfer Protocol Secure (HTTPS) secure provisioning, and deployment in conference service provider and IP PBX environments.

- [Dolby Conference Phone software application](#)
- [Network configurations](#)
- [Operation modes of Dolby Conference Phone](#)

2.1 Dolby Conference Phone software application

The Dolby Conference Phone software consists of two basic components: the application that loads the phone software when the phone is first powered on and implements the phone functions and features, and the configuration files that contain the phone settings.

The phone software application manages the protocol stack, the DSP, the user interface (UI), and the network interaction. This application implements these functions and features:

- Voice over Internet Protocol (VoIP) signaling for a range of voice telephony functions using Session Initiation Protocol (SIP) signaling for call setup and control
- Industry-standard security techniques for ensuring that all provisioning, signaling, and media transactions are authenticated and encrypted
- Advanced audio signal processing for communications using a group of audio codecs
- Flexible provisioning methods to support various deployment scenarios

The phone software application also performs these tasks after you power up or reboot the phone:

- Enables you to set various network and provisioning options
- Formats the file system, and removes any application and configuration files if new updates are downloaded
- Determines which files on the provisioning server are required by the phone
- Downloads the application and configuration files
- Installs the application, and uploads an event log file from the boot cycle to the provisioning server

The phone software application is a binary file image and contains a digital signature that prevents tampering or the loading of rogue phone software images. Each new release of phone software has a new image file.

2.2 Network configurations

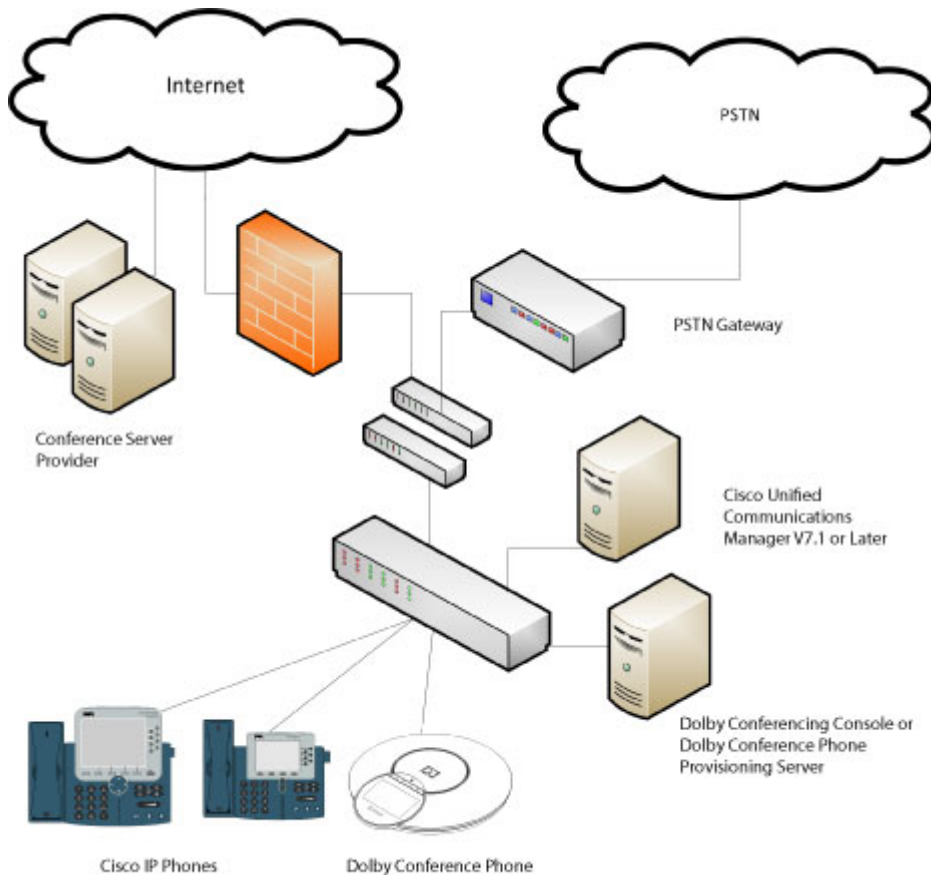
From an administrator's perspective, the Dolby Conference Phone is a SIP endpoint in a network that interoperates with other compatible hardware, such as application servers, media servers, gateways, switches, and other endpoints.

For the Dolby Conference Phone to operate as a SIP endpoint in the network, you require:

- A working IP network
- Network routers and switches configured for VoIP
- VoIP gateways configured for SIP
- Adequate firewall security
- The latest Dolby Conference Phone software image (or a compatible version)
- An active, configured call server to receive and send SIP messages
- Dolby Conferencing Console or a provisioning server on the IP network

You can set up a phone network using the Dolby Conference Phone in a number of ways. This figure shows one example of a network setup.

Figure 1: Dolby Conference Phone in a network



2.3 Operation modes of Dolby Conference Phone

The Dolby Conference Phone can integrate with the conferencing service provider as a conference endpoint, register to an IP PBX call control platform as a SIP audio-only endpoint, or do both simultaneously.

The operation mode determines which features are supported. For example, if the operation mode is set to **Conferencing Only Mode**, it cannot be configured to connect to an IP PBX, and features such as dialing a local extension, call forwarding, and call transferring are unavailable.

Similarly, if the operation mode is set to **PBX Only Mode**, the phone cannot be configured to connect to a conferencing service provider.

If the operation mode is set to **Dual Mode**, the phone can connect to both an IP PBX and a conferencing service enabled for Dolby Voice.

The default mode is **PBX Only Mode**.

For more information, see the configuration guide for your call control platform.

Related information

[Configuration guides](#) on page 8

[Setting the Dolby Conference Phone operational mode](#) on page 81

2.3.1 Integrating with your conferencing service provider

You can integrate the Dolby Conference Phone with your conferencing service provider. For information, see the configuration guide for your conferencing service provider.

Related information

[Configuration guides](#) on page 8

2.3.2 Supported IP PBX call control platforms

An IP PBX provides audio communications through a TCP/IP protocol stack for endpoints on internal networks and interconnects with a public switched telephone network (PSTN) for telephony connections for external endpoints.

Dolby Conference Phone supports these IP PBX call control platforms:

- Alcatel-Lucent Enterprise OXE
- Avaya Aura Platform
- BroadSoft BroadWorks
- Cisco Unified Communications Manager
- Edgewater Networks EdgeMarc 2900
- Mitel MiVoice Business
- NEC UNIVERGE 3C
- RingCentral
- Unify OpenScape Enterprise Express

Standard SIP call control platform

Dolby Conference Phone is a SIP-compliant endpoint and interoperates with call control platforms designed to comply with RFC 3261. For Transport Layer Security (TLS) transport, the phone is limited by the existing security configuration. Keep in mind:

- If the phone transport is configured as AUTO, TLS transport is attempted only if the phone fails to connect to Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) transports.
- The CA certificate of the SIP server must be downloaded to the phone through provisioning.
- TLS 1.0, 1.1, and 1.2 are supported. TLS 1.2 is the preferred protocol, but version 1.0 or 1.1 can be used, depending on peer support.
- Because the phone acts as a client, the server determines whether or not mutual authentication is required. (Normally, SIP servers are configured only with client authentication.)
- Certificate revocation lists (CRLs) are not supported.
- Simple Certificate Enrollment Protocol (SCEP) and Online Certificate Status Protocol (OCSP) are not supported.
- The phone device certificate cannot be replaced at this time.
- If you try to set up IP PBX with explicit TLS transport instead of using the **Auto** transport feature, you must set the transport port explicitly. If you change your configuration from TLS to **Auto**, you must set the port back to the correct default port.

Contact your network administrator to find out which port is used for the TLS connection in your environment.

When TLS is operational on the phone, Secure Real-time Transport Protocol (SRTP) is enabled; when TLS is not operational on the phone, SRTP is disabled. By default, if TLS is enabled, SRTP is set to Mandatory. Dolby Conference Phone offers a predefined list of cipher suites.

Related information

[TLS cipher suites](#) on page 98

Preparation and requirements

You can set up a Dolby Conference Phone in a conference room or an office. Before you use the phone, read the guide shipped with your phone package, and review the following requirements for room size, power, and network connectivity.

Before you begin, read the *Dolby Conference Phone Quick Start Guide*.

If you plan to support one or two Dolby Satellite Microphones with the phone, first read the *Dolby Satellite Microphone Quick Start Guide*. The phone must be running Dolby Conference Phone 2.1 or later software.

- [Conference room requirements](#)
- [Power requirements](#)
- [Network requirements](#)
- [Supported DHCP parameters](#)
- [Network ports used](#)
- [Bandwidth planning and codec selection](#)
- [Primary and secondary VLANs](#)

3.1 Conference room requirements

Before you install Dolby Conference Phone, make sure that the room does not exceed the maximum recommended room size.

From an acoustics point of view, choosing a Dolby Conference Phone solution based on the size of the room is important because it ensures natural and effective conversational flow for the usual seating positions in the room. Participant behavior and environmental factors influence the overall voice quality in the room and impact your experience; these factors include:

- Participant seating position
- Participant voice volume
- Ambient noise level
- Reverberation level
- Floor, wall, and ceiling materials
- Ceiling fans
- Projector placement

When Dolby Conference Phone is deployed in acoustically challenging rooms, we recommend testing the setup to determine whether the actual results meet your requirements.

Related information

[Optimizing Dolby Conference Phone for small rooms](#) on page 72

3.2 Power requirements

The Dolby Conference Phone is a class 3 Power over Ethernet (PoE) device that obtains power through a network port supporting PoE or a PoE adapter from a wall socket.

 **Note:** PoE and AC power adapters are not included in the Dolby Conference Phone package. If you require an adapter, contact your phone reseller.

The phone consumes a maximum of 12.95 W, reported through Link Layer Discovery Protocol (LLDP) type length value (TLV) as 13 W.

Follow the instructions in the *Dolby Conference Phone Quick Start Guide* to connect the phone to the network using a PoE source.

If a PoE source is not available, connect the phone to an AC power source. You need an AC power kit that includes a power adapter, a regional power cord, and an Institute of Electrical and Electronics Engineers (IEEE) 802.3af-compliant power injector cable (for example, the Phihong Single Port 15.4 W PoE Adapter PSA16U IEEE 802.3af compliant). Follow the power injector cable instructions carefully.

3.3 Network requirements

The Dolby Conference Phone is a VoIP device. It connects physically through a Cat5 cable to a standard office twisted-pair (IEEE 802.3) 10/100/1,000 Mbps Ethernet local area network (LAN), and it sends and receives all data using the same packet-based technology.

IP network requirements differ, depending on the partner application used.

You should have a Dynamic Host Configuration Protocol (DHCP) server and Domain Name System (DNS) server already set up in your organization.

Network connectivity must be available to all locations where you intend to deploy the Dolby Conference Phone.

Connecting the Dolby Conference Phone to the LAN initiates this sequence of events:

1. The phone establishes network connectivity.
2. DHCP negotiations occur between the network and IP address, network addressing options, default network gateway address, and time server.

 **Note:** If the Dolby Conference Phone reboots, it attempts to use the same IP address in the DHCP discovery request that was already assigned in the DHCP discovery before the reboot.

3. Appropriate security and Quality of Service (QoS) settings are applied.
4. The phone connects to the provisioning server.
5. The phone provisioning process starts.

The Dolby Conference Phone will not function until this link is established.

The phone is configured to automatically negotiate the Ethernet speed and duplex mode so that special configuration is not required. Although you can change the line speed and duplex configuration on the switch network port, we recommend that you keep the autonegotiation settings. If you change the settings, do so before deploying the phones.

If your organization uses an Hypertext Transfer Protocol (HTTP) or HTTPS proxy server to manage Internet traffic, see the requirements for proxy authentication.

If your organization separates voice and data traffic by using virtual LAN (VLAN)s, see the requirements for primary and secondary VLANs.

Related information

[Configuration guides](#) on page 8

[Primary and secondary VLANs](#) on page 19

[Proxy authentication](#) on page 48

3.4 Supported DHCP parameters

When the Dolby Conference Phone has established network connectivity, it needs to acquire several IP network settings to be fully functional. These settings are typically obtained from a DHCP server.

If some DHCP parameters are unavailable, you can set them manually from the phone user interface or in the phone web interface.

This table details the settings that are configurable through the DHCP protocol and where they are used.

Table 1: DHCP network parameters

Parameter	DHCP option
Subnet mask	1
IP gateway	3
DNS server IP address	6
DNS domain	15
Network Time Protocol (NTP) servers	42
Proxy autoconfiguration	252

 **Tip:** For more information on DHCP options, see RFC 2131 and RFC 2132.

Related information

[Automatic provisioning method](#) on page 25

[Accessing the Dolby Conference Phone web interface](#) on page 79

3.5 Network ports used

The network port used by the Dolby Conference Phone depends on protocol, data type, and whether it is sent by UDP or TCP.

Table 2: Network ports used

Port number	Protocol	Data	UDP or TCP
22	SSH	Diagnostics	TCP
80 ^[a]	HTTP	Not applicable	TCP
443	HTTPS	Not applicable	TCP
5060	SIP	SIP signaling	UDP or TCP
5061	SIP over TLS	Secure signaling	TCP
16400–16431 ^[b]	RTP/RTCP	Media, audio	UDP

[a] Port 80 is always redirected to port 443.

[b] Real-time Transport Protocol (RTP) ports are always even numbers; RTP Control Protocol (RTCP) ports are always odd numbers. The numbers always start from 16,400/16,401 and progress to 16,402/16,403 and so forth until they reach the end of the range. At this point, they revert to 16,400/16,401 and start over again.

3.6 Bandwidth planning and codec selection

Bandwidth planning depends on the audio codecs used.

Table 3: Audio codecs supported by the phone

Algorithm	Raw bit rate	IP bit rate ^[a]	Sampling rate	Supported payload size	Effective audio bandwidth
G.711μ-law RFC 1890	64 kilobits per second (kbps)	80 kbps	8 kHz	10 ms 20 ms (default) ^[b] 30 ms 40 ms 50 ms 60 ms 80 ms	4 kHz
G.711a-law RFC 1890	64 kbps	80 kbps	8 kHz	10 ms 20 ms (default) ^[b] 30 ms 40 ms 50 ms 60 ms 80 ms	4 kHz

Table 3: Audio codecs supported by the phone (continued)

Algorithm	Raw bit rate	IP bit rate ^[a]	Sampling rate	Supported payload size	Effective audio bandwidth
G.722 RFC 3551	48, 56, 64 kbps	64, 70, 80 kbps	16 kHz	10 ms 20 ms (default) ^[b] 30 ms 40 ms 50 ms 60 ms 80 ms	8 kHz
G.729ab	8 kbps	26.8 kbps	8 kHz	10 ms 20 ms (default) ^[b] 30 ms 40 ms 50 ms 60 ms 80 ms	4 kHz
DVC-2 in 2.0 and later (proprietary)	48 kbps	179 kbps (peak) 64 kbps (average)	16 kHz	20 ms (default) ^[b]	8 kHz
Opus 8 kHz (20ms)	8-48 kbps	24-64 kbps	16 kHz	20 ms (default) ^[b]	8 kHz

[a] The IP bit rate primarily depends on the tunnel type used.

[b] The Dolby Conference Phone always sends 20 ms payloads to the peer.

 **Note:** Opus is currently not supported in standalone and IP-PBX modes.

 **Note:**

During a call, the LED halo glows blue if the DVC-2 codec is being used, and glows green if a different codec is being used.

3.7 Primary and secondary VLANs

The Dolby Conference Phone system requires internet access to connect to the conferencing service. In some environments, voice and data traffic are handled by different VLANs.

If the voice VLAN lacks Internet connectivity, modify firewall policies to allow the system to connect to your conferencing service. Alternatively, you can set up the system on the data VLAN. If neither of these solutions works for you, then you need a dual-VLAN setup. After you enable dual VLANs, communication over the Internet with servers for your conferencing service provider passes through the secondary VLAN. With separate VLANs, the system has two IP addresses, one for each VLAN, but the same media access control (MAC) address.

When the secondary VLAN is enabled, the primary VLAN is designated as the voice VLAN, and the secondary VLAN is designated as the data VLAN. When the secondary VLAN is disabled, the system can be placed on

either the data VLAN or the voice VLAN. If your organization has a strict policy on segregating voice and data VLANs, and you want to configure the system for both IP PBX and conferencing service, enable the secondary VLAN.

Check the following to verify VLAN status:

DNS Server Status

Displays status of DNS for the voice and data VLAN.



Note: If the same DNS server is configured on both the primary VLAN and secondary VLAN, all DNS queries go through the primary VLAN.

In detached mode, dual VLAN cannot be enabled for Dolby Conference Phone.

The **Status** screens with voice and data VLAN settings are shown in these figures.

Figure 2: Status screen displaying Voice VLAN settings

Network

Host Name

VCS8000-E0001178

Ethernet

Ethernet Duplex

full

Ethernet Speed (Mbps)

1000

Total Packets Received

13122

Total Packets Transmitted

6648

DHCP

DHCP Server Address

10.0.0.1

Virtual LAN

VLAN ID In Use

0

Alarms and Status

DHCP Status

✓

DNS Server Status

✓

Default Gateway Status

✓

Hostname Resolvable

!

IEEE 802.1X Authentication Method

Unknown

IEEE 802.1X Authentication Phase 2 Method

Unknown

IEEE 802.1X Authentication Status

●

IEEE 802.1X Client Certificate

Dolby Factory Installed Certificate

IP Address Conflict

Link Layer State

✓

NTP Server Discovery

✓

PAC Status

●

PDS Connection Status

●

Voice VLAN Status

●

WPA2 Status

●

Figure 3: Status screen displaying Data VLAN settings

+
Network

-
Network - Secondary VLAN

Alternative DNS Server Address
DNS Server Address
Default Gateway
Domain Name
IP Address
Subnet Mask

+
DHCP

-
Status

DHCP Status

DNS Server Status

Default Gateway Status

IP Address Conflict

PAC Status

Secondary VLAN Status

WPAD Status

Network traffic is allocated onto different VLANs based on the traffic type, as listed in this table.

Table 4: Traffic-type-to-network-interface mapping

Traffic type	Network interface	Comment
DHCP	Primary and secondary	Works independently on both
Web Proxy Automatic Discovery (WPAD)	Primary and secondary	If dual VLAN is enabled, the secondary is supported.
Proxy automatic configuration (PAC)	Primary and secondary	If dual VLAN is enabled, the secondary is supported.
DNS ^[a]	Primary and secondary	DNS queries made on the network interface where the DNS requestor application runs
Conferencing services	Secondary	
Lightweight Directory Access Protocol (LDAP)	Secondary	
Web server	Primary and secondary	
SSH server	Primary and secondary	
Syslog	Primary	
NTP	Primary	

Table 4: Traffic-type-to-network-interface mapping (continued)

Traffic type	Network interface	Comment
Provisioning	Secondary	
SIP call	Primary	

[a] If the primary VLAN shares the same DNS settings, then DNS traffic runs only on the primary virtual LAN.

Dual VLAN limitations

Dual VLAN has these limitations:

- In attached mode, if dual VLAN is enabled, a voice VLAN must be assigned to .
- In detached mode, Dolby Conference Phone connects to the Dolby Voice Hub data VLAN IP address.
- The secondary VLAN must be untagged on the Dolby Voice Hub.
- On Dolby Conference Phone, provisioning requests can only be performed on the secondary VLAN.
- If dual VLAN is enabled for Dolby Conference Phone in a 802.1x environment and the Cisco ethernet switch, you must use multi-host mode on the Cisco switch.
- If dual VLAN is enabled, the primary VLAN proxy is disabled.
- Secondary VLAN IP assignment is only supported through DHCP, user cannot assign static IP to the secondary VLAN.

Voice VLAN checklist

Check the following to determine whether your network is ready for dual VLAN deployment.

Set up the Dolby Conference Phone on the tagged voice VLAN with the correct IP PBX configuration, and confirm that IP PBX calls can be successfully made:

- The Dolby Conference Phone can discover voice VLAN through LLDP.
- For IP PBX configuration instructions, see the configuration guide for your call control platform.

Data VLAN checklist

Keep the Dolby Conference Phone on the same Ethernet switch port, and then set up the Dolby Conference Phone on the data VLAN (usually, the untagged VLAN):

- If the Dolby Conference Phone discovers the voice VLAN using LLDP, you can disable LLDP from **Administrative Settings > Network Configuration > Ethernet > LLDP enabled**.
- Manually enable the VLAN.
- After the device is successfully set on the data VLAN, configure the settings for connecting to your conference service provider.

If both checklist tests work

If both tests work, you can proceed with the dual VLAN configuration.

 **Note:** When the dual-VLAN feature is used in conjunction with a Cisco switch port with 802.1X authentication enabled, the Ethernet port on the switch must be configured to Multiple-Hosts mode. If the port is set to Multidomain Authentication or Multiauthentication mode, dual VLAN does not work on the system.

Related information

[Configuration guides](#) on page 8

[Configuring the secondary VLAN](#) on page 46

[Troubleshooting](#) on page 100

4

Configuring the provisioning method

There are several different methods for provisioning the Dolby Conference Phone.

Depending on your requirements, use one of the following methods.

- [Automatic provisioning method](#)
- [Plug-and-play provisioning method](#)
- [Manual provisioning method](#)

4.1 Automatic provisioning method

We recommend using the automatic provisioning method. Using this method, the out-of-the-box setup wizard can discover the Dolby Conferencing Console or the provisioning server on your network. The automatic provisioning method is an on-premises, bulk method that simplifies deployment and minimizes configuration errors for multiple-site deployment scenarios.

Before you deploy the phone using the automatic provisioning method, you must prepare as described in this chapter.

Related information

[Deploying the phone](#) on page 33

[How the plug and play provisioning method works](#) on page 27

4.1.1 Preparation: DNS server administrative access

To use the automatic provisioning method, you must have administrative access to the DNS server on your network.

When the DNS server is enabled, the setup wizard can discover your File Transfer Protocol (FTP), HTTP, or HTTPS provisioning server using a Domain Name System service record (DNS SRV).

Syntax

The DNS SRV record is an entry in the forward zone file and has this syntax:

```
_service._protocolName IN SRV priority weight portNO target subdns.domain.com
```

_service._protocolName

_dccc._tcp if you want the phone to connect to Dolby Conferencing Console with HTTPS, or *_dcc._tcp* if you want the phone to connect to Dolby Conferencing Console with HTTP. If records for both of these protocols are present, HTTPS takes precedence.

priority

Numeric. Lower gets first priority.

weight

Numeric. Higher gets used more often.

portNO

The port used.

443: The typical port used by the HTTPS provisioning server.

80: The typical port used by the HTTP provisioning server.

subdns.domain.com

The fully qualified domain name (FQDN) of the Dolby Conferencing Console on your network.

Examples

DNS SRV record for HTTPS connection to Dolby Conferencing Console:

```
_dccc._tcp IN SRV 1 1 443 dcc.domain.org.
```

DNS SRV record for HTTP connection to Dolby Conferencing Console:

```
_dcc._tcp IN SRV 1 2 80 dcc.domain.org.
```

Related information

[Setting up the phone using the provisioning method](#) on page 37

4.1.2 Out-of-the-box discovery of the provisioning server

If you complete the necessary network preparation for the automatic provisioning method, the first time you plug in the Dolby Conference Phone, the setup wizard detects the provisioning server and completes setup.

The setup wizard displays on the phone user interface. You can use the setup wizard to find and connect the phone to Dolby Conferencing Console or any provisioning servers that you have set up on the network.

Note this information about provisioning phones using the automatic provisioning method:

- The phone uses the DNS SRV record to discover Dolby Conferencing Console and any provisioning servers that you have set up on the network.
- If your network requires IEEE 802.1X authentication, you can rediscover the provisioning server after you configure IEEE 802.1X settings and connect the phone to the network.

When you use the automatic provisioning method, the setup wizard discovers either the Dolby Conferencing Console or a provisioning server, and the phone automatically connects to the provisioning server. If you do not complete the preparation work, the phone cannot discover the on-premises Dolby Conferencing Console or provisioning server, and you will need to manually configure the phone with the provisioning server.

When you use the Dolby Conferencing Console, the setup wizard and provisioning procedures are different than if you use a provisioning server. For more information, see the .

Related information

[Deploying with 802.1X authentication](#)

4.1.3 Rediscovering the provisioning server

If needed, you can rediscover the Dolby Conferencing Console or the provisioning server from the phone user interface or phone web interface. You do not need to reset the phone to factory defaults.

Procedure

1. To rediscover the provisioning server from the phone user interface:
 - a) Tap this sequence:
 
 - b) When prompted, enter the administrative password (1739 by default).
 - c) Update provisioning settings as needed, and then tap the confirm button.
2. To rediscover the provisioning server from the phone web interface:
 - a) From the **Settings** tab, navigate to **Provisioning > Server**.
 - b) Turn off manual provisioning by clicking the **On** button next to **Manual Server Configuration**.
 - c) If the provisioning server is detected, confirm that you want to connect to it.

4.2 Plug-and-play provisioning method

The plug and play provisioning method is a fast and easy way to set up phones using your conferencing service provider's provisioning service. It makes trials and small-scale deployments of Dolby Conference Phones easier for administrators.

If the administrator completed the preparation work required for the automatic provisioning method, the Plug-and-Play screens do not appear. The phone discovers the on-premise Dolby Conferencing Console or provisioning server, and the screen that prompts you to choose a provider does not display.

Related information

[Configuration parameters](#) on page 51

4.2.1 How the plug and play provisioning method works

When you perform the plug and play provisioning method, you are provisioning the Dolby Conference Phone using your conferencing service provider's provisioning service.

If you use this method, you do not need to set up and use an on-premises instance of Dolby Conferencing Console or an on-premises provisioning server.

These actions occur when the phone boots up:

 **Note:** If you completed the preparation work for the automatic provisioning method, the screens and prompts in the setup wizard are different than described here. Instead, you are prompted to connect to your on-premises instance of Dolby Conferencing Console or your on-premises provisioning server.

1. If the phone has Internet access, the phone connects to the Dolby Voice Console.

The Dolby Voice Console is a provisioning discovery service that recognizes and redirects phones to provisioning services.

2. One of the following happens:

- If the phone is recognized by the Dolby Voice Console, the phone is redirected and connects to your conferencing service provider's provisioning service.
- If the phone is not recognized by the Dolby Voice Console, you can select your conferencing service provider. The phone is then redirected and connects to your conferencing service provider's provisioning service.

You can skip the plug and play provisioning method and perform manual setup if needed.

Related information

[Automatic provisioning method](#) on page 25

[Server addresses](#) on page 27

[Setting up the phone using the plug and play provisioning method](#) on page 35

4.2.2 Server addresses

With the plug and play provisioning method, the Dolby Conference Phone is provisioned automatically by your conferencing service provider's provisioning service.

The plug and play provisioning method requires that the phone have Internet access to the following servers using HTTPS protocol (port 443):

- The Dolby Voice Console at <https://console.dolbyvoice.com/>
- Your conferencing service provider's provisioning server
- Your conferencing service provider's setup app

 **Note:** For help with your provisioning server or setup app, refer to the appropriate configuration guide for your conferencing service provider.

For the plug and play provisioning method, the phone must be able to connect to your conferencing service provider's setup app through the Internet. After the phone connects to the setup app, screens and workflows specific to your conferencing service provider display on the phone. After out-of-the-box setup is complete, the phone uses your conferencing service provider's conferencing application, which is separate from the setup app.

The phone can utilize your HTTP proxy server to connect to the aforementioned servers. During the plug and play provisioning method, the phone automatically discovers the proxy server on your network.

The phone supports 802.1x as the authentication server. You can enter the authentication server credentials during out-of-the-box setup.

If your network does not implement proxy autodiscovery using WPAD, you can configure a PAC URL from the setup wizard. The phone supports NTLMv2, digest, and basic for proxy authentication methods.

Related information

[How the plug and play provisioning method works](#) on page 27

[Configuration guides](#) on page 8

[Deploying with 802.1X authentication](#)

4.3 Manual provisioning method

Use the manual provisioning method if the phone did not discover the Dolby Conferencing Console or a provisioning server, or if you want manual control.

The manual provisioning method begins when you skip the selection of a service provider during phone setup.

Configuring the phone with Dolby Conferencing Console

You can easily set up multiple Dolby Conference Phones by using the Dolby Conferencing Console as the provisioning server. We recommend using Dolby Conferencing Console if you have more than five Dolby Conference Phones to manage.

- [Reasons for using Dolby Conferencing Console](#)
- [Supported provisioning protocols with Dolby Conferencing Console](#)
- [Updating phone software using Dolby Conferencing Console](#)

5.1 Reasons for using Dolby Conferencing Console

A provisioning server allows for flexibility in installing, upgrading, maintaining, and configuring the phone. You must have a provisioning server to set up and upgrade Dolby Conference Phones in bulk.

You have these choices for provisioning phones:

- Use your conferencing service provider's provisioning service.
For more information, see [Plug-and-play provisioning method](#) on page 26.
- Set up and use the Dolby Conferencing Console as your provisioning server.
- Set up and use an FTP, HTTP, or HTTPS server as your provisioning server.

For more information, see [Appendix A: Configuring the provisioning server](#) on page 123.

You can use the Dolby Conferencing Console to provision devices, group them into device pools for ease of management, obtain analytic information about them, and monitor device status on both an individual and group level.

We recommend using Dolby Conferencing Console for provisioning if:

- You have more than five phones.
- You want full control of how your phones are managed.

With the plug and play provisioning method, your phones are managed by someone else.

- You want to allow calls to and from a conferencing service and from regular (IP PBX) phone calls.

5.1.1 The purpose of a provisioning server

A provisioning server contains configuration files, which contain configuration parameters. It may also contain logs from phones if the phones have been configured to upload logs to the provisioning server. When a phone reboots, configuration parameters from the provisioning server are downloaded to the phone.

This server allows the administrator to centrally manage phones from one location. Without a provisioning server, the administrator must configure settings on one phone at a time.

5.1.2 Overview of Dolby Conferencing Console

Dolby Conferencing Console is device-management software for the Dolby Conference Phone. We recommend using it because it provides additional features to simplify the administrator's work.

Dolby Conferencing Console accelerates Dolby Conference Phone deployment by enabling bulk provisioning, and removes management complexity with monitoring, support, and remote control capabilities. Key capabilities and benefits include:

Phone provisioning and updates

Simple provisioning and updating of large numbers of Dolby Conference Phones, utilizing device pools and profiles, saves time and improves IT productivity.

Device management

Allows for real-time tracking of assets using various attributes to improve planning and reduce costs.

Monitoring, diagnostics, and remote control

View Dolby Conference Phone status in real time, troubleshoot problems, and control devices remotely, speeding issue resolution.

Usage statistics and analytics

Get statistical intelligence about phone usage with detailed device and network analytics and reports to optimize your resource decisions.

System dashboard

Get a quick view into the overall system status to make informed decisions.

Web portal with popular browser support

Access anywhere, anytime, enabling participants to connect from their desks or remotely via PCs and mobile devices.

Data exportability

Highly extensible platform with RESTful application programming interface (API) and Simple Network Management Protocol (SNMP) version 2 (SNMPv2) support lets you easily export data and integrate with management and business intelligence solutions.

You can download the latest Dolby Conferencing Console software and documentation at <http://download.dolbyvoice.com/dcc>.

5.1.3 How the phone connects to Dolby Conferencing Console

You can connect the Dolby Conference Phone to Dolby Conferencing Console automatically or manually. The automatic provisioning method requires some preparation of the DNS server on the network.

If you complete the preparation for the automatic provisioning method, the first time you plug in the phone, the setup wizard automatically detects Dolby Conferencing Console and any provisioning servers that you have on the network.

For instructions about how to connect the phone to Dolby Conferencing Console, see the sources listed in the table.

Task	For more information
Setting up Dolby Conferencing Console	
Completing preparations that allow the phone to discover Dolby Conferencing Console	Automatic provisioning method on page 25
Connecting the phone to Dolby Conferencing Console automatically	Setting up the phone using the provisioning method on page 37
Connecting the phone to Dolby Conferencing Console manually	Connecting the phone to Dolby Conferencing Console or the provisioning server using the manual provisioning method on page 39

5.2 Supported provisioning protocols with Dolby Conferencing Console

HTTP and HTTPS provisioning protocols are supported with Dolby Conferencing Console.

If you connect the Dolby Conference Phone to Dolby Conferencing Console, the recommended protocol is HTTPS. If you connect the phone to Dolby Conferencing Console over an HTTP proxy, you must use the HTTPS protocol.

Both digest and basic authentication schemes are supported when using HTTP or HTTPS. Custom certificates, including those from certificate authorities, can be added to the phone for Transport Layer Security server authentication on HTTPS provisioning servers.

5.3 Updating phone software using Dolby Conferencing Console

Dolby Conferencing Console supports the use of multiple firmware releases, so you have the option of uploading new firmware releases and pushing these new releases out, as needed, to groups of phones (device pools).

Make sure that you reference the correct documentation when attempting to update software on phones. This varies depending on whether you use Dolby Conferencing Console or a provisioning server to provision phones.

Task	For more information
Using Dolby Conferencing Console to update software on phones	<i>Dolby Conferencing Console Administrator's Guide</i>
Using the provisioning server to update software on phones	Updating preinstalled software using a provisioning server on page 129

Deploying the phone

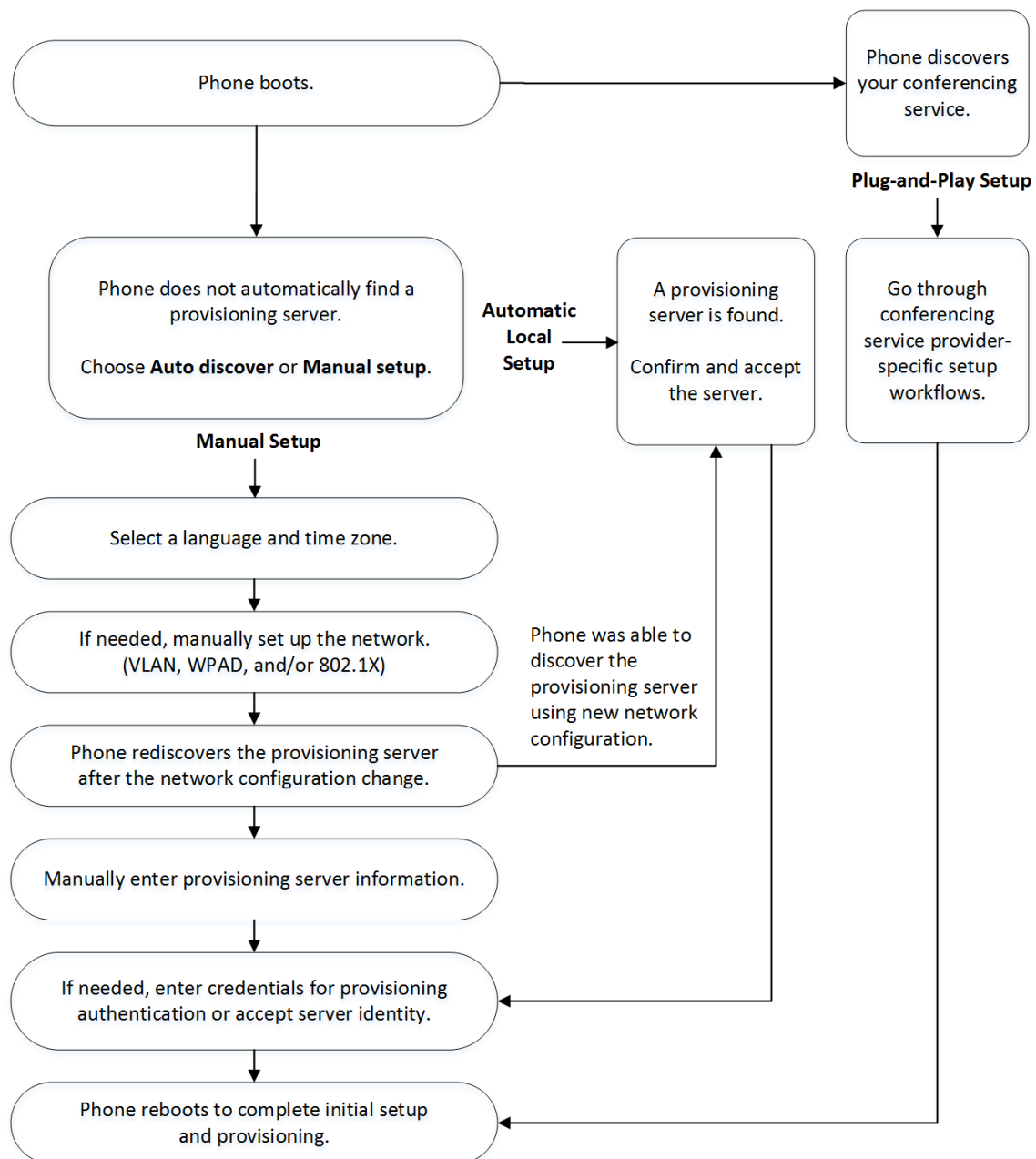
Deploying the phone involves setting it up for first-time use and connecting it to a provisioning server. The setup wizard screens vary depending on the setup methods available and the choices you make. The wizard also guides you through network setup, if needed.

Deploying the phone starts with unboxing the Dolby Conference Phone and physically connecting it to a PoE Ethernet port on the network using an Ethernet cable. When you do this, the setup wizard displays on the phone user interface.

The setup wizard behaves differently depending on:

- If you use the plug and play provisioning method and your phone is recognized by the Dolby Voice Console
- If you use the plug and play provisioning method and your phone is not recognized by the Dolby Voice Console
- If you prepared your network for the automatic provisioning method
- If you perform the manual provisioning method

For non-plug and play provisioning method deployments, the network setup is an important part of the process and the first half of the setup wizard covers network setup. You must connect the phone to the network before you can connect it to Dolby Conferencing Console or the provisioning server.



- [Setting up the phone using the plug and play provisioning method](#)
- [Setting up the phone using the provisioning method](#)
- [Setting up the phone using the manual provisioning method](#)
- [Deploying the phone in a secure network](#)
- [Changing network configuration after out-of-the-box setup](#)

6.1 Setting up the phone using the plug and play provisioning method

Use the plug and play provisioning method if you want to provision the Dolby Conference Phone using your conferencing service provider's provisioning service instead of setting up and using an on-premise instance of Dolby Conferencing Console or an on-premise provisioning server.

About this task

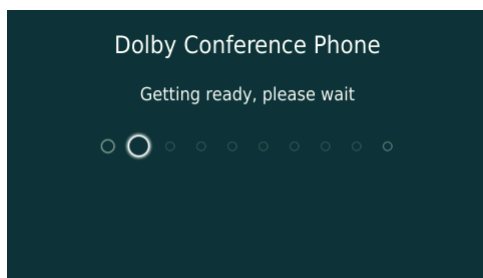
Be aware that if you completed the preparation work for the automatic provisioning method, the screens and prompts in the setup wizard are different than described here. Instead, you are prompted to connect to your on-premise instance of Dolby Conferencing Console or your on-premise provisioning server.

Procedure

1. Make sure that the Ethernet port that you plan to use has Internet access to the server addresses as described in [Server addresses](#) on page 27.
2. Connect the phone to a PoE Ethernet port on the network using an Ethernet cable.

The setup wizard displays on the phone user interface.

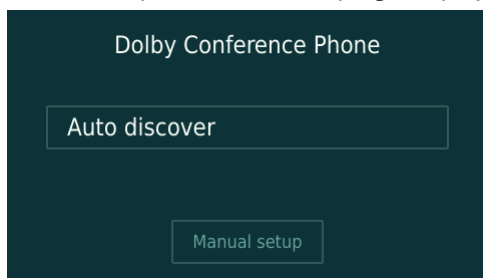
3. Wait for the phone to connect to the network.



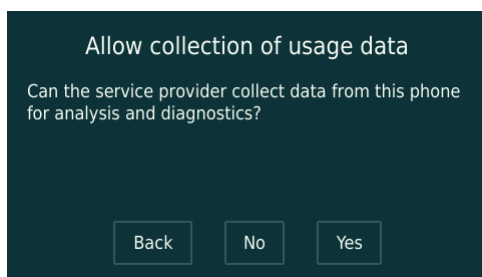
If the phone connects to the Dolby Voice Console and is recognized, the phone connects to your conferencing service provider's provisioning service.

If the phone connects to the Dolby Voice Console and is not recognized, the **Select your Dolby Voice service provider** screen displays.

If the phone fails to connect to the Dolby Voice Console, you must choose **Manual setup**. After that, you are prompted to choose a language and time zone. Next, an error screen displays letting you know that there was a problem with the plug and play provisioning method.

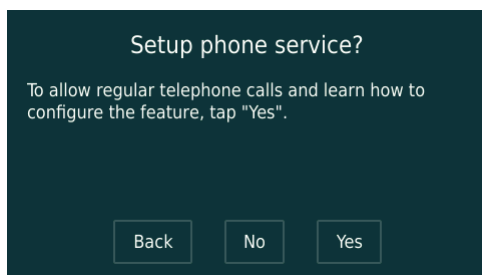


4. If the phone connected to the Dolby Voice Console and you still want to perform the plug and play provisioning method, choose your conferencing service provider from the list and then complete these steps:
 - a) From the **Welcome** screen, enter a name for the phone and then tap **Next**.
 - b) From the **Allow Dolby to collect usage data** screen, tap **Yes** or **No**.

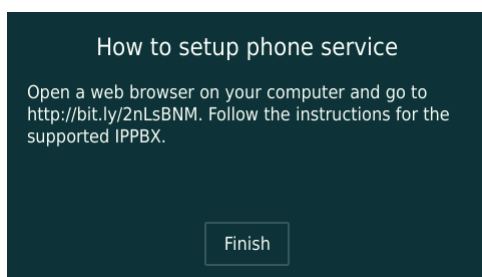


For more information, see [Collection of usage data](#) on page 92.

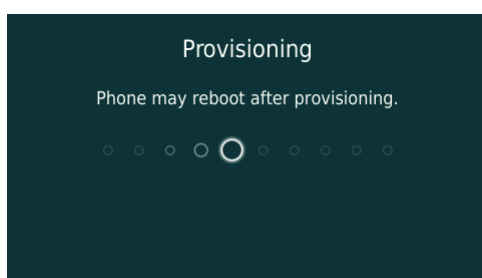
- c) From the **Setup phone service**, choose **Yes** if you want to allow regular telephone (IP PBX) calls to and from the phone.



Write down the URL provided on this screen. From your computer, go to [Dolby Support](#) for more information.



- d) Tap **Finish**.
e) Wait while the phone automatically reboots.



If the phone fails to provision using your conferencing service provider's provisioning service, instead you must either:

- Change the network configuration to allow the plug and play provisioning method.
- Set up the phone using the automatic provisioning method or manual provisioning method.

Results

Your setup is now complete and you can instruct users to start making calls.

If you chose to allow regular phone calls to and from the phone, the phone icon appears next to the conferencing icon, but appears dimmed until you configure the IP PBX settings.

 **Note:** After the plug and play provisioning method successfully completes, if you need to place the phone on a specific VLAN (such as a voice VLAN), you must enable LLDP in the **Settings** menu or manually set up the VLAN.

What to do next

If desired, go to [Dolby Support](#) for further instructions on IP PBX setup.

Related information

[How the plug and play provisioning method works](#) on page 27

[Collection of usage data](#) on page 92

[Setting up the phone using the provisioning method](#) on page 37

[Plug and play provisioning method issues and solutions](#) on page 113

6.2 Setting up the phone using the provisioning method

The first part of out-of-the-box setup involves network setup. You can use the setup wizard to connect the Dolby Conference Phone to the network, discover the Dolby Conferencing Console or the provisioning server, and complete the phone setup.

About this task

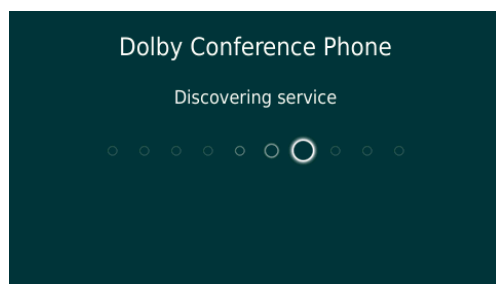
This procedure assumes that you have completed the preparation described in [Automatic provisioning method](#) on page 25. If you have not, go back and complete the preparation.

Procedure

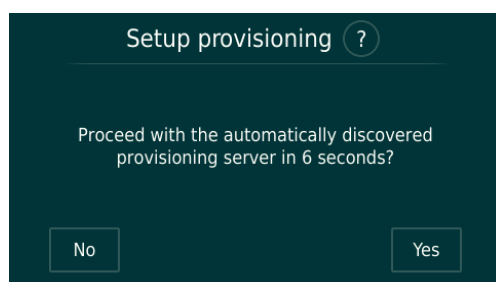
1. Connect the phone to a PoE Ethernet port on the network using an Ethernet cable.

The setup wizard displays on the user interface.

2. Wait for the phone to connect to the network.



3. After the phone successfully discovers a provisioning server, tap **Yes** on the **Setup Provisioning** screen for the phone to connect to the discovered provisioning server.



The setup wizard connects the phone to the provisioning server.

4. If prompted, enter a user name and password for the provisioning server.

This depends on your specific environment and network setup. You may not be prompted for this information at all.

5. If you are connecting to an HTTPS provisioning server, when prompted, confirm that you accept the provisioning server certificate.

If you are connecting to a different provisioning server, such as FTP or HTTP, this step is skipped.

6. Wait for the phone to connect to the provisioning server, and then reboot.
7. Confirm that the Dolby Voice software and the configuration files on the provisioning server are now on the phone.

From the phone user interface, tap  >  > **About** > **Status** > **Provisioning server**. Scroll down to **Configuration files**.

8. Monitor the log files uploaded to the provisioning server.

Log files are uploaded to the folder specified in `Provisioning.Server.UploadDir`.

Results

You can now instruct users to start making calls.

Related information

[Preparation: DNS server administrative access](#) on page 25

[Setting up the phone using the plug and play provisioning method](#) on page 35

6.3 Setting up the phone using the manual provisioning method

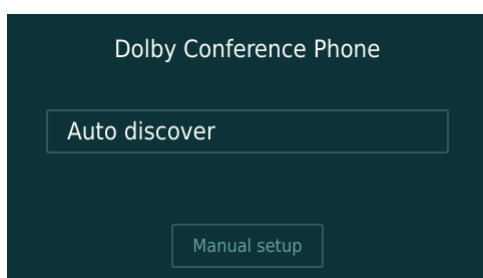
The first part of out-of-the-box setup involves network setup. You can use the setup wizard to connect the phone to the network manually, if desired. After you complete this portion and have connected to the network, the next step is connecting to the provisioning server.

Procedure

1. Connect the phone to a PoE Ethernet port on the network using an Ethernet cable.

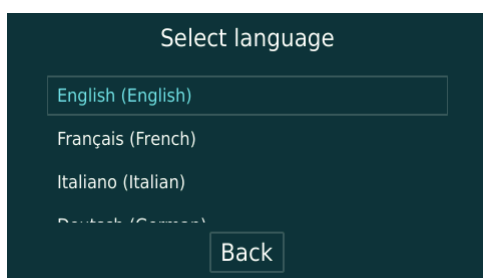
The setup wizard displays on the phone user interface.

2. If the phone did not discover a provisioning server, this screen displays on the phone user interface. Tap **Manual setup**.

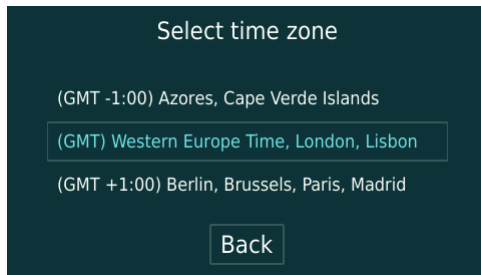


If the phone is not recognized by the Dolby Voice Console, the **Select your Dolby Voice service provider** screen displays.

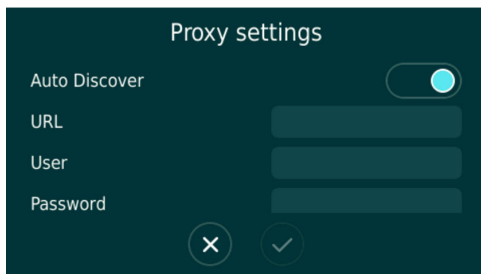
3. From the **Select language** screen, choose a language. If needed, scroll up or down to find the language.



4. From the **Select time zone** screen, choose a time zone. If needed, scroll up or down to find the time zone.



5. **Optional:** From the **Network setup** screen, you can change your VLAN settings, if needed.
6. From the **Network setup** screen, you can change your proxy settings, if needed.



7. If challenged for 802.1X authentication, tap **Confirm certificate**, and then follow the instructions in [Deploying with 802.1X authentication](#).
8. Wait for the phone to connect to the network.

What to do next

After you have connected to the network, continue through the setup wizard to connect the phone to the provisioning server.

To reenable Link Layer Discovery Protocol, you must do it manually in **Settings** either on the phone user interface or with the Dolby Conferencing Console.

6.3.1 Connecting the phone to Dolby Conferencing Console or the provisioning server using the manual provisioning method

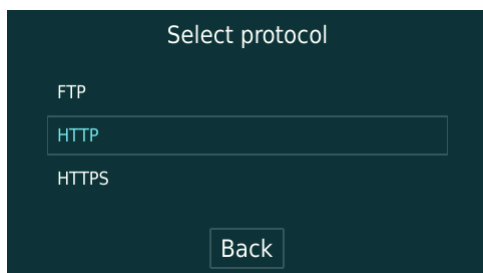
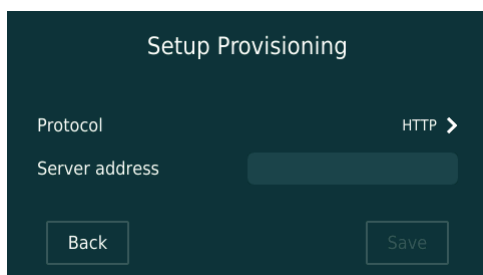
We recommend that you connect to Dolby Conferencing Console or the provisioning server using the automatic provisioning method, but you can connect using the manual provisioning method, if desired.

Procedure

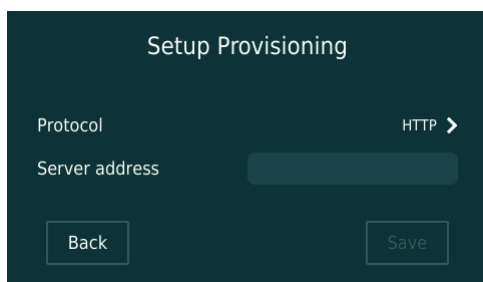
1. Connect the phone to the Dolby Conferencing Console or the provisioning server using one of these procedures:
 - [Setting up the phone using the provisioning method](#) on page 37
 - [Setting up the phone using the manual provisioning method](#) on page 38
2. From the **Setup Provisioning** screen, choose **Manual setup**.

 **Note:** If your network is not configured to support the automatic provisioning method, you will see only one option listed here: **Manual setup**.

3. From the **Setup Provisioning** screen, tap **Protocol**, choose **FTP**, **HTTP**, or **HTTPS**, and then go back to the previous screen.



4. From the **Setup Provisioning** screen, tap **Server address** and enter an IP address or host name, and then tap **Save**.



5. If prompted, enter a user name and password for the provisioning server.
This depends on your specific environment and network setup. You may not be prompted for this information at all.
6. Wait for the phone to connect to the provisioning server, and then reboot.
7. If you are connecting to an HTTPS provisioning server, when prompted, confirm that you accept the provisioning server certificate.
If you are connecting to a different provisioning server, such as FTP or HTTP, this step is skipped automatically.
8. Confirm that the Dolby Voice software and the configuration files on the provisioning server are now on the phone.
From the phone user interface, tap  >  > **About** > **General**.
9. Monitor the log files uploaded to the provisioning server.
Log files are uploaded to the folder specified in `Provisioning.Server.UploadDir`.

Results

You can now instruct users to start making calls.

What to do next

Reenable the LLDP manually in **Settings** on the phone user interface.

6.4 Deploying the phone in a secure network

During out-of-the-box setup, you may be prompted to confirm the certificate for the 802.1X authentication server, and later to confirm the certificate for the provisioning server.

802.1X authentication prevents unknown devices from connecting to the network. For example, if a guest were visiting the office and plugged their computer into a network port without permission, 802.1X authentication prevents them from being able to connect.

6.4.1 802.1X

If your network uses 802.1X authentication, then you must understand the restrictions placed on devices that do not have 802.1X credentials. You must obtain 802.1X credentials for every Dolby Conference Phone system.

You can customize these Dolby Conference Phone supported authentication methods:

- EAP-PEAP: Requires the device to validate the CA certificate, and the user to enter a user name and/or password, for network authentication.
- EAP-TLS: Provides for certificate-based and mutual authentication of the client and the network. It relies on client-side and server-side certificates to perform authentication.
- EAP-MD5: Requires the creation of an MD5 hash of a random value from the network, and the user to enter a password, to authenticate.
- EAP-TTLS: The only supported secondary authentication method is MSCHAPv2.
- EAP-FAST with automatic PAC provisioning: The supports anonymous or authenticated in-band PAC provisioning. Manual PAC provisioning and the "do not use PAC" provisioning modes are not supported.

6.4.2 Configuring 802.1X during out-of-the-box setup

If your network requires 802.1X during the network setup portion of the setup wizard, you are required to complete 802.1X setup. By default, 802.1X authentication is disabled during out-of-the-box setup.

About this task

You can use configuration parameters, the phone UI, or the web interface to enable 802.1X authentication.

Once you have enabled 802.1X authentication for Dolby Conference Phone, you can customize the protocol methods you want to use.

Procedure

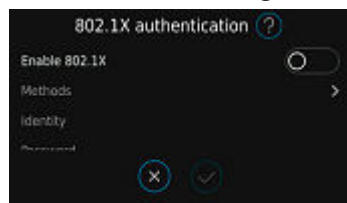
- Set the configuration parameter as described in the following table.

Configuration parameter name	Configuration parameter description
Network.EapOl.Enabled	Enables/disables 802.1X authentication. Set to false.

- To enable 802.1X authentication from the web interface:

 **Note:** If you need a Wi-Fi connection to access your device, see *Web administration portal access over Wi-Fi* in the *Troubleshooting* chapter.

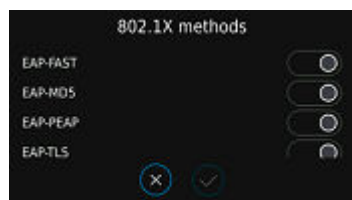
- Access the Dolby Conference Phone web interface.
 - Navigate to **Settings > Network > IEEE 802.1X**.
 - Toggle **Enable IEEE 802.1X Authentication** to turn 802.1X authentication on.
- To enable 802.1X authentication from the phone UI:
 - Access the Dolby Conference Phone **Settings** screen.
 - Tap this sequence:

Administrative settings > Network configuration > 802.1X

- c) Toggle **Enable 802.1X** to turn 802.1X authentication on.

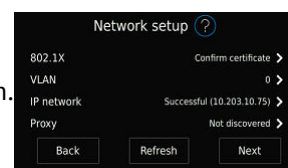
After enabling 802.1X, you can customize protocol methods, by selecting **Methods** from the **802.1X authentication** screen.

- d) Enable the protocol methods you want to use, and tap the confirm button.



Note: Depending on the authentication server configuration, you may be prompted to enter authentication credentials before you accept the server certificate.

The **Network setup** screen returns, showing successful authentication.



- e) Tap **Next** to continue provisioning.

What to do next

After you complete these steps, return to the provisioning task that you were working on before you were prompted to confirm this certificate.

6.4.3 Configuring 802.1X after out-of-the-box setup

You can configure 802.1X authentication after you have completed the out-of-the-box setup.

About this task

If a guest VLAN is not configured, you must use the embedded UI to enable 802.1X authentication.

If a guest VLAN is permitted, you can use the embedded UI and web UI to enable 802.1X authentication.

Procedure

- Set the configuration parameters as described in the following table.

Configuration parameter description	Configuration parameter name
Enables or disables 802.1X authentication	Network.Eapol.Enabled
Sets the IEEE 802.1X user identity	Network.Eapol.Identity

Configuration parameter description	Configuration parameter name
Sets the IEEE 802.1X password	Network.Eapol.Password
Sets the 802.1X authentication protocol method	Network.Eapol.EnableEapFast
	Network.Eapol.EnableEapMd5
	Network.Eapol.EnableEapPeap
	Network.Eapol.EnableEapTls
	Network.Eapol.EnableEapTtls

- Use the web interface:
 - a) Access the Dolby Conference Phone web interface.
 - b) Navigate to **Settings > Network > IEEE 802.1X**.
 - c) Toggle **Enable IEEE 802.1X Authentication** to turn 802.1X authentication on.

Go to **Settings > Network > IEEE 802.1X**, and enter the parameters or toggle the buttons to enable the protocol methods.

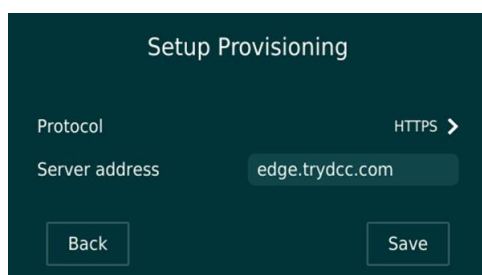
- Use the phone UI:
 - a) Access the Dolby Conference Phone **Settings** screen.
 - b) Tap this sequence: **Administrative settings > Network configuration > 802.1X**
 - c) Toggle **Enable 802.1X** to turn 802.1X authentication on.
 - d) Select **Methods**
 - e) Enable the protocol methods you want to use, and tap the confirm.

6.4.4 HTTPS provisioning

If you are using an HTTPS provisioning server, you are prompted to confirm the certificate of the provisioning server. This happens during the provisioning setup portion of the out-of-the-box setup wizard.

HTTPS encrypts communication and provides a secure connection between the phone and the HTTPS provisioning server. Using an HTTPS provisioning server prevents impostor devices from connecting to the network. This is accomplished by using certificates.

For example, by default, the phone does not trust any provisioning servers. When you perform out-of-the-box setup, you are prompted to accept the provisioning server certificate.



After you tap **Save**, this screen displays. Tap **Yes** to accept the server identity.



6.5 Changing network configuration after out-of-the-box setup

After you complete out-of-the-box setup, you may need to change the phone network configuration parameters. You can change network configuration parameters from the phone user interface, web interface, or the configuration files on Dolby Conferencing Console or the provisioning server.

You can change provisioning server parameters only from the phone user interface.

 **Note:** Certain parameters are dependent on other settings. For example, if DHCP is enabled, the Phone IP Address and Subnet Mask parameters are not editable, because the DHCP server automatically supplies the values.

6.5.1 Configuration and rediscovery of the provisioning server

You can modify these configuration parameters from the **Provisioning server** menu on the phone user interface:

- Type

If the provisioning type is set to **Manual**, you must set the provisioning server address.

If the provisioning type is set to **Automatic**, you can rediscover the provisioning server from the menu.

- Server
- Protocol
- User name
- Password

For the user name and password, if you are connecting to a Dolby Conferencing Console server, contact your system administrator for device pool access control credentials. If you are using an FTP provisioning server, see the values you set in [Setting up an FTP server](#) on page 125.

For more information about these parameters see “Configuration parameters.”

 **Note:** Changing the provisioning server IP address, type, user, and password will not cause the phone to reboot; the phone will be reprovisioned.

6.5.2 Network configuration

When configuring the network parameters on the Dolby Conference Phone, you must change certain parameters based on your particular network settings, whereas others can be left at their default values. Your organization’s IT administrator will be able to assist you.

6.5.3 Configuring IP Network options

From the **IP network** menu, you can modify certain network configuration parameters.

About this task

You can modify these configuration parameters:

- DHCP flag: If enabled, the DHCP server provides the IP address and subnet mask of the phone. Otherwise, you must enter those values.
- IPv4 address: If DHCP is disabled, you must enter the IP address for the phone.
- Subnet mask: If DHCP is disabled, you must enter the subnet mask for the phone.
- Gateway.
- Manual DNS flag: If enabled, you must configure DNS domain name and server addresses manually, even if DHCP network configuration is enabled.

- Domain name. If DHCP is disabled, or if manual DNS is enabled, you must enter the domain name.
- DNS server 1: If DHCP is disabled, or if manual DNS is enabled, you must enter the organization's primary DNS server IP address.
- DNS server 2: If DHCP is disabled, or if manual DNS is enabled, you must enter the organization's secondary DNS server IP address.
- DHCP VLAN option.

There are several ways to configure these options:

Procedure

- Set the configuration parameters:

Configuration parameter description	Configuration parameter name
Whether or not to enable DHCP (default true)	<code>Network.DHCP</code>
IP address of the phone	<code>Network.IpAddress</code>
Subnet mask of the network interface (available only if <code>Network.DHCP</code> is false)	<code>Network.Subnet</code>
Default gateway	<code>Network.DefaultGateway</code>
Whether or not to enable override of DHCP-assigned DNS settings (default false)	<code>Network.ManualDns</code>
Domain name	<code>Network.DomainName</code>
Primary DNS server IP address (available only if <code>Network.DHCP</code> is false or if <code>Network.ManualDns</code> is true)	<code>Network.DNS1</code>
Secondary DNS server IP address (available only if <code>Network.DHCP</code> is false or if <code>Network.ManualDns</code> is true)	<code>Network.DNS2</code>
User-configured VLAN ID	<code>Network.Vlan.ManualVlanId</code>

- Use the phone web interface:

Go to **Settings** > **Network** and edit the desired parameters.

- Use the phone user interface:

Tap  >  > **Administrative Settings** > **Network Configuration** > **IP network** and edit the desired parameters.

6.5.4 Configuring LLDP

From the **Ethernet** menu, you can modify the VLAN ID configuration parameter.

About this task

A VLAN can be used to separate and assign higher priority to a voice VLAN as a way of minimizing latency.

The phone will tag all outbound Ethernet packets, except the LLDP advertisement, with an 802.1Q VLAN header when:

- A valid VLAN ID is obtained from LLDP voice VLAN network policy type-length values; see [LLDP and supported type-length values](#) on page 94.
- A valid VLAN ID is specified in the phone network configuration.

There are several methods by which the phone can be configured to work on a particular VLAN:

- LLDP is a vendor-neutral Layer 2 protocol that allows a network device to advertise its identity and capabilities on the local network. LLDP-MED capabilities are also supported.
- Static: The VLAN ID can be manually set from the phone user interface or from a configuration file.

If the phone receives a VLAN setting from multiple of the above methods, the priority is as follows (from highest to lowest):

- LLDP (and LLDP-MED)
- Static VLAN ID

There are several ways to enable or disable LLDP discovery:

Procedure

- Set the configuration parameter:

Configuration parameter description	Configuration parameter name
Enables or disables LLDP discovery	Network.Services.LLDPEnabled

- Use the phone web interface:

Go to **Settings > Network > Services > Enable LLDP** and select **On** or **Off**.

- Use the phone user interface:

Tap  >  > **Administrative Settings > Network configuration > Ethernet > LLDP enabled.**

6.5.5 Configuring the secondary VLAN

Some organizations use a secondary VLAN to isolate voice and data traffic.

About this task

If you need information about setting up a secondary VLAN, continue reading, but also make sure that you review [Primary and secondary VLANs](#) on page 19.

If you need to isolate voice and data traffic by using different VLANs, you can enable the secondary VLAN. The secondary VLAN does not support static IP configuration and must be configured using DHCP.

After the secondary VLAN is enabled, the Dolby Conference Phone will create a secondary virtual network interface that is on the secondary VLAN ID. Then the phone would have two IP addresses, one on the primary (or default) VLAN and the other on the secondary VLAN.

 **Note:** WPAD and PAC work independently on the primary and secondary VLANs.

There are several ways to enable and configure the secondary VLAN:

Procedure

- Set the configuration parameters:

Configuration parameter description	Configuration parameter name
Whether or not to enable the secondary VLAN	Network2.Enabled
Enables WPAD	Network2.Proxy.AutoDiscover
Specifies the PAC configuration URL	Network2.Proxy.PacUrl
Specifies the proxy server login name	Network2.Proxy.UserName
Specifies the proxy server login password	Network2.Proxy.Password

- Use the phone web interface:

- a) Go to **Settings > Network > Secondary VLAN** and select **On**.
- b) To set the user-configured VLAN ID, go to **Settings > Network > Secondary VLAN > Virtual LAN** and enter a valid VLAN ID.
- Use the phone user interface:
 - Tap  >  > **Administrative Settings > Network configuration > Secondary VLAN** and change the **On/Off** option to **On**.

6.5.6 Configuring the time server

The Dolby Conference Phone uses a time server to synchronize the clocks on all of the phones on the network. If your network has a local NTP server, we highly recommend that you configure it.

About this task

The Dolby Conference Phone uses DHCP option 42 to configure the time server. If the phone cannot configure the time server using DHCP option 42, and the phone is connected to an IP PBX, the IP PBX supplies the time, and the NTP status is shown as **Synchronized with SIP** on the Dolby Conference Phone UI and web interface.

If you perform a factory reset, and DHCP option 42 is not received and the NTP servers are not provisioned manually, the time servers are set by default to *time.nist.gov* as the primary NTP server and *pool.ntp.org* as the secondary NTP server.

 **Note:** The network must have a DNS server for the phone to resolve the host names and connect to the default NTP servers. In addition, your firewall policy must permit access to these servers.

You can set and check the time server parameters in several ways:

Procedure

- Set the configuration parameters:

Configuration parameter name	Configuration parameter description
Network.Ntp.AutoDiscover	Enables NTP auto discovery (default true). When set to false, DHCP option 42 is ignored and you can manually set NTP server addresses with specified IP addresses or hostnames.
Network.Ntp.Server1	Sets the primary NTP server IP address or hostname
Network.Ntp.Server2	Sets the secondary NTP server IP address or hostname

- Use the web interface:
 - a) Go to **Settings > Network > NTP** and enter the parameters.
- Use the phone UI:
 - a) To review the time server status, access the **Settings** screen.
 - b) Tap  > **About > Status > Network time synchronization**.

6.5.7 Using IEEE 802.1p/Q

The Dolby Conference Phone supports two types of configuration for Quality of Service (QoS) and differentiated services tagging, in priority order: provisioning configuration and LLDP.

The phone will use the values in the configuration file. However, if provisioning configuration is unavailable, LLDP configures QoS settings.

For each voice/signaling packet, layer 2 and layer 3 priorities (Differentiated Services Code Point (DSCP) and link layer class of service) are set from LLDP.

Configuration parameter description	Configuration parameter name
Specifies the priority and link-layer priority of voice and signaling channels	Network.QoS.*

6.5.8 Configuring PAC and WPAD

PAC allows network administrators to automatically configure complex proxy settings across the network.

About this task

- PAC: Specify the URL of a JavaScript file that contains a function to be used to determine the appropriate proxy for each outgoing HTTP and HTTPS connection. The PAC functions are used for conferencing and provisioning connections. The maximum size of a PAC file is 64 KB. The phone supports the ECMA-262 specification and does not support proprietary extensions.

If the phone user interface becomes slow or unresponsive when you implement PAC, you can configure a PAC cache to improve performance. When you enable the cache, the Dolby Conference Phone uses the protocol name, host name, and port number to create a cache table key and save the PAC result in the cache. This reduces the frequency with which the Dolby Conference Phone accesses the PAC file and thereby reduces lag time.

The cache entry is removed automatically 12 hours after it is created.

- WPAD protocol: Let the system automatically discover the URL of the PAC file through a DHCP option and DNS A record lookup.



Note: If `Dvms.Proxy` is configured, `Network.Proxy.*` is ignored when connecting the phone to the conferencing service. If `Dvms.Proxy` is configured and the secondary VLAN is enabled, `Network2.Proxy.*` is also ignored. This allows existing configurations to remain valid after a Dolby Conference Phone software upgrade to 2.1 or later.

You can set up automatic proxy configuration in two ways:

Procedure

- Set the configuration parameters:

Configuration parameter description	Configuration parameter name
Enables WPAD	Network.Proxy.AutoDiscover
Specifies the PAC configuration URL	Network.Proxy.PacUrl
Specifies the proxy server login name	Network.Proxy.UserName
Specifies the proxy server login password	Network.Proxy.Password
Enables a cache composed of protocol name, host name, and port number, and saves the PAC result in the cache.	Network.Proxy.EnableCache

- Use the phone web interface:
 - To set the PAC or WPAD parameters, go to **Settings > Network > Proxy** and enter the parameters.
 - To check the PAC or WPAD status, go to **Status > Network**.

6.5.9 Proxy authentication

The Dolby Conference Phone can utilize the HTTP proxy server in a customer's network to connect to services on the Internet.

These methods are supported for proxy server authentication:

- No authentication
- Digest
- Basic
- NTLMv2

Related information

[Network requirements](#) on page 16

7

Configuring phone features

As the administrator, you can configure common features that are intended for end users of the Dolby Conference Phone (for example, call forwarding, automatic answer, and do not disturb). You can also set passwords and connect the phone to your corporate directory.

- [Configuration parameters](#)
- [Setting and viewing configuration parameters](#)
- [Configuring certificates](#)
- [Configuring preferences](#)
- [Provisioning options](#)
- [Phone access control](#)
- [Directory options](#)
- [Configuring IP PBX](#)
- [Optimizing Dolby Conference Phone for small rooms](#)
- [Disabling Bluetooth advertisement](#)

7.1 Configuration parameters

A large set of configuration parameters controls Dolby Conference Phone features and settings. The phone receives and stores the majority of configuration parameters from the provisioning server. You can manually set other configuration parameters using one of the interfaces that the phone provides.

A detailed description of all configuration parameters, along with the default or permitted values for those parameters, is available in the **Help** section of the web interface.

 **Note:** If a configuration parameter is set by the provisioning server, you cannot change it manually.

Default values

If a configuration parameter has not been set manually and is not present in the configuration files on the provisioning server, a default value is used.

Configuration parameters requiring reboot

Some configuration parameters require a reboot when changed. If the phone requires a reboot, a reboot prompt displays on the user interface and web interface. If you do not cancel the reboot within 20 seconds, the phone reboots automatically. Active calls are not interrupted; the reboot is delayed until those calls have ended. You can turn off the reboot prompt if desired.

Device pools and profiles

With Dolby Conferencing Console, you can easily assign multiple configuration settings to multiple devices in a single operation. You can define device pools, which are collections of Dolby Conference Phones, and you can create profiles, which are collections of configuration parameters and their values. You can then apply profiles to pools. See the *Dolby Conferencing Console Administrator's Guide* for details.

 **Note:** If you use the plug and play provisioning method, you cannot configure features for a pool of systems, because your systems are provisioned by your conferencing service provider. You can still configure features one phone at a time (for example, you can configure IPPBX settings, which are not provisioned by your conferencing service provider's provisioning service).

Deprecated configuration parameters

Dolby may occasionally deprecate configuration parameters in new releases of the Dolby Conference Phone software.

If you are using the automatic provisioning method, depending on the provisioning server in use, you might have an administration task to perform when a configuration parameter is deprecated.

- When using Dolby Conferencing Console

If you use Dolby Conferencing Console to manage your phones, no action is needed. Deprecated parameters are removed from the phone when you upgrade and do not appear in Dolby Conferencing Console.

- When using an on-premises provisioning server

If a configuration parameter is deprecated, we recommend that you remove the parameter from your provisioning server. Upgrading your phones to the latest version does not remove deprecated configuration parameters from the provisioning server. You must search the configuration files on your provisioning server for the parameter and then manually delete it.

If you choose not to do this, Dolby Conference Phone and other components (such as Dolby Conferencing Console) continue to work as expected. Deprecated parameters have no negative impact on functionality.

Configuration parameters on an on-premises provisioning server

If you are using an on-premises provisioning server, this information about configuration parameters applies:

 **Note:** If you are using Dolby Conferencing Console, this information does not apply. Skip this list.

- For Boolean configuration parameters, the values allowed in the configuration templates are case sensitive; only the values `false` and `true` are acceptable.
- If a parameter value is invalid, the value is ignored. Invalid parameter values can occur when enumerated-type parameters do not match a predefined value, when numeric parameters are set to nonnumeric values or outside of their valid range, when string parameters are either too long or too short, or when using null strings in numeric fields. All such situations are logged in the phone log files.
- Null should be interpreted as the empty string—that is, `attributeName=""` when the file is viewed in a JavaScript Object Notation (JSON) editor.
- If you accidentally set a configuration parameter two or more times in the same configuration file or in different configuration files, the parameter will take the last value being parsed by the program.
- Passwords are stored as encrypted strings on the phone.

Related information

[Plug-and-play provisioning method](#) on page 26

[Setting and viewing configuration parameters](#) on page 52

7.2 Setting and viewing configuration parameters

There are a few ways to set and view configuration parameters.

Procedure

- a) Tap the **Settings** button .
- b) **Optional:** For advanced settings tap .

 **Note:** If a setting is dimmed, it means that this configuration parameter is set by the provisioning server and cannot be changed manually.

- Use the web interface.

 **Note:** If a setting is dimmed, it means that this configuration parameter is set by the provisioning server and cannot be changed manually.

- Use the **configure** command in the diagnostic shell to set a parameter.
For example, to configure the system to automatically answer calls, enter this command:

```
configure Preferences.AutoAnswerEnabled true
```

To set the logging level to DEBUG for the SIP protocol, enter this command:

```
configure Logging.Level.Sip DEBUG
```

- Use the **show configuration** command in the diagnostic shell to view a parameter.
For example, to see how long a web interface or diagnostic shell session remains active before login is required again, enter this command:

```
show configuration Authentication.Local.SessionTimeout
```

To show all configuration parameters, enter this command:

```
show configuration all
```

To get a list of related configuration parameters, such as all parameters related to PBX, enter the first portion or first two portions of the parameter name, up to the period, and press Tab.

show configuration Sip.Pbx. (Tab)

```
Sip.Pbx.CallFailoverMode
Sip.Pbx.CallFailoverTimeout
Sip.Pbx.Codecs
Sip.Pbx.ContentShareEnabled
Sip.Pbx.Domain
Sip.Pbx.EnableP2PSIPVideo
Sip.Pbx.EnableURIDialingProxy
Sip.Pbx.InBandDTMF
Sip.Pbx.OnlyAcceptCallServerEvents
Sip.Pbx.OutboundProxy1
Sip.Pbx.OutboundProxy1Port
Sip.Pbx.OutboundProxy2
Sip.Pbx.OutboundProxy2Port
Sip.Pbx.RegistrationTimeout
Sip.Pbx.SipUriScheme
Sip.Pbx.TLSCertNameMatchingPolicy
Sip.Pbx.URIDialingEnabled
Sip.Pbx.URIDialingProxy
Sip.Pbx.URIDialingTransport
```

Related information

[Accessing the Dolby Conference Phone web interface](#) on page 79

[Configuration parameters](#) on page 51

[Administering with the diagnostic shell](#) on page 89

7.3 Configuring certificates

The collection of certificate authority (CA) and device certificates installed on the Dolby Conference Phone is a security feature and is used for authentication with features such as HTTPS provisioning, LDAP TLS connection, TLS SIP signaling encryption, 802.1X, and conferencing services that support Dolby Voice.

The phone can trust certificates issued by widely recognized certificate authorities when trying to establish a connection to a server. To establish new trust relationships, you can add new certificates to the phone through Dolby Conferencing Console or the web interface.

7.3.1 Device certificate

A device certificate is installed on each Dolby Conference Phone at the factory.

The device certificate is an X.509 digital certificate signed by the Dolby root certificate authority. The digital certificate and associated private key are stored on the phone in its nonvolatile memory as part of the manufacturing process. For more information on digital certificates, see www.ietf.org/html.charters/pkix-charter.html and IETF RFC 2459.

The Dolby Conference Phone has one of the following certificates installed:

Issuer name	Expiration date	CA Certificate download link
Dolby-PSM	Expires in 256 years	https://certificates.dolbyvoice.com/dolby_psm_ca_chain.cer
.Dolby.Comms.Devices.CA.1	Expires before 2050	https://certificates.dolbyvoice.com/dolby_comms_devices_ca1_chain.cer

Note:

If you have a Dolby Conference Phone that is installed with the Dolby-PSM issued certificate, updating to version 4.1 does not update the factory installed certificate.

Related information

[Viewing the device certificate issuer](#) on page 54

[Adding certificate authority certificates](#) on page 55

[Viewing certificate authority certificates status](#) on page 55

7.3.2 Viewing the device certificate issuer

You can view information about the device certificate issuer from the Dolby Conference Phone settings screen.

About this task

There are two different types of device certificates that can be issued on the Dolby Conference Phone.

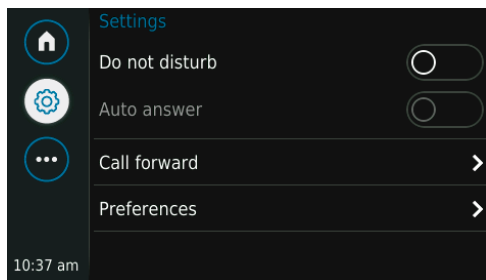
The Dolby Conference Phone displays one of the following certificate issuers:

- **DOLBY-PSM:** The issuer is the DOLBY-PSM server.
- **.Dolby.Comms.Devices.CA.1:** The issuer is the new Dolby infrastructure that supports SHA-2 certificates.

You can view the device certificate issuer from the phone user interface or from the web interface.

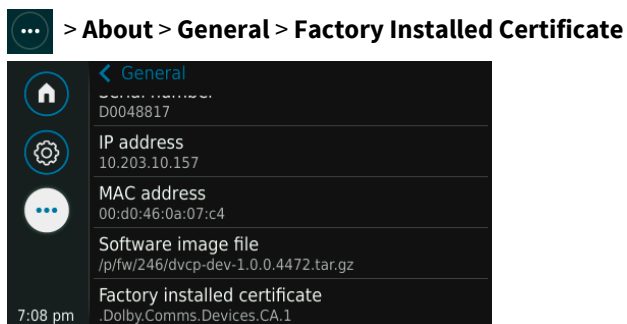
Procedure

- To view the device certificate issuer from the phone user interface:
 - a) Access the Dolby Conference Phone **Settings** screen.



Ask your conferencing service provider for instructions.

- b) Tap this sequence:



- To view the device certificate issuer from the web interface:
 - a) Access the Dolby Conference Phone web interface.
 - b) From the **Settings** tab, click **Device**, then **Factory Installed Certificate Issuer**.

Dolby® Satellite Microphones	No Microphones Connected
Bluetooth Module Initialized	✓
Factory Installed Certificate Issuer	.Dolby.Comms.Devices.CA.1
Application status	●

Related information

- [Device certificate](#) on page 53
- [Adding certificate authority certificates](#) on page 55
- [Viewing certificate authority certificates status](#) on page 55

7.3.3 Adding certificate authority certificates

A certificate that is issued by a certificate authority establishes a point of trust between two entities, for example, the Dolby Conference Phone and the LDAP directory or the provisioning server.

If you want the phone to trust any CAs, you must add their certificates to the phone in a CA certificate directory. This file must be in Privacy-Enhanced Electronic Mail (PEM) encoded format.

You can add CA certificates from:

- Dolby Conferencing Console
 - For details, see the *Dolby Conferencing Console Administrator's Guide*.
- The web interface
 - Access the Dolby Conference Phone web interface.
 - From the **Tools > Certificates > Import CA certificates**.

A standard file-selection dialog box displays.

- Select the CA certificate file and click **Open**.

 **Note:** Certificates imported through the web interface are deleted after a factory reset. You must re-import the certificates.

Related information

- [Device certificate](#) on page 53
- [Viewing the device certificate issuer](#) on page 54
- [Viewing certificate authority certificates status](#) on page 55
- [Dolby Conference Phone web interface main page](#) on page 80

7.3.4 Viewing certificate authority certificates status

With the CA certificates status screen, you can view the number of certificates that are expired, are about to expire, or are not yet valid.

About this task

All certificates are validated every time the certificate store changes, as well as every 24 hours if scheduled provisioning is enabled. Warnings and error messages are available from the UI and the web interface. If a certificate expires, the **Expired** count increases on the CA certificates status screen. If a certificate is set to expire within seven days, the **Expiring soon** count increases on the CA certificates status screen. You must replace the certificate before expiry, or certain features will stop working on the phone.

Procedure

- To view CA certificates status from the web interface:
 - From the **Status** tab, click **System**, then **Device**.

- b) **Optional:** To see a list of certificates, including a list of expired certificates, click the link next to **CA Certificates Details**.

 **Tip:** A list of expired certificates appears before the full list of certificates. You can select the hexadecimal number in the error message, copy it to the clipboard, and then use your browser's find command to find the matching certificate in the full list of certificates.

Related information

[Device certificate](#) on page 53

[Viewing the device certificate issuer](#) on page 54

[Adding certificate authority certificates](#) on page 55

[Accessing the Dolby Conference Phone web interface](#) on page 79

7.4 Configuring preferences

You can set up these preferences for your users, or you can allow users to set them themselves.

7.4.1 Using the recent calls list

The Dolby Conference Phone can hold a list of the 100 most recently missed, placed, and received calls if it is configured as an IP conference phone.

About this task

This list contains call information such as remote party identification, time and date of the call, and call duration. The recent calls list is enabled by default. The list is downloaded from the phone to the provisioning server after each call is complete. A count of the missed calls is displayed adjacent to the recent calls list icon. If the list is disabled, the count is not displayed.

The list can be used to redial previous outgoing calls, return missed calls, and save contact information to the contact directory. The list can be cleared manually by individual phone users. The list is retained when the phone reboots or when the phone software is upgraded.

There are several ways to configure the recent calls list.

Procedure

- Set the configuration parameter to enable or disable the recent calls list:

Configuration parameter description	Configuration parameter name
Enable or disable the recent calls list.	Features.RecentCallsEnabled

- Use the phone web interface to enable or disable the recent calls list:
Go to **Settings > Features**, and select the recent calls list **Off/On** option.
- Use the phone user interface to clear the recent calls list:

- a) Tap this sequence:



> Preferences > Clear personal data > Clear recent calls

- b) Tap **Yes** to confirm.

7.4.2 Enabling user feedback

You can set up the Dolby Conference Phone to solicit feedback from users about their experience with the phone, from a simple thumbs-up to requesting a call to provide more detailed feedback. By default, this feature is enabled.

Procedure

- Set these configuration parameters to enable user feedback.

Configuration parameter name	Configuration parameter description
Features.UserFeedback.Enabled	Enable or disable user feedback.
Features.UserFeedback.HotLine	Call user feedback hotline.

7.4.3 Configuring call forwarding

The phone provides a flexible call-forwarding feature that enables you to forward incoming calls to another party.

About this task

There are three ways to apply call forwarding:

- To all calls
- When the system is busy
- When a call is not answered after 40 seconds

There are several ways to configure call forwarding:

Procedure

- Use the web UI:
 - Navigate to **Settings > User Preferences**.
 - Enter the phone number in the appropriate forwarding options.
- Use the phone UI:
 - Access the **Settings** screen .
 - tap **Call Forward**.
 - Enter the phone number in the appropriate forwarding option.
- Set these configuration parameters:

Configuration parameter name	Configuration parameter description
Preferences.ForwardAllNumber	Enter the number where all calls are forwarded.
Preferences.ForwardBusyNumber	Enter the number where the call is forwarded if the system is busy.
Preferences.ForwardNoAnswerNumber	Enter the number where the call is forwarded if the user does not answer.

To disable any of these call-forwarding options, specify a null string:

```
configure Preferences.ForwardAllNumber ""
```

7.4.4 Enabling automatic answer

In certain circumstances, you may be unable to physically answer an incoming call. This feature allows you to configure the system to answer incoming calls automatically after the first ring. The call is answered with the microphone muted.

About this task

There are several ways to configure automatic answer:

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.AutoAnswerEnabled	Enable or disable automatic answer. Set to true or false.

- Use the webUI:
 - Go to **Settings > User Preferences**.
 - Select the automatic answer **Off/On** option.
- Use the phone UI:
 - Access the **Settings** screen.
 - Select the automatic answer **Off/On** option.

7.4.5 Enabling do not disturb

You use the do not disturb feature to temporarily stop incoming calls and turn off audio alerts.

About this task

While do not disturb is turned on, incoming calls are sent a busy signal and are logged as missed.

There are several ways to configure the do not disturb feature.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.DndEnabled	Enable or disable do not disturb behavior. Set to true or false.

- Use the web UI:
 - Go to **Settings > User Preferences**.
 - Select the Do Not Disturb **Off/On** option.
- Use the phone UI:
 - Access the **Settings** screen.
 - Select the do not disturb **Off/On** option.

7.4.6 Setting the language

You can change the language displayed on the phone UI. By default, English is shown.

About this task

There are several ways to set the language.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Preferences.Localization.Language	Possible values include: • English • French • German • Italian • Spanish • Portugal Portuguese • Brazilian Portuguese • Simplified Chinese • Traditional Chinese • Japanese • Korean • Russian • Czech • Hebrew • Arabic

- Use the web interface:
 - a) Navigate to **Settings > User Preferences > Localization**.
 - b) Select the language from the **Language** list.
- Use the phone UI:
 - a) Access the **Settings** screen.
 - b) Tap **Preferences > Language**
 - c) Select the language from the **Language** list.

7.4.7 Setting the time and date display

A clock and calendar are enabled by default. You can display the time and date for your time zone in several formats. The phone will synchronize with the configured time server.

About this task

The available date formats may change, depending on the language selection.

There are several ways to set the time and date display.

Procedure

- Set the configuration parameters:

Configuration parameter name	Configuration parameter description
Preferences.Localization.Timezone	Time zone. Permitted values are: • Pacific/Kwajalein • Pacific/Pitcairn • Pacific/Samoa • Pacific/Honolulu • America/Anchorage • America/Los_Angeles • America/Denver • America/Mazatlan • America/Phoenix • America/Chicago • America/Guatemala • America/Mexico_City • America/Bogota • America/New_York • America/Asuncion • America/Caracas • America/Campo_Grande • America/Santiago • America/Sao_Paulo • America/St_Johns • America/Argentina/Buenos_Aires • Atlantic/South_Georgia • Atlantic/Azores • UTC • Europe/London • Africa/Lagos • Europe/Paris • Africa/Johannesburg • Asia/Jerusalem • Asia/Baghdad • Asia/Tehran • Asia/Muscat • Asia/Kabul • Asia/Karachi • Asia/Kolkata • Asia/Kathmandu • Asia/Dhaka • Asia/Rangoon • Asia/Bangkok • Asia/Singapore • Asia/Tokyo • Australia/Adelaide • Australia/Darwin • Australia/Lord_Howe • Australia/Sydney • Asia/Magadan • Pacific/Auckland • Pacific/Tongatapu
Preferences.Localization.TimeFormat	Time format. Permitted values are: • <i>h:mm ap</i> (For 12-hour format. Enclose in quotes when specifying this value in the configure command.) • <i>h:mm</i> (For 24-hour format)
Preferences.Localization.DateFormat	Date format. Permitted values are:

- Use the web interface:
Go to **Settings > User Preferences > Localization**, and adjust the time and date parameters.
- Use the phone UI:
 - To change the time zone, access the **Settings** screen, tap **Preferences > Time zone**, and select the desired time zone from the list.
 - To change the time format, access the **Settings** screen, tap **Preferences > Time format**, and select the desired format from the list.
 - To change the date format, access the **Settings** screen, tap **Preferences > Date format**, and select the desired format from the list.

7.5 Provisioning options

You can select different provisioning options when preparing the phone for use.

7.5.1 Scheduling a software update

A phone software update initiates whenever there is new phone software on the provisioning server. This triggers a reprovisioning of your phone software. You can also trigger a reprovisioning of the Dolby Conference Phone manually.

About this task

Use any of these methods to trigger a reprovisioning:

Procedure

- Use Dolby Conferencing Console to push a configuration to multiple devices.
Select the devices and then click the push configuration button .
- Set a recurring provisioning time through the configuration files or phone web interface.
- Power off and on the phone.
- Use the phone user interface by tapping **Reboot Phone**.
- Use the phone web interface by selecting **Tools** and then selecting the reboot or reprovisioning link.
- Use the diagnostic shell command system provision.

7.5.2 Setting recurring provisioning

You can set a recurring time for the phone to perform a provisioning check. For example, if you set **Provisioning.Scheduled.StartTime** to 5 and **Provisioning.Scheduled.Duration** to 30, then a provisioning check will randomly occur every day between 5:00 a.m. and 5:30 p.m. By default, there is no scheduled check.

About this task

You can use configuration parameters or the phone web interface to schedule provisioning. You cannot schedule provisioning from the phone user interface.

Procedure

- Set the configuration parameters:

Configuration parameter description	Configuration parameter name
Enables or disables scheduled provisioning	Provisioning.Scheduled.Enable
Sets the provisioning request start time	Provisioning.Scheduled.StartTime
Sets the provisioning delay	Provisioning.Scheduled.Duration

- Use the phone web interface:
Tap **Settings > Provisioning > Scheduled** and set the appropriate parameters.

7.5.3 Disabling the reboot prompt

The reboot prompt is disabled by default, prompting you to reboot due to configuration changes. If you take no action in 20 seconds—for example, cancel the reboot—the phone reboots automatically.

About this task

This procedure disables both the reboot prompt.

You can use either configuration parameters or the web interface to disable the reboot prompt.

Procedure

- Set the configuration parameter:

Configuration parameter name	Configuration parameter description
Provisioning.CfgChange. UserAcknowledgementEnabled	Enables/disables the reboot. Set to false.

- Use the web interface:
 - a) Go to **Settings > Provisioning > On Configuration Change**.
 - b) Set **User acknowledgment enabled** to **Off**.

7.6 Phone access control

The administrative settings can be changed to suit the needs of your organization. You can control access to the phone from the command-line interface (CLI) shell and the phone web interface, as well as the phone itself.

7.6.1 Changing the administrative password

The phone **Administrative Settings** menu is protected with an administrative password. The phone displays a password prompt before it grants access to the subsequent menu options.

About this task

The default administrative password is 1739. Change the administrative password from the default value as soon as possible. If you leave the administrative password unchanged or use weak passwords, the phone is vulnerable to various network attacks.

The Dolby Conference Phone web interface is also protected by the administrative password.

Use one of these methods to change the administrative password.

 **Note:** We recommend that you do not enter passwords in configuration files, as it poses a security risk.

Procedure

- Use the configuration parameter.
Authentication.Local.Admin sets the local administrative password for the phone.
- Use the phone web interface.
Go to **Settings > Authentication > Local Account** and enter a new administrative password.
- Use the phone user interface.
 - a) Tap this sequence:
 >  > **Administrative Settings**
 - b) Enter the current password, and then tap **Change password**.
 - c) Enter the new password, and then confirm the new password.

7.7 Directory options

For making IP PBX calls, Dolby Conference Phone enables users to access a corporate directory and to create a local directory.

7.7.1 Corporate directory

Dolby Conference Phone can connect to a directory server that supports the LDAP version 3.

About this task

You can configure LDAP directory parameters to make your corporate directory available. After it is set up, users can search the corporate directory.

If you are required to access the LDAP server in a secure manner, you must download the appropriate CA certificate to the phone (see “Adding certificate authority certificates”).

Only Microsoft Active Directory 2003 SP2 and later is supported at this time. Microsoft Active Directory supports server-side sorting.

Tip:

Consult your LDAP administrator when changing any corporate directory configuration information. For more information on LDAP attributes, see IETF RFC 4510.

Related information

[Local directory](#) on page 63

[Configuring the corporate directory](#) on page 64

[Creating a local contact directory from the provisioning server](#) on page 65

[Clearing the local contact directory](#) on page 65

[Adding certificate authority certificates](#) on page 55

7.7.2 Local directory

A local directory on the phone can hold up to 1,000 contacts.

You can enter contacts to a local contact directory manually or save recent calls and corporate directory search results.

Note: For saved entries to work, you must format them in a manner that is consistent with your call-control platform. For example, you may need to include the number 9 or a plus sign (+) at the beginning of the entry.

A local directory can also be downloaded from the provisioning server. Users can then manually add entries to the downloaded directory.

Related information

[Corporate directory](#) on page 63

[Configuring the corporate directory](#) on page 64

[Creating a local contact directory from the provisioning server](#) on page 65

[Clearing the local contact directory](#) on page 65

7.7.3 Configuring the corporate directory

Use the web interface or set configuration parameters to configure an LDAP corporate directory.

About this task

You need this information to connect the phone to the LDAP directory:

- The IP address or host name of the LDAP server
- The log-in name and password for the LDAP server
- The base domain name for the search of the LDAP server

 **Note:** Certain parameters are set by default (for example, search filter, scope, time-out, limit, and attributes).

Procedure

1. Do one of the following :

a) Specify LDAP parameters with the web interface:

Navigate to **Settings > Corporate Directory > LDAP**.

b) Set LDAP configuration parameters, either with the diagnostic shell or from Dolby Conferencing Console.

Parameter	Description
Directory.Ldap.URIList	Server Uniform Resource Identifier (URI). Use a comma-separated list to add more than one server. For example: ldap://10.101.21.5,ldap://10.120.21.6
Directory.Ldap.LoginName	Log-in name
directory.Ldap.Password	Password
Directory.Ldap.BaseDN	Base domain name. For example, dc=example, dc=com
Directory.Ldap.SecureConnection	Set to <i>true</i> or <i>false</i> , depending on the security settings on the LDAP server.

2. Reboot the phone.

Results

After the reboot is complete, users can search for contacts in the LDAP server.

Related information

[Corporate directory](#) on page 63

[Local directory](#) on page 63

[Creating a local contact directory from the provisioning server](#) on page 65

[Clearing the local contact directory](#) on page 65

[Rebooting the Dolby Conference Phone](#) on page 87

7.7.4 Creating a local contact directory from the provisioning server

On the provisioning server, you can create a local contact directory for users.

When the phone reboots, the local contact directory is downloaded from the provisioning server. The user can then modify the downloaded directory by adding contacts, either manually or by saving recent calls or corporate-directory search results.

See the *Dolby Conferencing Console Administrator's Guide* for details.

 **Note:** Dolby Conferencing Console version 1.2 and later allows you to provision a local contact directory.

Related information

[Corporate directory](#) on page 63

[Local directory](#) on page 63

[Configuring the corporate directory](#) on page 64

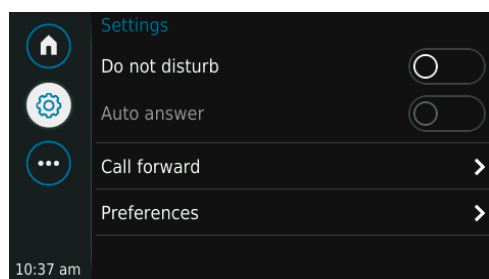
[Clearing the local contact directory](#) on page 65

7.7.5 Clearing the local contact directory

Use the phone user interface to clear the local contact directory.

Procedure

1. Access the Dolby Conference Phone **Settings** screen.



Ask your conferencing service provider for instructions.

2. Select **Preferences > Clear personal data > Clear local contacts**.
3. Tap  to confirm.

Related information

[Corporate directory](#) on page 63

[Local directory](#) on page 63

[Configuring the corporate directory](#) on page 64

[Creating a local contact directory from the provisioning server](#) on page 65

7.8 Configuring IP PBX

To use to make IP audio calls, you must configure IP PBX settings.

About this task

Use the web interface or the Dolby Conferencing Console to configure the account settings for your IP PBX and register the device. This involves modifying the **Account**, **Server**, and **Credential** settings in the **IP PBX** section.

Results

After you finish setting up the device for IP PBX, the phone icon on the home screen no longer appears dimmed.

7.8.1 Prerequisites for configuring IP PBX

Configuring IP PBX for the Dolby Conference Phone requires that you have information about the SIP domain name and the outbound proxy server address.

Your IP PBX service provider or IT department may provide the following information:

- One of the following SIP server/service connection information:
 - SIP domain name only.
 - SIP outbound proxy server address and/or SIP domain name. There may be two addresses for redundancy.
 - SIP domain name and SIP transport type.
 - SIP domain name, outbound proxy server address, and SIP transport type.
- SIP user account (required).
- SIP authentication user name; also known as the digest authentication user name (required).
- SIP authentication user password (required).

7.8.2 Support for SIP high availability

You can configure IP PBX settings for the Dolby Conference Phone to provide uninterrupted server availability and DNS-based redundancy.

The Dolby Conference Phone supports the following for SIP high availability:

- Server failover/failback
- DNS redundancy
- DNS-based load balancing
- NAPTR record support

Server failover has two modes: `ACTIVE_SERVER` and `PREFERRED_ORDER`. The `ACTIVE_SERVER` mode switches/fails over to a secondary server after ending the call. The `PREFERRED_ORDER` mode attempts to fail over to the preferred list of IP addresses during the call. You can specify the server failover mode in the `Sip.Pbx.CallFailoverMode` field.

If your IP PBX service provider configures the Dolby Conference Phone with only the SIP domain, the Dolby Conference Phone performs the DNS SRV query on the domain host name. After four attempts to register to the IP address, the Dolby Conference Phone attempts to register to the next IP in the DNS response.

If your IP PBX service provider configures the Dolby Conference Phone with the SIP domain name and outbound proxy server address configured, the Dolby Conference Phone performs the DNS SRV query on the outbound proxy server. After four attempts to register to the IP address, the Dolby Conference Phone attempts to register to the next IP in the DNS response.

You can specify NAPTR record as the transport type in the `Sip.Account.TransportType` field.

The following table lists the various high-availability configuration options available:

Table 5: High availability deployment configuration options

IP PBX server configuration options	Outcome
Sip.Pbx.Domain as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.Domain field.
Sip.Pbx.Domain as <i>hostname</i> :5060	SIP requests are sent to the IP address resolving the host name configured in the Sip.Pbx.Domain field.
Sip.Pbx.Domain as <i>hostname</i> and Sip.Pbx.OutboundProxy1 as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.OutboundProxy1 field.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as IP address, and Sip.Pbx.OutboundProxy2 as IP address	SIP requests are sent to the IP address configured in the Sip.Pbx.OutboundProxy1 field or the IP address configured in the Sip.Pbx.OutboundProxy2 field.
Sip.Pbx.Domain as <i>hostname</i> or <i>hostname</i> :0 only and Sip.Account.TransportType set to NAPTR record	NAPTR record query (to discover transport) followed by SRV query (to discover port) followed by A query.
Sip.Pbx.Domain as <i>hostname</i> or <i>hostname</i> :0 only and Sip.Account.TransportType set to AUTOSRV	SRV query (to discover port), followed by A query.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as <i>hostname</i> , Sip.Pbx.OutboundProxy1Port as 0, and Sip.Account.TransportType as AUTO	SRV query (to discover port) followed by A query.
Sip.Pbx.Domain as <i>hostname</i> , Sip.Pbx.OutboundProxy1 as <i>hostname</i> , Sip.Pbx.OutboundProxy1Port as 0, Sip.Pbx.OutboundProxy2 as <i>hostname</i> , Sip.Pbx.OutboundProxy2Port as 0, and Sip.Account.TransportType as AUTO.	SRV query (to discover port) followed by A query.

7.8.3 Configuring IP PBX using the web interface

You can use the web interface to configure IP PBX for the Dolby Conference Phone.

Procedure

1. Access the Dolby Conference Phone web interface.
2. Ensure that IP PBX calling is enabled. On the **Settings** tab, navigate to **Features** and choose **Dual Mode** or **PBX Only Mode** for the **Operation Mode**. Changing the operation mode causes the phone to reboot.
3. On the **Settings** tab, navigate to **IP PBX Settings > Account** and enter this information:
 - In the **Display Name** field, enter the device name.
 - In the **Extension number/Address** field, enter the extension or SIP user account.
 - If the dial-in number is different from the extension number/address, enter the dial-in number in the **Display Number** field.
 - If your IP PBX service provider has specified the transport type, choose the appropriate setting from **Transport Type**.
4. On the **Settings** tab, navigate to **IP PBX Settings > Server** and enter this information:
 - If a SIP domain is available, enter it in the **SIP Domain Name** field.
 - If an outbound proxy server is available, enter it in the **Primary Call Server/Outbound Proxy** field. To enable redundancy, enter an alternative outbound proxy server address in the **Secondary Call Server/Outbound Proxy** field.
 - If your IP PBX service provider requires a port other than 5060 or 5061, enter the port numbers in the **Primary Server/Outbound Proxy Port** and **Secondary Server/Outbound Proxy Port** fields.

- To restrict or reorder the audio codecs used, modify the list in the **PBX Codecs List**. By default, the list value is *G722, G711U, G711A, G729AB*. For example, to only use only G.711, edit the list to include only *G711U, G711A*.

For specific instructions, see the configuration guide for your conferencing service provider or call control platform.

- Expand the **Credential** section, and in the **User Credential Name** and **User Credential Password** fields, enter the SIP authentication user name and password, respectively.

If your IP PBX service provider requires a specific realm for authentication, enter it in the **Server Realm** field.

For specific instructions, see the configuration guide for your call control platform.

7.8.4 Configuring IP PBX using the Dolby Conferencing Console

You can use the Dolby Conferencing Console to configure IP PBX for the Dolby Conference Phone.

Procedure

- Create a profile, and configure these parameters:

- In **Features > Operation Mode**:

Configuration parameter	Setting
Features.OperationMode	PBX Only Mode or Dual Mode

- In **IP PBX Settings > Server**:

Configuration parameter	Setting
Sip.Pbx.Domain	The name of the SIP domain, if one is available.
Sip.Pbx.OutboundProxy1	The outbound proxy server address.
Sip.Pbx.OutboundProxyPort1	The outbound proxy server port number. Change the default value of 5060 if your IP PBX service provider requires a different port.
Sip.Pbx.OutboundProxy2	An alternative outbound proxy server address, if one is available.
Sip.Pbx.OutboundProxyPort2	The alternative outbound proxy server port number. Change the default value of 5060 if your IP PBX service provider requires a different port.
Sip.Pbx.SipUriScheme	Configures the URI scheme. Options are SIP and SIPS. The default is On = SIP.
Sip.Pbx.OnlyAcceptCallServerEvents	Configuration to allow SIP events from the call server only. If set to True, allows SIP events from the call server only. If set to False, allows SIP events from any address.
Sip.Pbx.CallFailoverMode	Configures call failover mode. Select one of these: <i>Active Server</i> - Call is always sent to the active server. <i>Preferred Server's Order</i> - Call is sent to the server in the order of preference.
Sip.Pbx.RegistrationTimeout	SIP registration time out. The range is from 4-59. The default is 32.
Sip.Pbx.CallFailoverTimeout	Call failover timeout. Range is 4-60. The default is 4.

Configuration parameter	Setting
Sip.Pbx.Codecs	Audio codecs list. Audio codecs used: <i>G722, G711U, G711A, G729AB</i> by default. Modify the list to specify allowable codecs or change order of preference. For example, to use only G.711, edit the list to include only <i>G711U, G711A</i> .
Sip.Pbx.InBandDTMF	Default is off. When set to <i>Off</i> out-of-band dual-tone multifrequency (DTMF) is enabled.
Sip.Pbx.EnableP2PSIPVideo	Enables Peer-to-peer SIP video calls through IP-PBX server. Default is off.
Sip.Pbx.ContentShareEnabled	Configures content share for SIP video calls is enabled or not. The default is <i>On</i> .
Sip.Pbx.URIDialingEnabled	Enables calls by URI dialing.
Sip.Pbx.URIDialingTransport	URI dialing transport. Select one of these: • <i>None</i> • <i>TLS</i> • <i>TCP</i> • <i>UDP</i>
Sip.Pbx.URIDialingProxy	URI dialing proxy server address.
Sip.Pbx.EnableURIDialingProxy	URI Dialing Proxy server to route all the SIP URI calls, ignoring URIProxy parameter.
Sip.Pbx.TLSCertNameMatchingPolicy	Matches the DNS name with the certificate name using a wildcard rule. Select: • <i>Strict Wildcard Matching</i> - If the certificate name is <i>*.abc.com</i> and the DNS name is <i>1234.abc.com</i> they are matched, but DNS name <i>abc.com</i> does not. • <i>Lenient Wildcard Matching</i> - Includes <i>Strict Wildcard Matching</i> and allows DNS name <i>abc.com</i> to be matched with <i>*.abc.com</i> • <i>Ignore name matching</i> - No name matching at all. If the certificate is valid it is accepted. The default is <i>Strict Wildcard Matching</i> .

2. Assign the profile you created to the device pool that contains the phones you are configuring.

3. For each phone in the device pool, set these parameters:

- In **IP PBX Settings > Account:**

Configuration parameter	Setting
Sip.Account.DisplayName	The name of the device.
Sip.Account.UserName	The extension or SIP address.

Configuration parameter	Setting
Sip.Account.DisplayNumber	The dial-in number, if different than the extension number/address.
Sip.Account.TransportType	The transport type. Select <i>AUTO</i> <i>UDP</i> <i>TCP</i> <i>TLS</i> <i>NAPTR</i> The default value is <i>AUTO</i> .

- In **IP PBX Settings > Credential:**

Configuration parameter	Setting
Sip.Credential.Name	The SIP authentication user name
Sip.Credential.Password	The SIP authentication user password
Sip.Credential.Realm	The realm for authentication, if your IP PBX service provider requires a specific realm

For specific instructions, see the configuration guide for your call control platform and the *Dolby Conferencing Console Administrator's Guide*.

7.8.5 Adjusting codec preferences

Dolby Conference Phone allows you to adjust the IP PBX codec preferences using the `Sip.Pbx.Codecs` parameter.

About this task

When the device functions as a VoIP phone, the `Sip.Pbx.Codecs` parameter determines which audio codecs are used and their priority order. By default, this parameter is set to *G722*, *G711U*, *G711A*, *G729AB*. You can add or remove codecs and change the order of codecs in the list. For example, you might want to move *G729AB* to the beginning of the list to reduce VoIP traffic across the wide area network (WAN).

This parameter is not required when the phone is used in **Conferencing Only Mode**.

Procedure

- From the web interface:
 - On the **Settings** tab, expand **IP PBX Settings**.
 - Expand the **Server** section.
 - In the **PBX Codec List** field, add, remove, or change the order of codecs as needed.
- From the Dolby Conferencing Console:
 - From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - Select the device you want to edit.
 - In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - Expand **IP PBX Settings > Server**.
 - Drag the **PBX Codec List** to the device configuration.
The `Sip.Pbx.Codecs` parameter appears in the device configuration.
 - In the **Sip.Pbx.Codecs** field, add, remove, or change the order of codecs as needed.

7.8.6 Configuring voice activation detection

The Dolby Conference Phone by default signals G.729AB (G.729A with Annex B) to its peer. If your IP PBX configuration requires the device to signal G.729A, you can set the `Sip.Vad.SignalAnnexB` configuration parameter to `false` to override the default behavior.

About this task

The Dolby Conference Phone supports the G.729AB codec, but it also allows Annex B in its Session Description Protocol (SDP) to be turned off.

You can perform this procedure from the web interface or from Dolby Conferencing Console.

Procedure

1. From the web interface:
 - a) From the **Settings** tab, expand the **IP PBX Settings** section, and then expand the **VAD** section.
 - b) Turn off `Signal G729 Annex B`.
This option controls the `Sip.Vad.SignalAnnexB` parameter.
2. Alternatively, from Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Navigate to **IP PBX Settings > VAD**.
 - e) Drag `Signal G729 Annex B` to the device configuration in the upper right-hand side of the screen.
The `Sip.Vad.SignalAnnexB` parameter appears in the device configuration.
3. Turn the `SIP.Vad.SignalAnnexB` parameter off.

Results

The device now signals peers using G.729A.

7.8.7 Configuring IP PBX for SIP URI dialing

You can use the web interface or the Dolby Conferencing Console to configure IP PBX for dialing using a SIP URI.

About this task

By default, SIP URI dialing is enabled and the configuration parameter is set to **TRUE**.

Procedure

- To disable SIP URI dialing from the web interface:
 - a) On the **Settings** tab, expand **IP PBX Settings**.
 - b) Expand the **Server** section.
 - c) Turn off **Enable SIP URI Dialing**.
- To disable SIP URI dialing from the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters.
 - b) Select the device you want to edit.
 - c) In the left navigation bar, click the **Edit** icon.
The device configuration screen displays.
 - d) Expand **IP PBX Settings > Server**.

- e) Drag **Enable SIP URI Dialing** to the device configuration.
The `Sip.Pbx.URIDialingEnabled` parameter appears in the device configuration.
- f) Set `Sip.Pbx.URIDialingEnabled` to **FALSE**.

Making a call using SIP URI dialing

You can make a call using the SIP URI dialer after configuration through the web interface or the Dolby Conferencing Console. SIP URI dialing works for unregistered and unauthenticated SIP lines.

About this task

The Dolby Conference Phone keyboard field supports a maximum of 128 characters.

Procedure

1. On the Dolby Conference Phone UI, tap the **URI** button.
The onscreen keyboard displays with **sip:**.
2. Enter the URI, and tap the **Call** button to join the call.

7.9 Optimizing Dolby Conference Phone for small rooms

In some cases, particularly in small or huddle conference rooms, participants stationed closer to the can sound much louder on the remote end than participants who are located further away from the phone.

About this task

If some conference participants in a small room sound much louder than others on the remote end, you can configure the room size to make voice volumes more uniform. To adjust microphone pickup for small or huddle conference rooms, follow these instructions:

Procedure

- From the web interface:
 - a) On the **Settings** tab, navigate to **Dolby Voice Conferencing Service > Audio**.
 - b) Set **Room size** to **Small**.
This option controls the `Dvms.Audio.RoomSize` parameter.
- From the Dolby Conferencing Console:
 - a) From the home screen, select the device pool that contains the device for which you want to edit parameters, and select the device to configure.
 - b) In the left navigation bar, click the **Edit** icon.
The **Device configuration** screen displays.
 - c) Navigate to **Dolby Voice Conferencing Service > Audio**.
 - d) Drag **Room size** to the device configuration.
 - e) Set the `Dvms.Audio.RoomSize` parameter to **Small**.

Related information

[Conference room requirements](#) on page 16

7.10 Disabling Bluetooth advertisement

By default, the Dolby Conference Phone is configured to allow the phone to advertise a Bluetooth low energy (Bluetooth LE) signal that can be detected by other Bluetooth-enabled devices. When used in conjunction with other Bluetooth-enabled devices, such as mobile phones, this feature can allow the phone to be controlled remotely. You can turn it off for security reasons, if needed.

About this task

Check with your conferencing service provider before disabling Bluetooth advertisement.

Procedure**1.** From the web interface:

- a) On the **Settings** tab, navigate to **Dolby Voice Conferencing Service > Service**.
- b) Turn **Allow bluetooth advertisement** off.

This option controls the `Dvms.Service.EnableBluetoothAdvertisement` parameter.

2. Alternatively, from Dolby Conferencing Console:

- a) From the home screen, select the device pool than contains the device for which you want to edit parameters.
- b) Select the device you want to edit.
- c) In the left navigation bar, click the **Edit** button.
The **Device configuration** screen displays.
- d) Navigate to **Dolby Voice Conferencing Service > Service**.
- e) Drag the **Allow bluetooth advertisement** to the device configuration.
The `Dvms.Service.EnableBluetoothAdvertisement` parameter appears in the device configuration.
- f) Turn the parameter off.

Related information

[Bluetooth LE initialization and advertisement](#) on page 91

[Confirming Bluetooth LE initialization and advertisement](#) on page 91

8

Administering the phone

You can view status information, diagnostic information, and error messages from the Dolby Conference Phone. You can perform administrative tasks, such as exporting a phone configuration or downloading logs, from the phone web interface.

- [Error messages](#)
- [Viewing the status menu](#)
- [Accessing the Dolby Conference Phone web interface](#)
- [Dolby Conference Phone web interface main page](#)
- [Setting the Dolby Conference Phone operational mode](#)
- [Viewing and managing event logs](#)
- [Exporting the Dolby Conference Phone configuration](#)
- [Rebooting the Dolby Conference Phone](#)
- [Call and audio statistics](#)
- [Running Dolby Conference Phone diagnostics](#)
- [Administering with the diagnostic shell](#)
- [Bluetooth LE initialization and advertisement](#)
- [Confirming Bluetooth LE initialization and advertisement](#)
- [Collection of usage data](#)
- [Phone memory resources](#)
- [Resetting to factory defaults](#)
- [LLDP and supported type-length values](#)
- [Trusted certificates and issuers](#)

8.1 Error messages

Most of the errors listed here are logged to the phone log file. However, if you are having trouble connecting to the provisioning server, the phone may not be able to upload every error.

Error message	What the administrator should do
Parameter does not exist.	You have added a configuration parameter that is not supported by the phone in the configuration file. Remove it, and try again.
Invalid value.	Enter a valid value.
Invalid value range.	Enter a value within the valid range.
Invalid value type.	You entered a value of a different type (for example, using an integer instead of a Boolean value). Enter a value of the correct type.
Invalid value length.	You entered a string that is too short or long. Enter a valid string.
Write access denied.	You are trying to set a configuration parameter in the configuration file that cannot be changed in that way. For information on parameters, see Configuration parameters on page 51.
Invalid numeric value.	You entered a numeric value that does not match the parameter range rule. (For example, it is larger than the maximum value or smaller than the minimum value.) Enter a valid numeric value.
Invalid alphabetic value.	You entered a value that contains invalid characters for an alphabetical parameter. Enter a valid alphabetical value.
Invalid alphanumeric value.	You entered a value that contains invalid characters for an alphanumeric parameter. Enter a valid alphanumeric value.
IPPBX service is not available. Try again later.	The IP PBX is not available, or the phone is not registered to the IP PBX. Contact the IP PBX administrator.
Invalid input.	You entered an invalid value. Verify and try again.
Invalid server address. Try again.	You entered an invalid provisioning server address. Verify and try again.
Reboot is required to apply new configuration, would you like to reboot now? Phone will reboot automatically in x seconds.	A change in the phone configuration parameter required the phone to reboot. If you want to delay rebooting the phone, tap No immediately.
The services application has stopped responding, press Debug to stop rebooting.	See Fatal error issues on page 112.
Unable to download software image file.	See Software update issues on page 112.
Invalid signature on software image file.	See Software update issues on page 112.
Unable to download meta file.	See Software update issues on page 112.
Invalid software image file.	See Software update issues on page 112.
Invalid meta file.	See Software update issues on page 112.

Error message	What the administrator should do
Software update failed. Contact system administrator.	See Software update issues on page 112.
Invalid IP.	You entered an IP address that does not conform to the IP address format requirement. Enter a valid IP address.
Invalid IP address. Try again.	You entered an invalid IP address. Verify the address, and enter a valid IP address.
Invalid domain value.	You entered a domain name that does not conform to the domain-name format requirement. Enter a valid domain name.
Invalid URL value.	You entered a URL that does not conform to the URL format requirement. Enter a valid URL.
Failed to connect to server.	The phone cannot connect to the provisioning server. Check network connections.
Failed to connect to the server due to SSL error.	The phone cannot connect to the HTTPS provisioning server. Check network connections, verify that the provisioning server CA certificate was downloaded to the phone, and confirm that the phone time is current.
Invalid credentials. Try again.	The phone cannot connect to the provisioning server. For an HTTPS provisioning server, verify that a CA certificate was downloaded to the phone from the provisioning server. Check provisioning server credentials. Check that phone credentials match the device pool credentials setup on Dolby Conferencing Console.
Could not find master configurations.	The master configuration files are not on the provisioning server.
Invalid configurations. Press the "More Options" button for more information.	See Provisioning issues on page 105.
Provisioning failed. Try again.	There was an error during the first-time provisioning of the phone.
Server address is required for static provisioning.	If you want to use static provisioning, you must enter a provisioning server IP address.
Write access not allowed for provisioning.	A <i>.json</i> file on the provisioning server contains a configuration parameter that is not writable to the current <i>.json</i> file on the phone. The parameter is not allowed to be configured from the provisioning server. See Provisioning issues on page 105.
Failed to get or set configuration.	The configuration files or certificates listed in the master configuration file are not on the provisioning server.
Not allowed to update provisioned configuration.	There was an attempt to update a configuration parameter that is presently in the configuration file on the provisioning server. Remove these parameters from the configuration files on the provisioning server, and try again.
Write not allowed for provisioning.	You specified a parameter in the configuration file on the provisioning server that cannot be modified by provisioned interface. Remove the parameter from the configuration file, and try again.

Error message	What the administrator should do
IEEE 802.1X is unconfigured but challenged.	You have to configure the IEEE 802.1X user identity and password to be able to use the phone on the current network.
Unsupported IEEE 802.1X authentication method required by RADIUS server.	The current network requires an authentication method that is not supported by the phone. To use the phone, you must move it to another network or get the network administrator to change the current authentication method to one that the phone supports. For a list of supported authentication methods, see Configuring 802.1X after out-of-the-box setup on page 42.
IEEE 802.1X authentication failed due to wrong credentials.	Verify that the correct credential has been provided for 802.1X authentication.
IEEE 802.1X authentication failed due to certificate validation.	Verify that the correct CA certificate is installed on the phone prior to 802.1X authentication if you are using EAP-PEAP authentication. Verify the installed certificate on the phone through the web interface. Select Status > System > Device and scroll down to CA Certificates Details. If the certificate is successfully installed, you should see the CA certificate for the 802.1X TLS authentication. If the certificate is not present, you must set up the configuration file for the phone to download the CA certificate. See Adding certificate authority certificates on page 55.
ERROR: Certificate <certificate fingerprint> is expired: <ddd mmm dd hh:mm:ssyyy>	This message displays on the CA certificate status screen only. This certificate has expired and needs to be replaced on the phone. Contact the issuer (for example, Dolby or your conferencing service provider) for a new certificate.
WARNING: Certificate <certificate fingerprint> will expire soon: ddd mmmdd hh:mm:ss yyyy>	This message displays on the CA certificate status screen only. This certificate will expire within the next seven days. Contact the issuer (for example, Dolby or your conferencing service provider) for a new certificate.
Unsupported authentication method.	An unsupported 802.1X authentication method was used in the configuration files. Change it to a supported method, and reboot the phone.
Failed to download PAC file.	The PAC file did not download. The administrator needs to check and, if necessary, correct the URL for downloading the PAC file; confirm that the phone can reach that URL; and then download the PAC file.
Invalid PAC file.	The downloaded PAC file is invalid.
PAC syntax error.	The PAC file has syntax errors and cannot be processed. Correct the errors and try again.
PAC file too large.	The PAC file size is greater than 64 kB. This is the maximum supported size.
Unresolvable hostname error	The proxy server host name is not resolvable. Verify the server name in the PAC file.
Unreachable error	The proxy server is unreachable. Verify the server name in the PAC file. Verify that the proxy server is online.
Authentication error	Proxy authentication failed. Verify the user name and/or password for the proxy server.

Error message	What the administrator should do
Higher logging level is enabled. Performance might be impacted due to excessive logging. Contact administrator to reset.	Reset logging level. See Log levels on page 82.
DHCP lease is not obtained on secondary VLAN.	Check whether the secondary VLAN ID is configured properly. If so, check whether the DHCP server is online on the secondary VLAN.
IP address conflict is detected on secondary VLAN.	As DHCP is supported only on the secondary VLAN, this is unlikely to happen. When it does happen, check the DHCP server logs to find out where the source of the conflict is.
Secondary VLAN failed.	General error.
Secondary VLAN is shadowed by primary.	Even though the secondary VLAN is enabled, it is the same as the primary VLAN. Informational error only.
Default gateway is unreachable on secondary VLAN.	As DHCP is supported only on the secondary VLAN, this is unlikely to happen. When it does happen, check if the assigned default gateway is online.
Not allowed to update provisioned configuration.	You attempted to make a local configuration change to a parameter that is provisioned, where local means either on the web interface or through a shell command. The phone user interface disables entry of the provisioned parameters, so you can view the value but not make a change.
Invalid PBX codec list.	You entered an invalid IP PBX codec list as the last entry in the table.
Invalid domain.	You entered an invalid SIP domain name.
Invalid URL.	You entered an invalid URL.
Duplicate parameter.	The provisioned configuration file contains multiple instances of the same parameter. Remove the duplication, and try again.

8.2 Viewing the status menu

Debugging of a single phone may be possible by examining the phone status menu.

Procedure

1. Tap the settings button on the home screen.



2. Tap the advanced settings button and then choose **About > Status** to view the **Status** menu:



3. Scroll to one of the **Status** menu items, and tap the desired item.



If  is displayed on the home screen and you tap it, you are taken directly to the **Status** menu.

Results

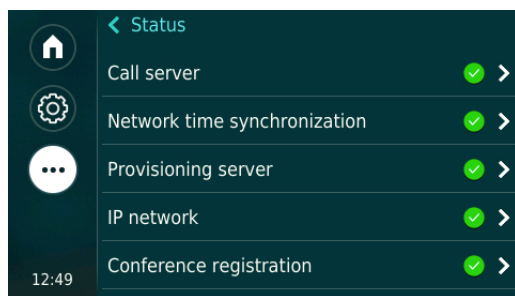
In the **Status** menu, you can view your phone communication status with external systems.

Status icons  and  display for following:

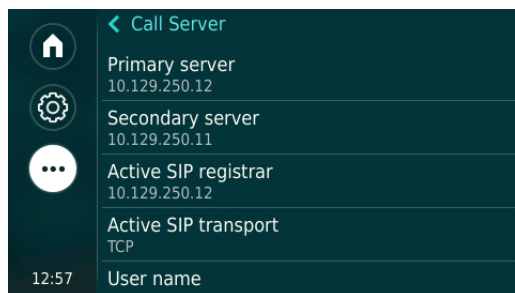
- DHCP server

- Provisioning server
- Call server
- Directory server
- Conferencing registration
- Network time synchronization
- IP network
- Secondary VLAN
- CA certificates
- Log file upload
- Logging impact
- Reboot required
- Dolby Satellite Microphones
- Bluetooth
- Phone software update (if an error occurred when downloading the image or the user ignored the request)

This figure shows good communications with all external systems.



If any communications have failed, they are marked as , and a message directing the user to contact the system administrator appears at the top of the **Status** list screen (shown in the previous figure).



 **Note:** Tap  to return to the **Settings** menu from the **Advanced settings** menu.

8.3 Accessing the Dolby Conference Phone web interface

Use the web interface to view and change Dolby Conference Phone settings.

About this task

You can access the Dolby Conference Phone web interface using any of these web browsers:

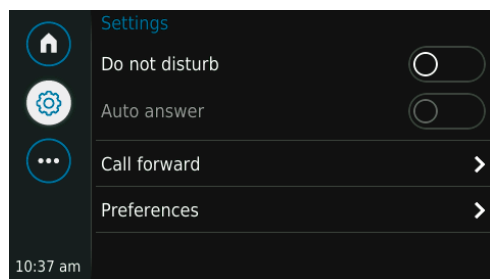
- Microsoft Internet Explorer 10.0.x or later on Windows-based computers (backward compatibility not supported)
- Google Chrome 31.0.x or later
- Apple Safari 7.0.2.x

Note: The Dolby Conference Phone web interface has been optimized for modern browsers. The best experience is obtained using one of the supported browsers. Versions 8 and 9 of Internet Explorer work in a degraded fashion; support for these browsers is deprecated and will be removed in a future release.

By default, access to the web interface is enabled. You can disable access only through the configuration files by changing `Network.Services.WebServerMode`.

Procedure

1. Determine the phone IP address:
 - a) Access the Dolby Conference Phone **Settings** screen.



Ask your conferencing service provider for instructions.

- b) Tap this sequence:



The IP address displays.

2. Open one of the supported web browsers.
3. Enter `http://your-phone-IP-address` in the browser address bar.
The log-in page displays.

Note: If the browser displays a certificate-related error message, such as **Site is not secure** or **Your connection is not private**, then take the required steps to bypass the error page and proceed to the log-in page.

4. Log in as the administrator. (Default credentials are *admin* and *1739*.)
The web interface main page displays. Expand the headings and hover the mouse over a parameter name to see a parameter description, along with the name of the corresponding configuration parameter.

Note: We recommend that you change the default administrative password.

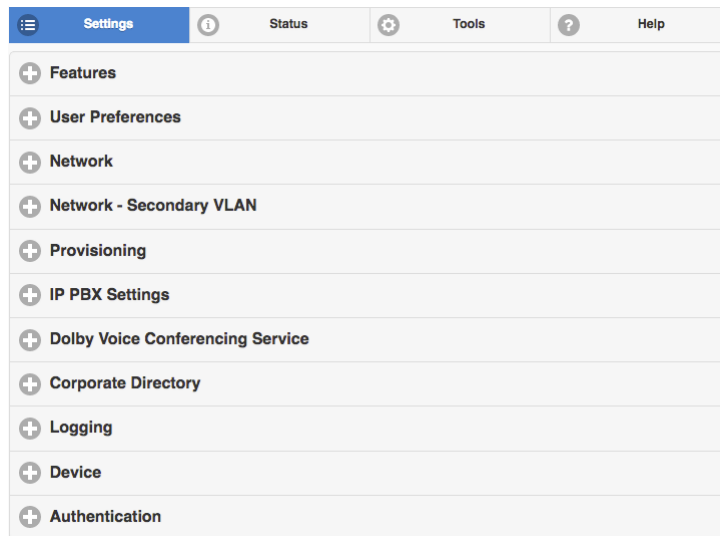
Related information

[Changing the administrative password](#) on page 62

8.4 Dolby Conference Phone web interface main page

This page provides you with specific information about the phone that you logged into.

Figure 4: Dolby Conference Phone web interface main page



Contextual help on other pages of the phone web interface provides you with instructions on how to navigate the menus and detailed explanations of the current configuration of the phone.

You can change the phone settings from the **Settings** tab. You can view the phone status from the **Status** tab, where you will see these icons:



Functioning normally



Requires attention



Not applicable

The warning icon displays on the phone web interface as shown in this figure.



You can perform other tasks from the **Tools** tab, such as exporting the configuration files, rebooting the phone, and viewing the log and help files.

You can view the details of CA certificates stored on the phone through the phone web interface. Select **Status > System > Device**, and scroll down to **CA Certificates Details**.

8.5 Setting the Dolby Conference Phone operational mode

Set the operational mode to control whether Dolby Conference Phone operates only as a conferencing endpoint, only as a SIP endpoint only for IP PBX calls, or both.

Procedure

- Use the web interface:
 - a) Access the Dolby Conference Phone web interface.
 - b) Go to **Settings > Features** and, for **Operation Mode**, select **PBX Only Mode, Conferencing Only Mode** or **Dual Mode**.
- Use the diagnostic shell:
 - a) Open the diagnostic shell.

- b) Use the **configure** command to set the `Features.OperationMode` configuration parameter to one of the following values:

- DUALMODE
- CONFONLYMODE
- IPPBXONLYMODE

Example: `configure Features.OperationMode DUALMODE`

Related information

[Operation modes of Dolby Conference Phone](#) on page 13

8.6 Viewing and managing event logs

The Dolby Conference Phone has a sophisticated event-logging system for diagnostics and troubleshooting.

8.6.1 Event logging

The Dolby Conference Phone generates event-log entries on a continuous basis, both for ordinary operation events and for error conditions. The event logs provide an essential troubleshooting tool. Dolby Customer Support or your conferencing service provider might ask you to share logs during problem resolution.

Log files are stored in a flash file system on the Dolby Conference Phone. You can use the web interface to download the logs directly from the device.

If you are using automatic provisioning with Dolby Conferencing Console, Dolby Conference Phone automatically uploads log entries to Dolby Conferencing Console, and you view the logs with the Dolby Conferencing Console web interface.

Dolby Conference Phone supports the syslog protocol. You can configure the phone to automatically upload logs to a syslog receiver.

For information about viewing logs on Dolby Conferencing Console, see the .

Related information

[Log levels](#) on page 82

[Uploading log files to a provisioning server](#) on page 83

[Audio logging](#) on page 84

[syslog logging](#) on page 85

[Viewing log files](#) on page 85

8.6.2 Log levels

Each of the components of the Dolby Conference Phone software is capable of logging events of different severity. This allows you to capture low-severity events in one component, and only high-severity events for other components.

Log-level configuration parameters enable you to indicate how much logging is done for each software component. The range of log levels, in order of detail and verbosity from lowest to highest, includes NONE, EMERGENCY, ALERT, CRITICAL, ERROR, WARNING, NOTICE, INFO, DEBUG, and DEBUG1. (DEBUG1 contains maximum verbose trace information.) By default, all log-level parameters are set to INFO.

 **Note:** You should not modify the default log levels unless directed to by Dolby Customer Support. Inappropriate logging levels can cause performance issues.

In the log files, messages are prefixed by timestamp and log level.

- `debug`: Indicates verbose traces generated only when log level is set to DEBUG
- `info`: Indicates normal phone behavior

- **notice**: Indicates minor issues or failures, but could be normal behavior
- **warn**: Indicates that an error may occur if action is not taken
- **err**: Indicates that a failure or loss of functionality occurred
- **crit**: Indicates that an unrecoverable failure of a semi-important component occurred
- **alert**: Indicates that an unrecoverable failure of an important component occurred
- **emerg**: Indicates that a complete unrecoverable system failure occurred

The **warn**, **err**, **crit**, **alert**, and **emerg** messages indicate problems of increasing severity.

A full description of all log-level parameters can be found under the **Help** section of the web interface.

To change the logging level of the SIP communication module to the **DEBUG** level, set `Logging.Level.Sip` to **DEBUG**. You can also make this change in the **Logging** section of the web interface.

 **Note:** If you set logging to a more verbose level for testing or troubleshooting, then when testing or troubleshooting is complete, remember to return logging levels to their default values.

Related information

[Event logging](#) on page 82

[Uploading log files to a provisioning server](#) on page 83

[Audio logging](#) on page 84

[syslog logging](#) on page 85

[Viewing log files](#) on page 85

8.6.3 Uploading log files to a provisioning server

If you are using the automatic provisioning method with a provisioning server other than Dolby Conferencing Console, you may decide to upload the logs from each Dolby Conference Phone to the provisioning server so that you can collect the logs from all devices onto one place.

 **Note:** This section does not apply if you are using plug and play provisioning or if you are using Dolby Conferencing Console. Dolby Conference Phone automatically uploads log entries to Dolby Conferencing Console, and Dolby Conferencing Console automatically manages storage for log entries.

If you want the log files written to the provisioning server, you must enable log uploading (set `Logging.Upload.Enable` to `true`), and specify an upload directory for the log files (set `Provisioning.Server.UploadDir`). The directory must exist. To use the web interface, turn on log uploading in the **Logging** section, and specify the upload directory in the **Provisioning > Server** section.

You can specify a directory as follows:

`/logs:`

The log files are uploaded to the provisioning server logs directory. If the provisioning server is set to `ftp://10.12.122.124`, the upload path will be `ftp://10.12.122.124/logs`.

`/logs/[MAC] or /logs/[SN]`

The log files are uploaded to the provisioning server `logs/mac-address` or `logs/serial-number` directory. Each phone will upload its logs to a separate folder. For example, a phone with serial number `D0230030` will upload its logs to `../logs/d0230030`.

Log files are saved on the provisioning server with the serial number of the phone prepended to the file name. By default, log files are uploaded every five minutes, but you can change the interval with `Logging`.

`Upload.Interval`. There can be up to two log files. Log messages are written to the files in a cyclic manner: *messages*, then rename *messages* to *messages.0*, and then back to *messages*. For example, *D0025436-messages* and *D0025436-messages.0* are uploaded to the provisioning server every five minutes. The maximum log file size is 2 MB.

Check the **Status** menu to see whether or not the log file upload to the provisioning server was successful (see [Viewing the status menu](#) on page 78).

Related information

[Event logging](#) on page 82

[Log levels](#) on page 82

[Audio logging](#) on page 84

[syslog logging](#) on page 85

[Viewing log files](#) on page 85

8.6.4 Audio logging

For the audio module of the Dolby Conference Phone software, you can enable logging through configuration files or the web interface.

 **Note:** Turn on audio logging only if requested by Dolby Customer Support.

Audio logging tracks these audio characteristics:

- Echo reference input
- Echo reference output
- Jitter buffer input
- Jitter buffer output

A full description of all audio logging configuration parameters can be found in the **Help** section of the web user interface.

By default, all audio logging is disabled. If you want the audio log files written to the provisioning server, you need to set the audio logging mode, `Logging.Audio.Mode`. The supported modes are `DISABLED`, `STATISTICS`, and `DIAGNOSTIC`.

If the audio logging mode is set to `STATISTICS` or `DIAGNOSTIC`:

- A call statistics log containing per-second statistical information (including jitter buffer and packet loss) is uploaded to the provisioning server. For example, the file *D0025454-call_stats_b502bb32-2bdd-49da-8ae7-0d89ff295a01.json* is uploaded from the device numbered D0025454.
- An audio dump of preprocessing statistics that contains metadata about the preprocessing algorithm is uploaded to the provisioning server. For example, the file *D0025454-vsv_stats_20140529163458.bin* is uploaded from the device numbered D0025454.

If the audio logging mode is set to `DIAGNOSTIC`, the recorded diagnostic data is uploaded to the provisioning server. For example, the file *D0025454-audiolog_20140529141935.gz* is uploaded from the device numbered D0025454. In this audio logging mode, a diagnostics icon displays during calls in the top-right corner of the screen. If users have any issues in any call, and the conferencing service provider is BT, they should tap the icon and select an issue from the displayed list.

 **Note:** The diagnostics icon on the Dolby Conference Phone display should be used only if BT is the conferencing service provider. Upload of diagnostic information fails when attempted with any other service provider.

 **Note:** The STATISTICS audio logging mode may impact the end of a call (when data is uploaded to the provisioning server). The DIAGNOSTIC audio logging mode impacts the entire call, adding approximately 10–15% overhead.

Related information

[Event logging](#) on page 82

[Log levels](#) on page 82

[Uploading log files to a provisioning server](#) on page 83

[syslog logging](#) on page 85

[Viewing log files](#) on page 85

8.6.5 syslog logging

syslog is a standard for forwarding log messages in an IP network. You can configure the Dolby Conference Phone to ship log entries to a syslog receiver.

Because syslog is supported by a wide variety of devices and receivers, syslog can be used to integrate log data from many different types of systems into a central repository.

The syslog messages contain much of the same information as the Dolby Conference Phone logs, with the exception of the events during the early stage of the phone initialization (as the device is not yet configured to send syslog messages). syslog data is sent in clear text.

By default, syslog is disabled. (`Logging.RemoteSysLog.Enable` is `false`.) To use syslog, you must set `Logging.RemoteSysLog.Enable` to `true` and set the syslog server address (`Logging.RemoteSysLog.Address`). Messages relating to developer logs and intermodule communication are sent to the DEBUG level. Messages pertaining to user action (if logged) and module initialization (if any) are available in the INFO level. Error and warning messages are logged in ERROR and WARNING, respectively.

For more information on the syslog protocol, see RFC 3164.

Related information

[Event logging](#) on page 82

[Log levels](#) on page 82

[Uploading log files to a provisioning server](#) on page 83

[Audio logging](#) on page 84

[Viewing log files](#) on page 85

8.6.6 Viewing log files

You can download and view log files stored in the Dolby Conference Phone or on Dolby Conferencing Console.

If logs are located here	Do this
Dolby Conferencing Console	Use the Dolby Conferencing Console web interface. See the .
Dolby Conference Phone flash memory  Note: All logs originate from the device flash memory. The flash memory always has the most recent log entries, some of which may not have been uploaded yet.	1. Access the Dolby Conference Phone web interface. 2. Select Tools, and then click Download Logs. 3. Save the <i>tar.gz</i> file, and unpack it with your preferred decompression utility. The TAR archive contains one or two log files, named <i>message</i> and <i>message.0</i>. The file named <i>message</i> contains the latest entries. 4. Open the log files with your preferred text editor.

Related information

[Event logging](#) on page 82

[Log levels](#) on page 82

[Uploading log files to a provisioning server](#) on page 83

[Audio logging](#) on page 84

[syslog logging](#) on page 85

[Accessing the Dolby Conference Phone web interface](#) on page 79

8.7 Exporting the Dolby Conference Phone configuration

You can export the files that represent the Dolby Conference Phone configuration by using the Dolby Conference Phone web interface.

About this task

This is primarily a tool to help you create multiple configuration files.

To export the current phone configuration:

Procedure

1. From the phone web interface home page, select **Tools** > **Export** from the navigation menu.

A web page similar to the one shown in this figure displays.

The screenshot shows the web interface of the Dolby Conference Phone. At the top, there is a navigation bar with four items: 'Settings' (with a list icon), 'Status' (with an 'i' icon and a red exclamation mark), 'Tools' (with a gear icon and highlighted in blue), and 'Help' (with a question mark icon). Below the navigation bar, there are four large circular icons: 'Download Logs' (with a download icon), 'Reboot' (with a power icon), 'Factory Reset' (with a circular arrow icon), and 'Provision' (with a refresh icon). Below these icons, there is a section titled 'Export' with a minus icon. The 'Export' section contains the following text: 'This wizard helps the system administrators to generate configuration files that can be placed on the Provisioning Server for bulk configuration. For more information on the configuration parameters, please refer to this [link](#).' Below this text, there is a prompt: 'Enter the following optional information for the master configuration file:'. There are two input fields: 'Software Image URL' with an information icon, and 'Certificate(s) URL' with an information icon. To the right of these fields is a button labeled 'Generate JSON files'. At the bottom of the 'Export' section, there are three expandable sections: 'Screenshot', 'Certificates', and 'Diagnostics', each with a plus icon.

- (Optional) Enter the fully qualified phone software image URL (for example, *dvcp-dev-2.2.0.123.tar.gz* or *http://10.203.131.56/something/dvcp-dev-2.2.0.38.tar.gz*).
- (Optional) Enter the certificate URLs separated by commas (for example, *cert.pem, ldap.pem*).

2. Click **Generate JSON files**.

The three configuration files are displayed and validated (as JSON).

3. Click **Download** for each file.

The files are downloaded to the default download directory on your computer. (Depending on your browser settings, you may be prompted to keep the first file.)

4. Copy these files to the appropriate location on your provisioning server, and modify them with a JSON editor.

Related information

[Accessing the Dolby Conference Phone web interface](#) on page 79

8.8 Rebooting the Dolby Conference Phone

You can reboot the Dolby Conference Phone by using either the Dolby Conference Phone web interface or the phone user interface.

About this task

Rebooting with the phone web interface is useful if the phone is in a different office or at a remote location.

To reboot the phone from the Dolby Conference Phone user interface:

Procedure

1. Tap this sequence:



2. Tap the confirmation button.



Related information

[Accessing the Dolby Conference Phone web interface](#) on page 79

8.9 Call and audio statistics

Through the Dolby Conference Phone web interface, you can view call statistics for active audio calls.

To view call statistics:

- Access the web interface and select **Status > Call Statistics**.

The call statistics page shows:

Call information:

- Calling party first and last name
- Calling party address (extension)
- Call duration in minutes and seconds
- Type of call (IP PBX or conferencing service provider)
- Audio codec type

Network statistics

- Packet count
- Average jitter
- Packet loss rate
- Packet loss burstiness (rate of audio packet losses occurring in a burst)

- Concealed packet count
- Jitter histogram

Audio diagnostics

- Average automatic gain control gain
- Voice capture level
- Voice rendering level
- Signal-to-noise level
- Signal-to-echo level
- Average round-trip time (RTT)
- Signal-to-noise level too low
- Signal-to-echo level too low
- Low input power detected
- Microphone input clipping

Related information

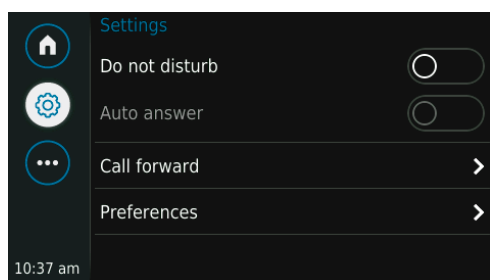
[Accessing the Dolby Conference Phone web interface](#) on page 79

8.10 Running Dolby Conference Phone diagnostics

The phone user interface includes a menu that enables you to test the various components of the Dolby Conference Phone, such as the microphones and speakers.

Procedure

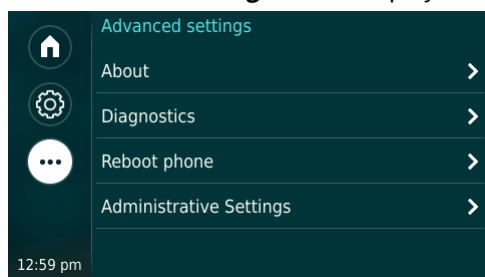
1. Access the Dolby Conference Phone **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap

The **Advanced settings** screen displays.



3. Tap **Diagnostics**, then **Test hardware**.
4. Select one of these menu items to perform a hardware diagnostic test:
 - **Touch screen:** Test the touch screen response area.

Tap the screen anywhere in the grid and determine whether the crosshair that appears is exactly where you touched. Try again in a different location. If there is a discrepancy, see the instructions below for calibration.

- **LED:** Test the LED halo on the phone.
- **Keys:** Press the three keys, one by one.
- **Microphones:** Test the phone microphones. If satellite microphones are connected, they can also be tested.
 1. Select a microphone (1, 2, or 3) and tap **Start test**.
 2. Speak into the phone and wait for the phone to play back the recording.
 3. Repeat with the other microphones.
- **Speakers:** Test the speakers.

Results

If the touch screen needs calibration:

1. Open the diagnostic shell and log in.
2. Type this command to connect directly to the Dolby Conference Phone:

```
ssh-to-dcp
```

3. Type *y* to bypass the trusted host message, and then enter the administrative password.
4. Enter this command to clear the calibration data:

```
system rmc calibration
```

5. Type this command to reboot the phone:

```
system reboot
```

6. When the calibration screen displays during the reboot process, follow the instructions to calibrate the touch screen.

8.11 Administering with the diagnostic shell

The diagnostic shell provides a command-line interface for administering Dolby Conference Phone.

8.11.1 Diagnostic shell

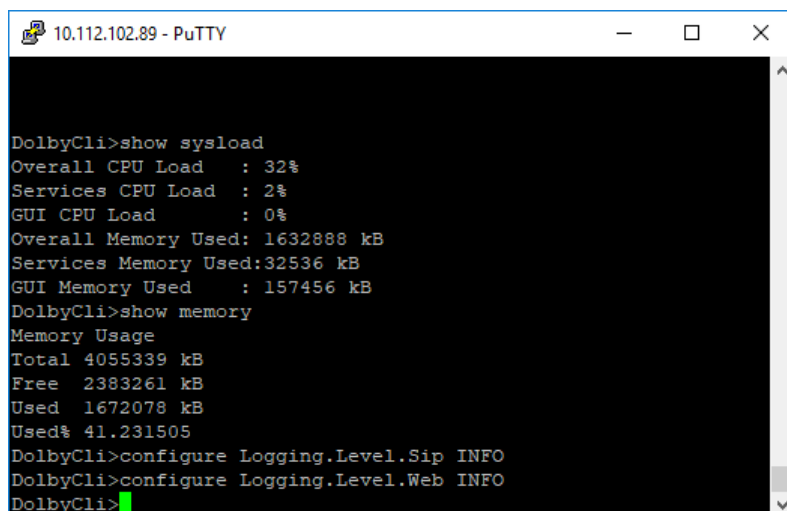
The diagnostic shell provides a command-line interface for you to query the phone state, run diagnostics, and configure selected parameters.

You access the diagnostic shell with the SSH protocol.

Only one instance of the shell service is supported. After a connection to the shell service is established, subsequent requests are denied until the current session is terminated. The session times out after a period of inactivity. The time-out period is set by the `Authentication.Local.SessionTimeout` configuration parameter. The shell service starts when the phone boots up, but it is accessible remotely only if the configuration parameter `Network.Services.SSHEnabled` is enabled, which is the default. You must enter the administrative user name and password to use the diagnostic shell.

Shell commands include localized command help, function calls, diagnostic tests, commands to view and set configuration parameters, and more. A list of commands is available in the **Help** section of the web interface.

Figure 5: Diagnostic shell command sample



```

10.112.102.89 - PuTTY

DolbyCli>show sysload
Overall CPU Load   : 32%
Services CPU Load  : 2%
GUI CPU Load       : 0%
Overall Memory Used: 1632888 kB
Services Memory Used: 32536 kB
GUI Memory Used    : 157456 kB
DolbyCli>show memory
Memory Usage
Total 4055339 kB
Free 2383261 kB
Used 1672078 kB
Used% 41.231505
DolbyCli>configure Logging.Level.Sip INFO
DolbyCli>configure Logging.Level.Web INFO
DolbyCli>

```

Tip: The shell provides help and shortcuts:

- Type *help* to see a list of commands.
- Press Up-arrow to view previous commands. Press Enter to submit.
- Press Tab to complete partially typed commands or to prompt for additional command information, such as command parameters. For example, if you enter *system*, and then press Tab, the possible system commands are listed and the command is repeated on the command line for you to finish command entry. If you enter *system traceroute* and press Tab, the system indicates that the command parameter is the IP address of the test host.

Related information

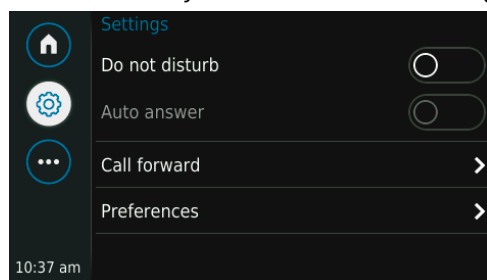
[Opening the diagnostic shell](#) on page 90

8.11.2 Opening the diagnostic shell

You access the diagnostic shell remotely with the SSH protocol or locally with an external keyboard. You access the diagnostic shell with the SSH protocol.

- Open the diagnostic shell remotely:
 1. Determine the phone IP address:

- a. Access the Dolby Conference Phone **Settings** screen.



This may be different for each service provider.

- b. Tap this sequence:

 > **About** > **General**

The IP address displays.

2. Open your preferred SSH client application.

3. Connect to the phone IP address using SSH (port 22).
4. Log in with the administrative user name and password. (Default credentials are *admin* and *1739*.)
5. Enter commands, and then type *exit* to exit the shell.

Related information

[Diagnostic shell](#) on page 89

8.12 Bluetooth LE initialization and advertisement

The Bluetooth LE module is a hardware component of Dolby Conference Phone. If this component is faulty or does not initialize, then the phone cannot transmit a Bluetooth LE advertisement.

You can confirm whether the phone is advertising a Bluetooth LE signal. However, be aware that the phone only advertises Bluetooth LE under certain circumstances. Lack of Bluetooth LE advertisement is a problem only if your conferencing application allows the phone to advertise Bluetooth LE.

In order for the phone to advertise a Bluetooth LE signal, these criteria must be met:

- The Bluetooth LE hardware module on the phone is working and initialized.
- You have not disabled Bluetooth LE advertisement.
- The conferencing application on the phone allows Bluetooth and has turned on Bluetooth LE advertisement.

The `Dvms.Service.EnableBluetoothAdvertisement` parameter controls whether the phone can advertise Bluetooth LE. However, the conferencing application controls whether the phone does or does not advertise Bluetooth LE.

Related information

[Disabling Bluetooth advertisement](#) on page 72

8.13 Confirming Bluetooth LE initialization and advertisement

From the phone user interface, you can confirm that the Bluetooth LE hardware is initialized and advertisement is active.

About this task

If the Bluetooth LE module fails to initialize, a red status alarm  displays on the phone user interface and Dolby Conference Phone web interface.

When the phone boots, it attempts to initialize the Bluetooth LE module (up to ten times). If initialization fails, each attempt is recorded in the `syslog` log file with this warning:

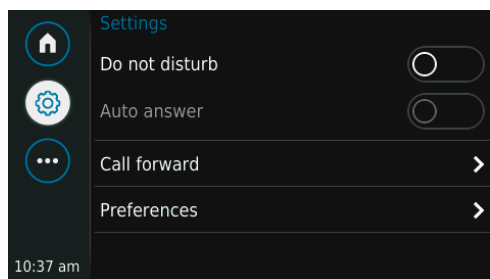
Bluetooth module initialization failed. Retrying...

If all initialization attempts fail, the phone records this warning in the `syslog` log file and quits Bluetooth LE module initialization:

Unable to initialize module. Quit application.

Procedure

1. Access the Dolby Conference Phone **Settings** screen.



Ask your conferencing service provider for instructions.

2. Tap this sequence:



3. Scroll down to view **Bluetooth** status.

A green  or red  status icon indicates the status of this feature.

4. Tap **Bluetooth** to check the initialization and advertisement statuses.

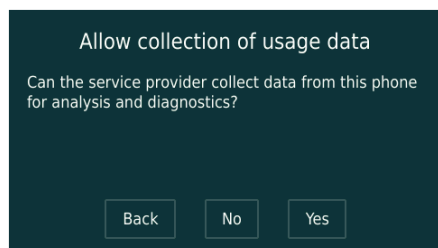
Related information

[Disabling Bluetooth advertisement](#) on page 72

8.14 Collection of usage data

Dolby Conference Phone can collect and report a variety of data to your conferencing service provider's provisioning service or your on-premises instance of the Dolby Conferencing Console. If you are using your conferencing service provider's provisioning service, you may wish to protect yourself by blocking the collection of usage data.

If you use the plug and play provisioning method, this screen may appear during initial setup:



If you choose **No**, **Privacy block** is enabled. The phone cannot share data with your conferencing service provider's provisioning service.

If you choose **Yes**, the phone shares data with your conferencing service provider's provisioning service. This includes diagnostics reports, core dump records, call detail records, log files, and screenshots of the phone user interface.

Related information

[Setting up the phone using the plug and play provisioning method](#) on page 35

8.14.1 Enabling Privacy block

When privacy block is disabled (the default), the Dolby Conference Phone system can share usage data with your conferencing service provider. You can enable or disable privacy block at any time from the UI, web interface, or CLI.

About this task

Privacy block is not a provisionable feature. You must enable or disabled it one phone at a time. The name of the configuration parameter that controls this feature is `Provisioning.Server.PrivacyBlock`.

From the Dolby Conference Phone web interface:

1. From the **Settings** tab, expand the **Provisioning** section.
2. Expand the **Server** section.
3. Turn on **Disable Privacy-related Uploads**.

The `Provisioning.Server.PrivacyBlock` configuration parameter is now `true`. The phone cannot share data with your conferencing service provider's provisioning service.

8.15 Phone memory resources

The Dolby Conference Phone is designed to operate in a variety of office environments and deployments.

New phone software releases may add new features that require varying amounts of phone memory resources. To ensure that the phone and its configured features operate easily, you will need to check that the phone has adequate available memory from time to time.

When the phone resources start to run low, you may notice that the phone reboots or freezes.

You can check the available memory on the phone by running the command **show memory through the diagnostic shell**, by viewing the log file, or by accessing the Dolby Conference Phone web interface. Typically, you want the phone to be using less than 95% of its available memory.

If you find that it is using more than 95%, you can reduce this amount by following these steps:

- Check how the corporate directory (LDAP) is set up, and make possible configuration parameter changes.
- Return the log configuration parameters to the default values when you are finished testing or troubleshooting issues.

The phone logs the current memory usage every hour as part of the `sysLog` files, which can be accessed via Dolby Conferencing Console or the Dolby Conference Phone web interface.

If the overall memory usage is 75 MB or higher, the logging level will change from `info` to `warning`. The logging level can be accessed on the phone web interface under **Settings > Logging > Log Level**, or on Dolby Conferencing Console.

The Boolean configuration variable `Logging.ReportFreeMemory` can be set to `True` through the Dolby shell so that the log will include free memory on every log entry.

Related information

[Accessing the Dolby Conference Phone web interface](#) on page 79

[Diagnostic shell](#) on page 89

[Corporate directory](#) on page 63

[Uploading log files to a provisioning server](#) on page 83

8.16 Resetting to factory defaults

Resetting the phone to factory defaults clears the user data—such as the recent calls list and contact directory—and removes log files and cached data.

Procedure

1. Select one of these methods to perform the factory reset:

Method	Instructions
Through the phone user interface	From the home screen, tap  >  , and then Administrative Settings > Factory reset . Enter the administrative password—by default, it is 1739—when prompted.
Through the diagnostic shell command	Enter the factory reset command, <code>system factoryReset</code> .
Through the Dolby Conference Phone web interface	Go to Tools and click Factory Reset .
Through the multiple key combination	Press the mute and volume-down buttons simultaneously for 10 seconds.  Note: You can also access the Settings and Diagnostics screen for advanced settings and system diagnostics.
Through the multiple key combination when the phone reboots	Press the mute and volume-down keys simultaneously for five seconds. The LED turns green, and then turns yellow to show the phone is reset to factory default.

2. If prompted, confirm the reset.

Results

If you confirm the option, the phone reboots to complete the reset process even if a call is active.

8.17 LLDP and supported type-length values

The LLDP is a vendor-neutral Layer 2 protocol that allows a network device to advertise its identity and capabilities on the local network.

The protocol was updated by [IEEE standard 802.1AB](#) in September 2009.

The LLDP feature supports VLAN and QoS discovery, and LLDP power management, but not power negotiation. LLDP has a higher priority than DHCP VLAN discovery.

These mandatory type-length values are supported:

- Chassis ID: Must be first type-length value
- Port ID: Must be second type-length value
- Time to live: Must be third type-length value, set to 120 seconds
- System capabilities: Set as telephone capability
- MAC/PHY configuration status: Detects duplex mismatch
- LLDP-MED capabilities: 1
- LLDP-MED network policy: VLAN, L2 QoS, L3 QoS
- LLDP-MED extended power via MDI type-length value: Power type, power source, power priority, power value
- End of LLDP DU: Must be last type-length value

The following optional type-length values are supported:

- Port description
- System name: Administrator-assigned name
- System description: Includes device type, phone number, hardware version, and Dolby Conference Phone software version
- Management address: Used for network discovery

An LLDP frame must contain all mandatory type-length values. The frame will be recognized as LLDP only if it contains mandatory type-length values. The Dolby Conference Phone supports LLDP frames with both mandatory and optional type-length values.

All outbound LLDP packets are sent to multicast MAC address 01:80:c2:00:00:0e.

Related information

[Supported type-length values](#) on page 95

8.17.1 Supported type-length values

The Dolby Conference Phone uses an LLDP frame and type-length values to determine the phone identity, its capabilities, and organizations to which the phone belongs.

The basic type-length value format is as follows:

- Type-length value type (7 bits) [06]
- Type-length value length (9 bits) [7–15]
- Type-length value information (0–511 bytes)

Each type-length value appears only once, except for the network policy type-length value, which may appear twice. The order of the four type-length values (chassis ID, port ID, time to live, and end of LLDP DU) are fixed; the others may appear in any order.

Name	Type (7 bits)	Length (9 bits)	Org. code (3 bytes)	Sub type	Value (sample)
Chassis ID	1	7	-	4: MAC address	(00:d0:46:00:01:9f)
Port ID	2	7	-	3: MAC address	(00:d0:46:00:01:9f)
Time to live	3	2	-	-	120 seconds ^[a]
Port description	4	4	-	-	eth0
System name	5	0 to 255	-	-	(DVCP-D0020473)
System description	6	0 to 255	-	-	(SN:D00XXXXXX/R evX/SW:X.X.X.X)
System capabilities	7	4	-	-	• Other: Not capable • Repeater: Not capable • Bridge: Not capable • WLAN access point: Not capable • Router: Not capable • Telephone: Capable • DOCSIS cable device: Not capable • Station only: Not capable • C_VLAN component of a VLAN bridge: Not capable • S-VLAN component of a VLAN bridge: Not capable • Two-port MAC relay: Not capable

Name	Type (7 bits)	Length (9 bits)	Org. code (3 bytes)	Sub type	Value (sample)
Management address	8	12	-	-	- Address string length: 5 - Address subtype: IPv4 - Management address: (example) 10.203.188.92 - Interface subtype: ifIndex - Interface number: X - OID string length: 0
Organizationally specific	127	9	0x00120fv: IEEE 802.3	1: MAC/PHY configuration status	- Auto negotiation support: 1 - Auto negotiation status: 0 or 1 - Auto negotiation advertised capability: Asymmetric and symmetric PAUSE: 0 - Symmetric PAUSE: 0 - Asymmetric PAUSE: 0 - PAUSE: 0 100Base-TX full duplex: 1 100Base-TX half duplex: 1 10BASE-T full duplex: 1 10BASE-T half duplex: 1
Organizationally specific	127	7	0x00120bb: TR-41	1: Media capabilities	- LLDP-MED capabilities: 1 - Network policy: 1 - Location identification: 0 - Extended power via MDI-PSE: 0 - Extended power via MDI-PD: 1 - Inventory: 0
Organizationally specific	127	8	0x00120bb: TR-41	2: Network policy	- Application type: Voice or voice signaling - Unknown policy flag: 0 or 1 - Tagged flag: 0 or 1 - VLAN ID: 0 to 4094 - Link layer priority: 0 to 7 - DSCP: 0 to 63
Organizationally specific	127	7	0x00120bb: TR-41	4: Extended power via MDI	- Power type: PD device - Power source: PSE - Power priority: Unknown - Power value: 13 watts^[b]
End of LLDP DU	0	0	-	-	-

[a] Time to live value is 0 seconds when the phone is powered on.

[b] The Dolby Conference Phone is a class 3 device that has a maximum power consumption of 12.95 watts.

8.18 Trusted certificates and issuers

The Dolby Conference Phone trusts the certificates listed here.

Certificate name	Issuer
*.lobby.pp.iconf.net	Entrust Certification Authority - L1K
Amazon Root CA 1	Starfield Services Root Certificate Authority - G2

Certificate name	Issuer
Amazon Root CA 1	Self-signed
Amazon Root CA 2	Self-signed
Amazon Root CA 3	Self-signed
Amazon Root CA 4	Self-signed
Baltimore CyberTrust Root	Baltimore CyberTrust Root
Baltimore CyberTrust Root	Baltimore CyberTrust Root
C=US/O=Starfield Technologies, Inc./OU=Starfield Class 2 Certification Authority	ValiCert
C=US/O=The Go Daddy Group, Inc./OU=Go Daddy Class 2 Certification Authority	The Go Daddy Group, Inc.
C=US/O=VeriSign, Inc./OU=Class 2 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network	VeriSign, Inc.
C=US/O=VeriSign, Inc./OU=Class 3 Public Primary Certification Authority - G2/OU=(c) 1998 VeriSign, Inc. - For authorized use only/OU=VeriSign Trust Network	VeriSign, Inc.
Cybertrust Public SureServer SV CA	Baltimore CyberTrust Root
DigiCert Assured ID Root CA	DigiCert Assured ID Root CA
DigiCert Assured ID Root G2	DigiCert Assured ID Root G2
DigiCert Assured ID Root G3	DigiCert Assured ID Root G3
DigiCert Global Root CA	DigiCert Global Root CA
DigiCert Global Root G2	DigiCert Global Root G2
DigiCert Global Root G3	DigiCert Global Root G3
DigiCert High Assurance EV Root CA	DigiCert High Assurance EV Root CA
DigiCert SHA2 Secure Server CA	DigiCert SHA2 Secure Server CA
DigiCert Trusted Root G4	DigiCert Trusted Root G4
Entrust.net Certification Authority (2048)	Entrust.net Certification Authority (2048)
Entrust.net Certification Authority (2048)	Entrust.net Certification Authority (2048)
Entrust Root Certification Authority	Entrust Root Certification Authority
Entrust Root Certification Authority - G2	Entrust Root Certification Authority - G2
GeoTrust Global CA	GeoTrust Global CA
GeoTrust Global CA 2	GeoTrust Global CA 2
GeoTrust Primary Certification Authority	GeoTrust Primary Certification Authority
GeoTrust Primary Certification Authority - G3	GeoTrust Primary Certification Authority - G3
GeoTrust Universal CA	GeoTrust Universal CA
GeoTrust Universal CA 2	GeoTrust Universal CA 2
GlobalSign	GlobalSign
GlobalSign Root CA	GlobalSign Root CA
Go Daddy Root Certificate Authority - G2	Go Daddy Root Certificate Authority - G2

Certificate name	Issuer
Go Daddy Root Certificate Authority - G2	Go Daddy Root Certificate Authority - G2
lync-DOLBYVOICE2-CA	lync-DOLBYVOICE2-CA
O=RSA Security Inc/OU=RSA Security 2048 V3	O=RSA Security Inc/OU=RSA Security 2048 V3
Starfield Root Certificate Authority - G2	Starfield Root Certificate Authority - G2
Starfield Services Root Certificate Authority - G2	C=US/O=Starfield Technologies, Inc./OU=Starfield Class 2 Certification Authority
TC TrustCenter Class 2 CA II	TC TrustCenter Class 2 CA II
TC TrustCenter Universal CA I	TC TrustCenter Universal CA I
thawte Primary Root CA	thawte Primary Root CA
thawte Primary Root CA - G3	thawte Primary Root CA - G3
Thawte Universal CA Root	Thawte Universal CA Root
VeriSign Class 3 Public Primary Certification Authority - G5	VeriSign Class 3 Public Primary Certification Authority - G5
VeriSign Universal Root Certification Authority	VeriSign Universal Root Certification Authority

8.18.1 TLS cipher suites

The Dolby Conference Phone offers a predefined list of cipher suites. This list cannot be changed at this time.

 **Note:** The current SRTP implementation supports the AES_CM_128_HMAC_SHA1_80 and AES_CM_128_HMAC_SHA1_32 cipher suites. It does not support the F8_128_HMAC_SHA1_8 cipher suite.

Table 6: TLS cipher suites

Cipher	Cipher suites
0x2f	TLS_RSA_WITH_AES_128_CBC_SHA
0x30	TLS_DH_DSS_WITH_AES_128_CBC_SHA
0x31	TLS_DH_RSA_WITH_AES_128_CBC_SHA
0x35	TLS_RSA_WITH_AES_256_CBC_SHA
0x36	TLS_DH_DSS_WITH_AES_256_CBC_SHA
0x37	TLS_DH_RSA_WITH_AES_256_CBC_SHA
0x3c	TLS_RSA_WITH_AES_128_CBC_SHA256
0x3d	TLS_RSA_WITH_AES_256_CBC_SHA256
0x3e	TLS_DH_DSS_WITH_AES_128_CBC_SHA256
0x3f	TLS_DH_RSA_WITH_AES_128_CBC_SHA256
0x41	TLS_RSA_WITH_CAMELLIA_128_CBC_SHA
0x42	TLS_DH_DSS_WITH_CAMELLIA_128_CBC_SHA
0x43	TLS_DH_RSA_WITH_CAMELLIA_128_CBC_SHA
0x84	TLS_RSA_WITH_CAMELLIA_256_CBC_SHA
0x85	TLS_DH_DSS_WITH_CAMELLIA_256_CBC_SHA
0x86	TLS_DH_RSA_WITH_CAMELLIA_256_CBC_SHA

Table 6: TLS cipher suites (continued)

Cipher	Cipher suites
0x9c	TLS_RSA_WITH_AES_128_GCM_SHA256
0x9d	TLS_RSA_WITH_AES_256_GCM_SHA384
0xa0	TLS_DH_RSA_WITH_AES_128_GCM_SHA256
0xa1	TLS_DH_RSA_WITH_AES_256_GCM_SHA384
0xa4	TLS_DH_DSS_WITH_AES_128_GCM_SHA256
0xa5	TLS_DH_DSS_WITH_AES_256_GCM_SHA384
0xc004	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA
0xc005	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA
0xc009	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA
0xc00a	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA
0xc00e	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA
0xc00f	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA
0xc013	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
0xc014	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA
0xc023	TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256
0xc024	TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384
0xc025	TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256
0xc026	TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384
0xc027	TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
0xc028	TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
0xc029	TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256
0xc02a	TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384
0xc02b	TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256
0xc02c	TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384
0xc02d	TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256
0xc02e	TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384
0xc02f	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
0xc030	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
0xc031	TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256
0xc032	TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384

9

Troubleshooting

If you encounter an issue with the Dolby Conference Phone, we recommend that you attempt basic troubleshooting before you contact Support. The information in this chapter describes likely causes of phone issues and corrective actions you can take to resolve them. Follow the available troubleshooting procedures to review logs and error messages, run diagnostics, and possibly upgrade the phone software.

- [Report to Dolby](#)
- [Phone issues and solutions](#)
- [Plug and play provisioning method issues and solutions](#)
- [Decommissioning and disposing of the phone](#)

9.1 Report to Dolby

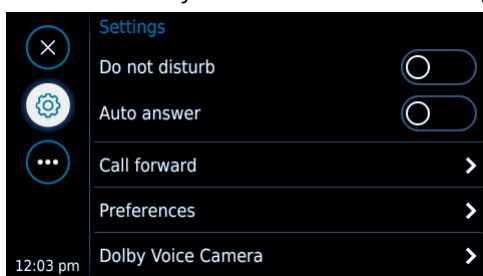
Report to Dolby allows you to easily send system logs to Dolby customer support.

About this task

You can send log files to *diagnostics.dolby.com*. Your service provider must work with Dolby customer support to retrieve this data.

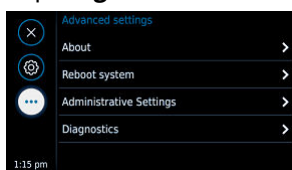
Procedure

1. Access the Dolby Conference Phone **Settings** screen.

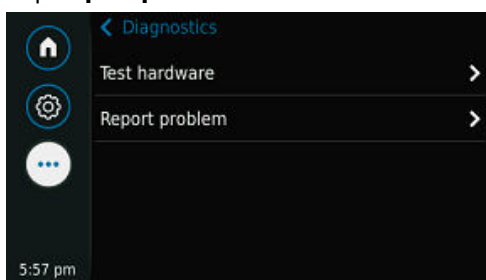


2. Tap  to call up the **Advanced Settings** screen.

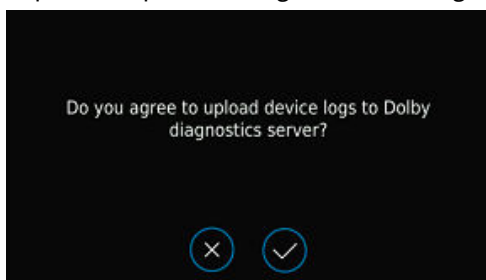
3. Tap **Diagnostics**.



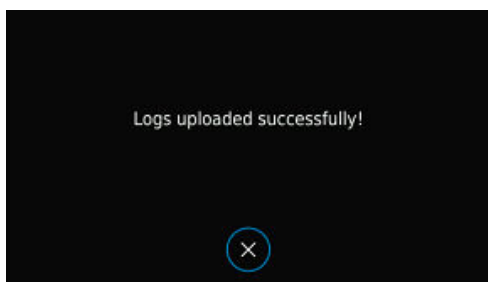
4. Tap **Report problem**.



5. Tap **Yes** to upload the log files to the diagnostics server.



If the logs were uploaded successfully, a dialogue box displays.



9.2 Phone issues and solutions

Use this information for likely causes and corrective actions for your phone.

9.2.1 Power and start-up issues

For power and start-up issues, see these possible solutions.

Issue

The phone has power issues or the phone has no power.

Solution

Determine whether the problem is caused by the phone, the AC outlet, or the PoE switch. Do one of the following:

- If using PoE, ensure that the port is a PoE port and not just a regular Ethernet port.
- If using a power injector, ensure that the injector is plugged into a functional AC outlet.
- Verify that no lights appear on the unit when it is powered up.
- Check for a faulty cable. Replace the cable and try again.

9.2.2 Hardware issues

These are possible solutions to hardware issues.

Issue	Solution
Buttons do not work. For example, when the phone is idle, pressing the volume button does not bring up the ringer widget, and you cannot hear the ringer tone.	Tap the phone icon on the home screen. Tap the keypad digits on the dialer screen: • If you can hear the DTMF tone, then there might be a problem with the hardware. To confirm whether the keys are working, access the Settings menu, then tap  > Diagnostics > Test hardware > Keys and follow the instructions on the screen. • If you cannot hear the DTMF tone, wait until the fatal error screen displays (see Fatal error issues on page 112) and the phone reboots. Log files from the error are uploaded to the provisioning server and will assist you in finding the problem.
Touch screen is not accurate. For example, when you touch the screen, the coordinates in the phone software do not compare to where you touched the screen.	It is possible that the touch screen is not accurately calibrated. Access the Settings menu, then tap  > Diagnostics > Test hardware > Touch Screen. Tap the screen anywhere in the grid and determine whether the crosshair that appears is exactly where you touched. Try again in a different location. If it appears that the phone touch screen is not accurately calibrated, then recalibrate. See Running Dolby Conference Phone diagnostics.

Issue	Solution
One or more speakers are not working.	To verify, access the Settings menu, then tap  > Diagnostics > Test hardware > Speakers and test each speaker.
One or more microphones are not working.	To verify, access the Settings menu, then tap  > Diagnostics > Test hardware > Microphones and test each microphone one at a time. When testing a microphone, speak a short phrase and wait for the phone to play back the recording.
The LED halo flashes red and the phone unexpectedly mutes and unmutes.	Check for and remove any HDMI devices, other than the satellite microphones, connected to the phone microphone ports.

Related information

[Fatal error issues](#) on page 112

9.2.3 Network access issues

Here are possible solutions to network access issues.

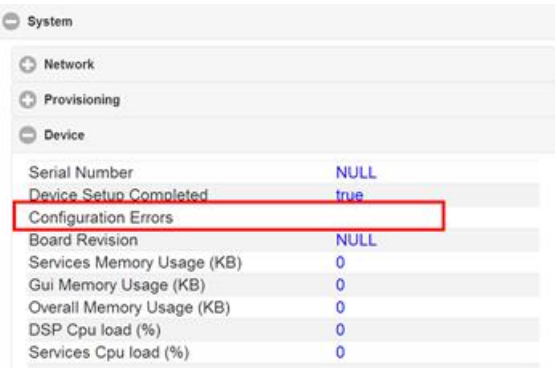
Issue	Solution
The web interface Status tab indicates an alarm, and an alarm icon appears for the network status . Position your mouse pointer over the alarm icon to view the network status code.	If one of these status codes appears, check whether the switch is correctly set up and if the Ethernet cable is acceptable: • Link is dormant: Network is pending authentication. • Link is down. • Unknown link error: There is an unknown network error. If one of these status codes appears, check whether there is an unmanaged switch connecting the LLDP switch and the Dolby Conference Phone: • LLDP Ethernet speed/duplex mismatched: Ethernet speed/duplex mismatch reported by LLDP. • Two or more LLDP neighbors detected: Device is connected to an unmanaged switch. • LLDP neighbor end device detected. Device is connected to an unmanaged switch: An LLDP end-device neighbor is detected. The phone is connected to an unmanaged switch. If this status code appears, check the LLDP VLAN settings on switch: LLDP voice and voice signaling VLAN ID mismatch detected: Voice and voice signaling ID conflict detected. If this status code appears, check the network status: FAIL_DHCP: Failed. If this status code appears, check whether another device on the network is using the same IPv4 address: mac-address: Another device on the network is using the same IPv4 address. If one of these status codes appears, check whether the default gateway is up: • Default gateway is unreachable. • Default gateway is unconfigured. If one of these status codes displays, check whether the DNS servers are up: • DNS servers are unreachable. • DNS servers are unconfigured. • Only secondary DNS server is reachable: The primary DNS server is unreachable, whereas the secondary DNS server is reachable. If one of these status codes displays, check that the dynamic DNS is correctly configured on the DNS servers and the DHCP server: • DNS hostname unresolvable: The host name DVCP-xxxxxxx is not resolvable. • DNS servers are unconfigured.
The phone does not obtain an IP address on the secondary VLAN.	The secondary VLAN was configured incorrectly. For example, the wrong VLAN ID was specified. Verify the VLAN ID, and try to provision the phone again.

Issue	Solution
The phone obtains an IP address on only one VLAN when the dual-VLAN feature is enabled in conjunction with the Cisco switch port with 802.1X authentication enabled.	Set the authentication host mode of the Cisco switch port to either single-host or multiple-host mode. For more information on Cisco 802.1X host modes, see http://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst4500/12-2/50sg/configuration/guide/Wrapper-46SG/dot1x.html#wp1240475 .
The phone does not obtain an IP address on the secondary (data) VLAN when the dual-VLAN feature is enabled and port security is enabled on the Cisco switch port.	Run this command on the Cisco switch: <pre>switchport port-security mac-address xx:xx:xx:xx:xx:xx vlan access</pre> The 00:d0:46:0a:00:29 is the MAC address of the selected phone. This command allows the phone with the specified MAC address to be on both the voice and data VLANs at the same time. For more information on Cisco switch port security, see http://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst6500/ios/12-2SX/configuration/guide/book/port_sec.html.
You see a Failed to load message from your conferencing service provider, or a service unavailable message from the conferencing service.	Your organization uses a dual VLAN, and the phone was unable to connect to your conferencing service provider over the Internet. See Primary and secondary VLANs on page 19 and Configuring the secondary VLAN on page 46.

9.2.4 Provisioning issues

Here are some possible solutions for first-time provisioning issues.

Issue	Solution
The phone does not obtain an IP address.	The phone usually takes about 50 to 60 seconds to boot up. If it takes much longer than that (for example, more than 90 seconds), the phone may not be obtaining its IP address. The phone waits up to 90 seconds at start-up for the DHCP response, and if it does not get the IP address, it uses a self-allocated link-local IP address (see RFC 3927) and continues with the boot. Make sure the phone has proper network connectivity and is connected to a port that allows the device to get DHCP IP configuration. There may also be a VLAN configuration issue, and in the case that the phone cannot obtain an IP address from the default VLAN, wait until the phone boots up, then modify the VLAN manually by accessing the Settings screen and then tapping  > Administrative Settings > Network configuration > Ethernet.
The phone cannot connect to the provisioning server.	After you enter the provisioning server IP and access credentials, the phone attempts to connect to the provisioning server. If the access fails, the phone displays a message that the connection to the provisioning server has failed. The following error conditions display on the phone user interface: • Network error • Authentication failure • Secure Sockets Layer (SSL) error Make sure to provide the correct: • IP address • Credentials
There are errors in the configuration files on the provisioning server.	If there is an error in the configuration file, the phone displays error messages similar to the following: • Invalid configuration files or parameters. Press the “More” button for more information. • A list of the parameters in error Correct the configuration files on the provisioning server based on the information provided on the phone screen. Possible errors are: • Write access not allowed for provisioning—The parameter is not allowed to be configured from the provisioning server. • Data validation error—The value supplied is not valid. • Invalid parameter—The parameter specified in the configuration file is not recognized by the phone. Configuration errors are also displayed in the Dolby Conference Phone web interface when you position the mouse pointer over the red circle in the Provision Successful field. (Status > System > Device)

Issue	Solution
The phone is unable to connect to an HTTPS provisioning server.	If you want to use an HTTPS provisioning server, the phone may not be able to connect to the provisioning server due to security settings issues if not configured properly. The phone must first be initially provisioned in a non-secure environment, in an FTP or HTTP protocol, to download the server CA certificate to the phone. Then it needs to be switched over to the HTTPS provisioning server. See Appendix A: Configuring the provisioning server on page 123.
The phone is having unexpected issues.	If you run into an unexpected issue (that is not described in this table), skip the first-time provisioning wizard and go directly to the phone home screen. Tap  for more information about the configuration failures. For example, you can get the device IP address and then gain access to the SSH shell to use the diagnostics shell commands or Dolby Conference Phone web interface.
The changed configuration contains one or more errors	On the home screen, you will see . Tap the icon; the provisioning error appears on the system status screen. Access the phone through the Dolby Conference Phone web interface. Go to the Status > System > Device section, and look for the Configuration Errors field. If there are configuration errors, a link appears that opens a complete list of configuration errors detected on the phone. 

Related information

[Configuring the phone with Dolby Conferencing Console](#) on page 29
[Appendix A: Configuring the provisioning server](#) on page 123

9.2.5 Authentication issues

Here are possible solutions to authentication issues.

Issue	Solution
You forgot the administrative password.	Perform a factory reset. The default administrative password (1739) is restored. After factory reset, you must reenter user preferences, such as language selection, date format, brightness, auto-answer, and so on.

Related information

[Resetting to factory defaults](#) on page 93

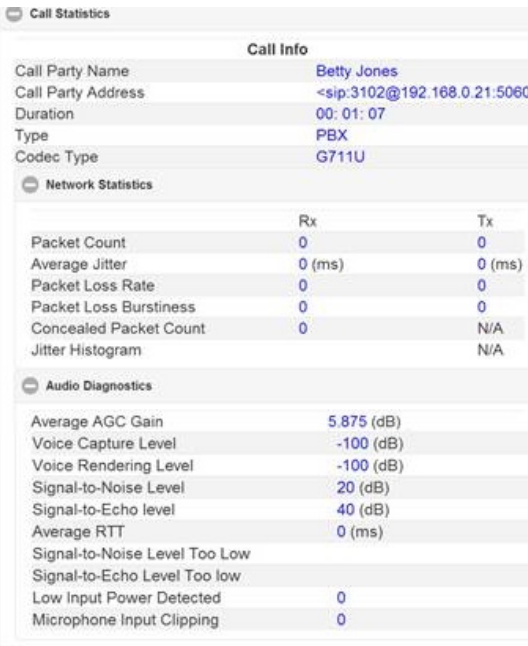
[Adding certificate authority certificates](#) on page 55

9.2.6 IP PBX calling issues

Here are possible solutions for IP PBX calling issues.

Issue	Solution
The user cannot make a call because the phone button is grayed out, and there are no alarms (red circle with exclamation point) relating to the call server.	The phone is not configured for IP PBX calling. Ensure that the phone is registered with the call server and check these configuration parameters: • <code>Sip.Account.DisplayName</code> • <code>Sip.Account.UserName</code> • <code>Sip.Account.DisplayNumber</code> • <code>Sip.Pbx.OutboundProxy1</code> • <code>Sip.Pbx.OutboundProxy1Port</code> You can view these parameters on the web interface here: • Settings > IP PBX Settings > Account • Settings > IP PBX Settings > Server
The user cannot make a call because the phone button is grayed out and the Status screen shows an alarm.	Access the Status screen to determine whether there is an alarm (red circle with exclamation point) adjacent to Call Server. If there is, then the phone is not registered with the call server or cannot connect to the call server. Through the Dolby Conference Phone web interface, go to Status > System > Device. Find SIP Registered, and position the mouse pointer over the alarm icon to obtain the failure details. The reason could be one of the following: • The phone cannot connect to the call server due to network errors. • The phone cannot connect to the call server due to authentication failures (incorrect credentials). • The phone cannot connect to the call server due to other configuration errors. You can review all configuration parameters listed under SIP on the web interface Help page.
The user always gets a busy tone after dialing a number.	The possible reasons for the failure are: • The dial plan on the server side is set incorrectly. • The phone the user is calling is no longer registered to the server. • The phone is using an incompatible codec configuration. Through the Dolby Conference Phone web interface, go to Settings > IP PBX Settings > Server and scroll down to the PBX Codec List. It is possible that the phone that the user is trying to call does not support the same codec. See Bandwidth planning and codec selection on page 18.

Issue	Solution
The user cannot receive a call.	The possible reasons for the failure are: • The do not disturb feature is enabled. To disable, access the Settings screen, then change the do not disturb Off / On option to Off. If do not disturb is enabled, a scrolling message displays on the home screen. • The call forwarding feature is enabled. To disable, access the Settings screen, tap Call Forward, and then delete the appropriate forwarded number. If call forwarding is enabled, a scrolling message displays on the home screen. • The phone is using an incompatible codec configuration. To see a list of supported codecs, in the web interface, navigate to Settings > IP PBX Settings > Server and scroll down to the PBX Codec List. It is possible that the phone the user is trying to call does not support the same codec. See Bandwidth planning and codec selection on page 18.

Issue	Solution
During a call, the user cannot hear or be heard by the other end of the call, but the call timer shows an increasing call length.	The possible reasons for the failure are: • Network connectivity is lost between the phone and the other end of the call. • The phone encountered a serious error. Go to Fatal error issues on page 112. • The signal processing on the phone does not get packets from the network or send packets to the network. Through the Dolby Conference Phone web interface, navigate to Status > Call Statistics to view the run-time call statistics as shown below. If the receiver (Rx) or transmitter (Tx) counts grow, this indicates that the signal processing is sending or receiving packets to or from the network. If the receiver counts do not grow, this indicates that the phone is not receiving any packets from the other end. 

Issue	Solution
The IP-PBX registration is successful but calls cannot be made and are not received.	Check the logs to see if the system is continuously trying to register with IP-PBX. If this occurs, you must set the Allow Contact Header Rewrite parameter to OFF. 1. Using the admin web page, navigate to Settings > IP PBX > Account. 2. Set Allow Contact Header Rewrite to OFF. 3. Click Save.
SIP registration is successful, but intermittent calls fail.	Check if the outbound proxy server hostname resolves to more than eight IP addresses. If this occurs, you must set the Set Active Server to Preferred Server's Order. 1. Using the admin web page, navigate to Settings > IP PBX > Server > Call Failover Mode. 2. Set Set Active Server to Preferred Server's Order. 3. Click Save.

Related information

[Fatal error issues](#) on page 112

9.2.7 Audio quality issues

If users experience audio quality issues, they can use the audio diagnostic feature to collect more information about the problem.

 **Note:** This information applies only when BT is the conferencing service provider.

Issue

Users have commented that there are audio quality or echo issues during calls.

Solutions

Complete these steps:

1. Set the audio logging mode configuration parameter `Logging.Audio.Mode` to *DIAGNOSTIC*. (Use upper case.)

A diagnostics icon () displays in the top-right corner of the phone display during calls.

2. Ask users to provide information by tapping  during a call and selecting an issue from the list provided on the phone. Possible issues are:
 - **Others can't hear me**
 - **Others hear me sound robotic**
 - **Others hear echo from my phone**
 - **I hear echo**
 - **I hear choppy/robotic audio**
 - **I can't hear anyone**
 - **I hear noises when I talk**
 - **Other problems**
3. Review the logs on the provisioning server.

See “Audio logging” for more information.

4. Set `Logging.Audio.Mode` back to `DISABLED`.

Related information

[Audio logging](#) on page 84

9.2.8 Software update issues

Here are solutions for Dolby Conference Phone software update issues.

Issue	Solution
The message Invalid signature on software image file appears on the .	The image is signed by Dolby and the phone performs a validation upon software upgrade. If you see this message, you are using an unauthorized image. Contact Dolby Customer Support if you received the image from Dolby. Otherwise, contact your conferencing service provider.
The message Invalid software image file appears on the .	After downloading the image, the phone performs a checksum validation. This message appears if the validation failed. Reasons for this include transmission errors or a corrupt image file. If the image file came from Dolby, contact Dolby Customer Support. Otherwise, contact your conferencing service provider. If the issue persists, check with your organization's IT department to investigate any network transmission errors.
The message Failed to complete software update for an unknown reason appears on the .	During the software upgrade, a possible flash error was detected. Retry the software update. If the message appears again, contact Dolby Customer Support.

9.2.9 Corporate directory issues

Here are solutions for corporate directory issues.

Issue	Solution
Cannot connect to the LDAP directory. The home screen shows  . The system status shows there is a directory error.	The possible reasons for the failure are: • An incorrect LDAP server IP address • Incorrect LDAP server access credentials • An incorrect or missing LDAP server root CA certificate (necessary when accessing the LDAP server using SSL) In the Dolby Conference Phone web interface, go to Status > System > Device . Find LDAP connected . . A red circle indicates an LDAP error. Position your mouse pointer over the red circle to see the specific error that occurred.

9.2.10 Fatal error issues

Here are possible solutions to fatal error issues.

Issue	Solution
This message appears on the display: The services application has stopped responding. Press Debug to stop rebooting.	You can choose to debug the problem. This stops the phone from performing the automatic reboot. If you do not tap Debug, the phone automatically reboots after five seconds. If the phone is configured to upload log files, the phone uploads the core dump (if there is any) to the provisioning server during the next reboot. The core dump file name is <i>SN-timestamp-core.appname</i>. For example, <i>D0025416-1402090840-core.services_app</i> is a core dump generated from the <i>services_app</i> application on the device numbered D0025416. Attach the uploaded log files as well as the core dump file, and create a bug report for Dolby Customer Support to perform further analysis.  Note: Uploading a core dump during booting prolongs boot time.
The user interface continues to display the Dolby logo and does not display the home screen, even after 90 seconds.	A combination of configuration and software issues have left the phone in a state from which it cannot recover. Try rebooting again by disconnecting the CAT6 cable from the phone and then reconnecting. If the reboot fails, then perform a factory reset. The LED halo flashes yellow when the factory reset is complete. The home screen should display. If the factory reset still stalls at the Dolby logo and the phone software was recently upgraded, reboot the phone again while pressing and holding the Mute and Volume Up keys until the LED halo shows blue, about 10 seconds. Then reboot the phone again. This downgrades the phone software. If the reboot still stalls at the Dolby logo, contact Dolby Customer Support.

Related information

[Hardware issues](#) on page 102

[IP PBX calling issues](#) on page 108

[Resetting to factory defaults](#) on page 93

9.2.11 Dolby Satellite Microphone issues

Here are possible solutions for Dolby Satellite Microphone issues.

Issue	Solution
The user plugs in a Dolby Satellite Microphone, but no tone plays.	Verify that the phone software is Version 2.1 or later. - If it is not, update the phone software. - If it is, check the microphone on another phone.

9.3 Plug and play provisioning method issues and solutions

Use this information for likely causes and corrective actions if you are using the plug and play provisioning method to deploy your phone. These messages appear on the Dolby Conference Phone screen.

Related information

[Setting up the phone using the plug and play provisioning method](#) on page 35

9.3.1 Dolby Voice Console error messages

These are possible solutions to Dolby Voice Console errors when deploying Dolby Conference Phone with the plug and play provisioning method.

These messages appear on the Dolby Conference Phone screen.

Error code	Message displayed	Solution
101	Plug and play setup failed because the phone has failed 802.1X authentication, which is required by your network. You can modify 802.1X settings in the phone according to your network security policy. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone has failed 802.1X authentication, which is required by your network. You can modify 802.1X settings in the phone according to your network security policy. 802.1X, also known as port guard, configuration on the phone often requires the security team in your IT organization to provide help and guidance. If you are not sure about the 802.1X policy, consult your security team for more information. You can modify basic 802.1X settings by tapping the Next button on the Dolby Conference Phone to navigate to the network settings screen. You can toggle between 802.1X authentication methods, turn off 802.1X authentication, enter 802.1X authentication credentials, or accept server certificates to get the device authenticated against your network. You may also need to import a server certificate into the phone through its web user interface.
102	Plug and play setup failed because the phone was unable to obtain a valid IP address. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to obtain a valid IP address for communicating with the Dolby Voice Console . The phone needs an IP address that can reach Internet in order to function properly. If there are multiple network ports in the room, you can try to plug the main network interface to a different port. If your network uses VLAN, the phone may need to be configured for a specific VLAN. The phone supports manual VLAN configuration. You can modify the VLAN settings by tapping the Next button on the Dolby Conference Phone. You may need to consult your IT help desk for assistance.
103	Plug and play setup failed because a DNS server was not configured for the phone. Contact your IT help desk for assistance.	The plug and play provisioning method failed because a DNS server was not configured for the phone. The phone requires a valid DNS server to resolve the host name to the IP address on the Internet. If your DHCP server did not supply a DNS configuration to the phone, you can configure the DNS server manually on the Dolby Conference Phone. Follow the instructions on the Dolby Conference Phone to provide the DNS server settings; for example you can enter the public DNS server 8.8.8.8.

Error code	Message displayed	Solution
104	Plug and play setup failed because the phone cannot complete proxy auto detection. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the PAC file to complete the setup. The phone can automatically detect proxy server on your network. It has detected a PAC file, but was unable to download the file to complete the auto discovery process. Follow the instructions on the Dolby Conference Phone to disable WPAD. You can either leave the PAC URL field blank, or enter a valid URL for the phone to download the PAC file.
105	Plug and play setup failed because the proxy on your network requires authentication. Provide the proxy authentication credentials on the next screen.	The plug and play provisioning method failed because the proxy on your network requires authentication. If your proxy server requires authentication, the phone must be configured with authentication credentials. If you do not know what credentials to use, contact your IT organization's security team. If you already know the authentication credentials, tap the Next button and move to the next screen to supply proxy authentication credentials and try again.
106	Plug and play setup failed because of an unknown proxy error. Contact your service provider for assistance.	The plug and play provisioning method failed because of an unknown proxy error. Contact your conferencing service provider for help.
107	Plug and play setup failed because the Dolby Voice Console (https://console.dolbyvoice.com) host name is not resolvable. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console host name is not resolvable. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is required for the plug and play provisioning method. The DNS server configured on phone must be able to resolve the Dolby Voice Console host name. Tap the Next button to review the network configuration. You can manually override the DNS server settings on the next screen. Configuring the public DNS server 8.8.8.8 may help resolve this issue.
108	Plug and play setup failed because the Dolby Voice Console (https://console.dolbyvoice.com) is unreachable from this network. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console is unreachable from this network. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the phone to connect to the Dolby Voice Console. Make sure that your firewall whitelists <i>*.dolbyvoice.com</i> to allow the plug and play provisioning method and service data traffic from the phone to pass through. Consult your conferencing service provider's network preparation guidelines for all addresses required by the service. For more information, see Server addresses on page 27.
109	Plug and play setup failed because of an unknown SSL error. Contact your service provider for assistance.	The plug and play provisioning method failed because of an unknown SSL error. The phone detected an unknown SSL error and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
110	Plug and play setup failed because of a communication error with the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because of a communication error with the Dolby Voice Console. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The phone experienced a communication error with the Dolby Voice Console and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.
111	Plug and play setup failed because of an error in discovering provisioning information for the phone. Contact your service provider for assistance.	The plug and play provisioning method failed because of an error in discovering provisioning information for the phone. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The phone experienced an error while trying to retrieve the provisioning service information from the Dolby Voice Console and was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.
112	Plug and play setup failed because the phone's provisioning information could not be found on the Dolby Voice Console (https://console.dolbyvoice.com). Contact your service provider for assistance.	The plug and play provisioning method failed because of an error in discovering provisioning information for the phone. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. The provisioning information for the phone was not found on the Dolby Voice Console and the phone was unable to complete the plug and play provisioning method setup. Contact your conferencing service provider for help.
113	Plug and play setup failed because the phone cannot reach the proxy server. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone cannot reach the proxy server. The phone has detected a proxy and is trying to connect to the Dolby Voice Console using that proxy. However, the proxy server is unreachable. Follow the instructions on the Dolby Conference Phone to disable WPAD and clear the setting for PAC configuration. If your network allows Internet access bypassing the proxy server, this may resolve the issue.

Error code	Message displayed	Solution
115	Plug and play setup failed because the phone encountered an SSL certificate issue connecting to Dolby Voice Console (https://console.dolbyvoice.com). Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone encountered an SSL certificate issue connecting to Dolby Voice Console. The phone is unable to verify the identity of the Dolby Voice Console. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the phone through its web user interface. See Accessing the Dolby Conference Phone web interface on page 79 and Adding certificate authority certificates on page 55 for information on connecting to the Dolby Conference Phone web interface using a web browser and importing the CA certificate into the device from the web interface.
116	Plug and play setup failed because the Dolby Voice Console is blocked by a firewall. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the Dolby Voice Console is blocked by a firewall. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the phone to connect to the Dolby Voice Console. Make sure that your firewall whitelists <i>*.dolbyvoice.com</i> to allow the plug and play provisioning method and service data traffic from the phone to pass through. For additional help, contact your conferencing service provider.
150	Plug and play setup failed because the main network interface of the Dolby Conference Phone is down. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the main network interface of the Dolby Conference Phone is down. Contact your IT help desk for assistance.
151	Plug and play setup failed because the phone detected another device on your network with a duplicate IP address. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone detected another device on your network with a duplicate IP address. Contact your IT help desk for assistance.
152	Plug and play setup failed because the default gateway is unreachable. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the default gateway is unreachable. Contact your IT help desk for assistance. Verify that the DHCP assigned default gateway address on the Dolby Conference Phone is valid. This error may be due to incorrect DHCP configuration. If your network uses VLAN for video conferencing equipment, you can enter the VLAN ID on the Dolby Conference Phone so that the Dolby Voice Hub can use a different IP address and default gateway.

9.3.2 Provisioning service error messages

These are possible solutions to provisioning service errors when deploying Dolby Conference Phone with the plug and play provisioning method.

These messages appear on the

screen.

Error code	Message displayed	Solution
201	Plug and play setup failed because the phone could not connect to the provisioning server. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone could not connect to the provisioning server. The phone needs to connect to the Dolby Voice Console to discover the provisioning service. This is a required step for the plug and play provisioning method setup. Your network must allow the phone to connect to the Dolby Voice Console. Log in to the phone user interface to determine which provisioning server the system is configured with. Make sure that your firewall whitelists *.dolbyvoice.com to allow the plug and play provisioning method and service data traffic from the phone to pass through. Consult your conferencing service provider's network preparation guidelines for all addresses required by the service. For more information, see Server addresses on page 27.
202	Plug and play setup failed because the phone encountered an SSL error while connecting to the provisioning server. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone encountered an SSL error while connecting to the provisioning server. The phone is unable to verify the identity of the provisioning server. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the phone through its web user interface.
203	Plug and play setup failed because the phone failed authentication with the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone failed authentication with the provisioning server. The phone is attempting to connect to a provisioning server that requires authentication. If you do not know the credentials, contact your conferencing service provider.
204	Plug and play setup failed because the phone cannot find the master configuration file from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone cannot find the master configuration file from the provisioning server. The master configuration file is the first file the phone downloads during the provisioning process. The phone was unable to download the master configuration file. Contact your conferencing service provider for help.
205	Plug and play setup failed because the phone cannot find the configuration file from the provisioning server. Contact your service provider for help.	The plug and play provisioning method failed because the phone cannot find the configuration file from the provisioning server. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
206	Plug and play setup failed because the phone found an invalid master configuration file from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone found an invalid master configuration file from the provisioning server. The master configuration file is the first file the phone downloads during the provisioning process. The phone found an invalid master configuration file. Contact your conferencing service provider for help.
207	Plug and play setup failed because the phone cannot download the upgrade package metadata file. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone cannot download the upgrade package metadata file. The upgrade package metadata file contains software upgrade information for the phone. The phone uses this file to determine whether it needs to upgrade. Contact your conferencing service provider for help.
208	Plug and play setup failed because the phone found an invalid upgrade package metadata file. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone found an invalid metadata file. The upgrade package metadata file contains software upgrade information for the phone. The phone uses this file to determine whether it needs to upgrade. Contact your conferencing service provider for help.
209	Plug and play setup failed because the phone detected an invalid configuration from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone detected an invalid configuration from the provisioning server. Contact your conferencing service provider for help.
210	Plug and play setup failed because the phone detected an invalid or empty certificate URL from the provisioning server. Contact your service provider for assistance.	The plug and play provisioning method failed because the phone detected an invalid or empty certificate URL from the provisioning server. Contact your conferencing service provider for help.

9.3.3 Dolby Conference Phone setup app error messages

These are possible solutions to setup app errors when deploying Dolby Conference Phone with the plug and play provisioning method.

For the plug and play provisioning method, the phone must be able to connect to your conferencing service provider's setup app through the Internet. After the phone connects to the setup app, screens and workflows specific to your conferencing service provider display on the phone. After out-of-the-box setup is complete, the phone uses your conferencing service provider's conferencing application, which is separate from the setup app.

Error code	Message displayed	Solution
301	Plug and play setup failed because the Dolby Conference Phone setup app encountered a program failure. Contact your service provider for assistance.	The plug and play provisioning method failed because the Dolby Conference Phone setup app encountered a program failure. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone detected a program error in the Dolby Conference Phone setup app and your conferencing service provider will need to examine the error. Contact your conferencing service provider for help.
302	Plug and play setup failed because the Dolby Conference Phone setup app has missing files. Contact your service provider for assistance.	The plug and play provisioning method failed because the Dolby Conference Phone setup app has missing files. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone detected missing files in the Dolby Conference Phone setup app and your conferencing service provider will need to examine the error. Contact your conferencing service provider for help.
303	Plug and play setup failed because the phone was unable to download the Dolby Conference Phone setup app. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the Dolby Conference Phone setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone was unable to download the Dolby Conference Phone setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the phone needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.
304	Plug and play setup failed because the server refused the Dolby Conference Phone setup app download request. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the server refused the Dolby Conference Phone setup app download request. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone was unable to download the Dolby Conference Phone setup app because the server has refused the download request. To determine which servers the phone needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.
305	Plug and play setup failed because the phone failed to download the Dolby Conference Phone setup app due to a network error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the Dolby Conference Phone setup app due to a network error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone was unable to download the Dolby Conference Phone setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the phone needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
306	Plug and play setup failed because the phone failed to download the setup app due to a DNS error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the Dolby Conference Phone setup app due to a DNS error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone was unable to download the Dolby Conference Phone setup app. This error can occur if the DNS server assigned to the phone is unable to resolve the server name. Contact your conferencing service provider for help.
307	Plug and play setup failed because the phone failed to download the Dolby Conference Phone setup app due to a network timeout. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the Dolby Conference Phone setup app due to a network timeout. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone was unable to download the Dolby Conference Phone setup app. This could mean that your firewall blocked access to your conferencing service provider's service. To determine which servers the phone needs to access, see the configuration guide from your conferencing service provider. Contact your conferencing service provider for help.
308	Plug and play setup failed because the phone failed to download the Dolby Conference Phone setup app due to a proxy error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the Dolby Conference Phone setup app due to a proxy error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone was unable to download the Dolby Conference Phone setup app. This error can occur if the proxy that the phone has automatically discovered is unable to process the setup app download request. Contact your conferencing service provider for help.
309	Plug and play setup failed because the phone failed to download the Dolby Conference Phone setup app due to an SSL error. Contact your IT help desk for assistance.	The plug and play provisioning method failed because the phone was unable to download the Dolby Conference Phone setup app due to an SSL error. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. This is usually caused by TLS/SSL intercepting proxy that is deployed in your network. You need to consult the security team of your IT organization for help. You will need to obtain the CA certificate of your intercepting proxy, and import the certificate into the phone through its web user interface.
310	Plug and play setup failed because the server rejected the phone from downloading the Dolby Conference Phone setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because the server rejected the phone from downloading the Dolby Conference Phone setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help.

Error code	Message displayed	Solution
311	Plug and play setup failed because of an authentication failure when the phone was downloading the Dolby Conference Phone setup app. Contact your service provider for assistance.	The plug and play provisioning method failed because of an authentication failure when the phone was downloading the Dolby Conference Phone setup app. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help.
312	Plug and play setup failed because the Dolby Conference Phone setup app encountered a program failure. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app encountered a program failure. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. The phone detected a program error in the Dolby Conference Phone setup app and your conferencing service provider will need to examine the error. Contact your conferencing service provider for help.
314	Plug and play setup failed because the Dolby Conference Phone setup app contains an invalid certificate. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app contains an invalid certificate. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help.
315	Plug and play setup failed because the Dolby Conference Phone setup app contains an invalid signature. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app contains an invalid signature. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help.
316	Plug and play setup failed because the Dolby Conference Phone setup app has contains an invalid checksum. Contact your service provider for assistance.	The plug and play provisioning method failed because the setup app contains an invalid checksum. The setup app is designed to assist you in completing the conferencing service provider designed plug-and play-experience. Contact your conferencing service provider for help.

9.4 Decommissioning and disposing of the phone

Ensure that faulty devices are disposed of in an appropriate way, according to local regulations.

About this task

The Dolby Conference Phone Support website includes a disposal process about how to properly decommission and dispose of the phone.

Procedure

1. Go to <https://www.dolby.com/us/en/support/conferencing-support.html>.
2. Under **Dolby Conference Phone Software and Documentation**, scroll down to **Decommissioning and Disposal Process** and click the link to the *Dolby Conference Phone Decommissioning and Disposal Process*.

Appendix A: Configuring the provisioning server

If you do not have Dolby Conferencing Console, you can connect the phone to a provisioning server instead.

- [Reasons for using a provisioning server](#)
- [Building the provisioning server](#)
- [Security for provisioning servers](#)
- [Setting up an FTP server](#)
- [HTTPS server setup](#)
- [Master configuration file](#)
- [Template files](#)
- [Configuration file \(JSON\)](#)
- [Recommended file and directory layout](#)
- [Redundant provisioning servers](#)
- [Log directory](#)
- [Updating preinstalled software using a provisioning server](#)
- [Sample configurations](#)

10.1 Reasons for using a provisioning server

The provisioning server allows for flexibility in installing, upgrading, maintaining, and configuring the Dolby Conference Phone.

 **Note:** If you are already using a provisioning server in your organization, you can continue to use these procedures to provision your phones. However, if you are using Dolby Conferencing Console for provisioning the phones, you can skip this section.

The provisioning server can be set up on the local LAN or anywhere on the Internet. Configuration, log, and directory files are normally located on the provisioning server.

If you allow the phone write access to your provisioning server, the phone can use the provisioning server to upload files (such as logs) and store the user's files. If it cannot locate a provisioning server when it boots up, the phone is designed to operate with internally saved parameters.

10.2 Building the provisioning server

Before you attempt to plug in the phone, review the provisioning server requirements, and decide what type of provisioning server to use (FTP, HTTP, or HTTPS). Make sure that the provisioning server is set up and ready for use.

Review these topics:

- [Requirements for provisioning servers](#) on page 124
- [Security for provisioning servers](#) on page 124
- [Setting up an FTP server](#) on page 125
- [HTTPS server setup](#) on page 125

10.2.1 Requirements for provisioning servers

You can use either a Microsoft Windows- or Linux-based server as your provisioning server. The FTP, HTTP, or HTTPS servers can reside on the same server. However, this should be a separate server from the server running your call control platform.

10.2.2 Supported provisioning protocols with provisioning servers

FTP, HTTP, and HTTPS provisioning protocols are supported with a provisioning server.

Both digest and basic authentication schemes are supported when using HTTP or HTTPS. Custom certificates, including those from certificate authorities, can be added to the phone for Transport Layer Security server authentication on HTTPS provisioning servers.

10.3 Security for provisioning servers

File permissions should provide the minimum access required, and the account used should have no other rights on the server.

 **Note:** For organization purposes, we recommend configuring a separate log file directory, though this is not required. The different directories can have different access permissions. For example, you can allow log files to have write access.

Log files provide backup copies of changes users make to the phone configuration. Centralization logging management is performed through `syslog`.

You can also download the log from the Dolby Conference Phone web interface. These log files greatly assist in the ability to provide customer support for diagnosing issues that may occur with the phone operation.

The phone needs to be able to write to the upload file directory and the provisioning directory. All other files that the phone needs to read, such as the application executable and the standard configuration files, should be made read-only through file server file permissions.

Each phone may open multiple connections to the server.

Related information

[syslog logging](#) on page 85

10.4 Setting up an FTP server

A simple provisioning operation uses File Transfer Protocol (FTP). Although most FTP servers are free, they require installation and use logins and passwords.

About this task

A free and popular server, FileZilla Server, is available for Windows at filezilla-project.org. For Unix, we suggest [Red Hat Enterprise Linux FTP](#). This section provides instructions for setting up FileZilla.

To set up an FTP server using FileZilla Server:

Procedure

1. Download and install the latest version of FileZilla Server.
2. After installation, a **Connect to Server** pop-up displays on your computer. Select **OK** to open the administrative user interface.
3. To configure a user, select **Edit > Users** in the status bar.
4. Select **Add**. Enter the user name for the phone (for example, *bill123*), and select **OK**.
Select the **Password** check box, and enter a password (for example, *1234*). The phone will use this password to log in.
5. Select **Page > Shared** folders to specify the server-side directory where the provisioning files will be located (and the log files uploaded).
6. Select **Add**, and pick the directory.
7. To allow the phone to upload logs onto the provisioning server, select **Shared Folders > Files**, select the **Write** and **Delete** check boxes, and then select **OK**.
8. Determine the IP address of the FTP server by entering `cmd` in the **Run** dialog on your **Start** menu, and `ipconfig` in the command prompt.
The IP address of the FTP server is shown.

10.5 HTTPS server setup

You can provision the Dolby Conference Phone with an HTTPS provisioning server.

You must ensure that the CA certificate of the HTTPS provisioning server certificate is installed on the phone before it attempts to communicate with your HTTPS server.

10.6 Master configuration file

The master configuration files can be used to configure the Dolby Conference Phone individually or to configure all phones at once.

Single phone master configuration file

If customization is required separately for each phone, the files should be named *dvcp-
<MACaddress>.cfg*, where *MACaddress* is the MAC address (or serial number) of the phone. For a-f hexadecimal digits, use lower case only, for example, *dvcp-0004f200106c.cfg*. The MAC address can be

viewed through the  >  > **About** > **General** menu. It is also printed on a label on the back of the phone. If this file cannot be downloaded, the phone will search for the default master configuration file (described in this section).

Default master configuration file

For systems in which the configuration is identical for all phones (that is, there are no unique *dvcp-<MACaddress>.cfg* files), the default master configuration file may be used to configure all phones. The file named *dvcp-default.cfg* is the default master configuration file, and we recommend that you store this file on the provisioning server. If a phone does not find its own *dvcp-<MACaddress>.cfg* file, it will use this one, and establish a baseline configuration. This file is part of the standard distribution of configuration files. It should be used as the template for the *dvcp-<MACaddress>.cfg* files.

The default master configuration file, *dvcp-default.cfg*, is shown here.

```
{
  "img_file" : "dvcp-1.0.0.<buildID>.tar.gz",
  "cfg_files" : [
    " "
  ]
}
```

Master configuration files must contain these JSON attributes:

img_file

The path name of the application executable. It can have a maximum length of 255 characters. This can be a URL with its own protocol, user name and password. For example, *dvcp-rel-2.20.212.tar.gz*.

cfg_files

A comma-separated list of configuration files. Each file name has a maximum length of 255 characters, and the list of file names has a maximum length of 2,047 characters, including commas and white space. Each configuration file can be specified as a URL with its own protocol, user name, and password (for example, *http://admin:password@prov.example.com/phone2034.cfg*). Note that the order of the configuration files listed in *cfg_files* is significant:

- The files are processed in the order listed (left to right).
- If the same parameter is included in more than one file or more than once in the same file, the last (right) parameter read will be used.

Master configuration files may also contain these JSON attributes:

certificates

A comma-separated list of certificate files. Each file name has a maximum length of 255 characters, and the list of file names has a maximum length of 2,047 characters, including commas and white space.

directory_file

A local directory (or phone book) that will be loaded into the phone. The file name has a maximum length of 255 characters. A sample master configuration file is shown in the following example file. The configuration parameters are split between two configuration files in this sample, namely *common.json* and *office.json*.

Here is an example file:

```
{
  "img_file" : "dvcp-1.0.0.1234.tar.gz",
  "cfg_files" : [
    "common.json",
    "office.json"
  ],
  "certificates": [
    "ca_pem.pem"
  ],
  "directory_file": [
    "vandirectory.json"
  ]
}
```

If you have a requirement for different behaviors on different phones on the same provisioning server, you can create a variable, [MAC], in the master configuration file that is replaced by the MAC address of each phone when it reboots. You can also create a variable, [SN], that is replaced by the serial number of each phone. The sample master configuration file shown here allows for different phones to use a different set of configuration parameters and a different directory. All MAC address-specific files must reside on the provisioning server.

```
{
  "img_file" : "dvcp-1.0.0.1234.tar.gz",
  "cfg_files" : [
    "common.json",
    "[MAC].office.json"
  ],
  "directory_file": [
    "[MAC].directory.json"
  ]
}
```

10.6.1 Setting up the CA certificate files for the phone to download

Update the master configuration file to specify the location of the CA certificate file so that it may be downloaded to the phones.

Procedure

1. Download the PEM file to the provisioning server where the configuration files reside.
2. Add the file name to the master configuration file.

```
{
  "img_file" : "dvcp-1.0.0.1234.tar.gz",
  "cfg_files" : [
    "dvcp-config1.json"
  ],
  "certificates": [
    "ca_certs.pem"
  ]
}
```

3. Reboot the phone.

10.7 Template files

Use these template files to create a *filename.json* file for the configuration parameters that you need to change to configure the desired phone features.

- *dvcpcfg-core.json*: Parameters that must be set for a phone to make and receive calls
- *dvcpcfg-user.json*: User-related parameters
- *dvcpcfg-common.json*: Site-specific parameters

Use *dvcpcfg-all.json* only as a reference, as all parameters are not required. A sample user-specific configuration file is shown in this figure.

```
{
  "Sip.Account.DisplayName": "Sarah Chu",
  "Sip.Account.UserName": "3105",
  "Sip.Credential.Name": "3105",
  "Sip.Credential.Password": "1234",
  "Sip.Credential.Realm": "*"
  "Preferences.AutoAnswerEnabled": true
}
```

10.8 Configuration file (JSON)

Administrators can use configuration parameters to dictate the behavior of the Dolby Conference Phone after it is running the executable specified in the master configuration file.

To verify JSON files to be used for provisioning the phone, go to <http://www.jsonlint.com> and follow the directions.

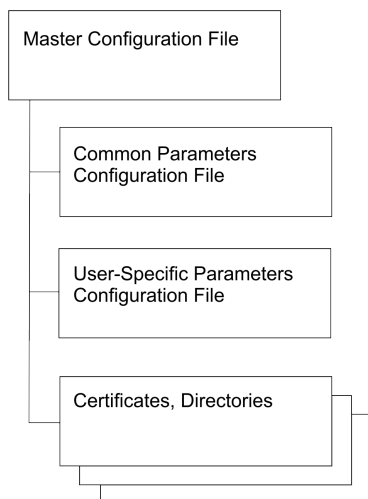
Related information

[Accessing the Dolby Conference Phone web interface](#) on page 79

10.9 Recommended file and directory layout

You must create at least two configuration files for the Dolby Conference Phone to boot up, and certain parameters will need to be changed for the phone to be usable within your organization. The Dolby Conference Phone software package comes with sample configuration files for you to use.

Figure 6: Sample grouping of configuration files for provisioning



For additional information related to recommended file and directory layout:

Related information	See
The use of multiple configuration files for the configuration parameters.	Master configuration file on page 125
A description of the template files you can use to create the common and user-specific configuration files.	Template files on page 127
Which parameters apply to each phone feature.	Configuring phone features on page 50
Information about having different features on some phones. You will need to create a number of configuration files, one for each phone that you want to deploy. You will also need the list of MAC addresses you created.	Conference room requirements on page 16

 **Note:** The configuration on the provisioning server overrides any existing configuration of the phone. For example, if there are no certificates on the provisioning server, but there are existing certificates on the phone, those certificates will be removed during the provisioning process.

10.10 Redundant provisioning servers

With large-scale deployments (for example, 10,000 devices or more), consider setting up multiple (redundant) provisioning servers (an active and a backup version). This ensures that the user does not experience service disruptions or performance problems.

If you have Dolby Conferencing Console, you can use it to set up redundant provisioning servers (see the).

If you do not have Dolby Conferencing Console, you can configure multiple (redundant) provisioning servers—one logical server with multiple addresses—by mapping the provisioning server DNS name to multiple IP addresses. The default number of provisioning servers is one, and the maximum number is two.

If you set up multiple provisioning servers, you must be able to reach all of the provisioning servers with the same protocol, and the contents on the provisioning servers must be identical.

10.11 Log directory

The Dolby Conference Phone is typically configured to upload log files to the provisioning server.

We recommend creating a subdirectory for uploading log files, so they are not confused with configuration and phone software files. The phone must be configured to upload log files to the new directory; see [Uploading log files to a provisioning server](#) on page 83.

10.12 Updating preinstalled software using a provisioning server

The Dolby Conference Phone software is available for download from the provider of your phone. Updates are issued periodically.

Make sure that you reference the correct documentation when attempting to update software on phones. This varies depending on whether you use Dolby Conferencing Console or a provisioning server to provision phones.

If your organization uses Dolby Conferencing Console, skip this section and see the following documentation instead:

- [Updating phone software using Dolby Conferencing Console](#) on page 31
- *Dolby Conferencing Console Administrator's Guide*

10.12.1 Available software packages

If your organization uses a provisioning server, download the software packages described here and then refer back to the instructions in [Appendix A: Configuring the provisioning server](#) on page 123. The Dolby Conference Phone software packages must be uncompressed on the provisioning server for the phone (or a temporary location if you have not set up your provisioning server yet).

The software package, `dvcp-rel-<x.y.z.buildID>.zip`, contains these files:

`dvcp-rel-<x.y.z.buildID>.tar.gz`

Dolby Voice application image file

`dvcp-rel-<x.y.z.buildID>.meta`

Dolby Voice application *.meta* file (checksums and digital signature)

 **Note:** The prefixes of the `.tar.gz` and `.meta` files must match, or the phone software will not download to the phone correctly.

`dvcp-default.cfg`

Master phone configuration file

Three configuration parameter template files

`dvcp-rel-<x.y.z.buildID>.cli-help.html`

Diagnostic shell command descriptions

`dvcp-rel-<x.y.z.buildID>.cfg-help.html`

Configuration parameter descriptions

`internalConf-rel-<x.y.z.buildID>.xml`

Configuration parameter definitions

`internalConfEnum-rel-<x.y.z.buildID>.xml`

Configuration parameter enumerations

When you download the phone software, a notice displays asking you to accept the download. Read the notice, click the button indicating that you have read the notice, and then click the **Submit** button to continue the phone software package `.zip` download.

See the latest release notes for a detailed description of each file in the distribution, issues that are fixed in this release, and issues that still remain (with possible workarounds).

10.12.2 Upgrading and downgrading phone software

You can simultaneously upgrade the phone software that is running on all Dolby Conference Phones in your organization, or you can upgrade the phones one at a time.

You may be prompted to confirm the upgrade operation. Tap **Update** or **Skip**. If you do not respond within 20 seconds, the upgrade will automatically proceed.

If you tap **Skip**, this icon displays:



 **Note:** While the phone software update is in progress, the progress displays at the top of the Dolby Conference Phone web interface page.

Related information

[Viewing the status menu](#) on page 78

10.12.3 Upgrading phones to the latest Dolby Conference Phone software

The Dolby Conference Phone software and configuration can be updated together using the centralized provisioning model.

Procedure

1. Back up your existing application (and configuration files).
2. Create new configuration files.
Configuration file changes and enhancements are explained in the release notes that accompany the phone software.
3. Save the new configuration files and phone software images on your provisioning server.
4. Reboot the phone.

From the phone user interface, tap  >  > **Reboot phone**, or, from the phone web interface, select **Tools** and click **Reboot**.

5. Ensure that the configuration process completed correctly.

From the phone user interface, tap  >  > **About** > **General**.

Confirm the Dolby Voice software version and the configuration files downloaded to the phone.

6. Monitor the log files uploaded to the provisioning server.

10.12.4 Downgrading phones to an earlier version of Dolby Conference Phone software

The Dolby Conference Phone software and configuration can be reverted to earlier versions if the software and configurations have been backed up.

Procedure

1. Back up your existing application (and configuration files).

2. Create new configuration files.

Configuration file changes and enhancements are explained in the release notes that accompany the phone software.

3. Save the new configuration files and phone software images on your provisioning server.

4. Reboot the phone.

The phone file system may be reformatted.

5. Ensure that the configuration process completed correctly.

From the phone user interface, tap  >  > **About** > **General**.

Confirm the Dolby Voice software version and the configuration files downloaded to the phone.

6. Monitor the log files uploaded to the provisioning server.

10.13 Sample configurations

Use the sample configuration files provided with the Dolby Conference Phone software package to create your own configuration file.

The best configuration reference is available in the phone software package:

- Configuration directives: *dvcp-rel-x.y.z.build.cfg-help.html*
- Shell commands: *dvcp-rel-x.y.z.build.cli-help.html* and in the phone web interface: **Tools** > **Help**.

10.13.1 Directory structure

The following examples each assume the same directory structure on the provisioning server. They assume a top-level directory that contains all of the configuration and phone software files, and a subdirectory called */logs* to contain the uploaded log files.

Directory	Purpose
/	Location of phone software and all configuration files
/logs	Destination for uploading files

10.13.2 Master configuration file

The master configuration file is common to all phones and specifies the phone software version that the phones should load, as well as the names of the configuration files and certifications that the phones should load.

```
{
  "img_file" : "dvcp-rel-2.0.1.116.tar.gz",
  "cfg_files" : [
    "common.json", "[MAC].office.json" ],
  "certificates": [
    "rootCA.cer",
    "otherRootCA.cer"
  ]
}
```

In this example:

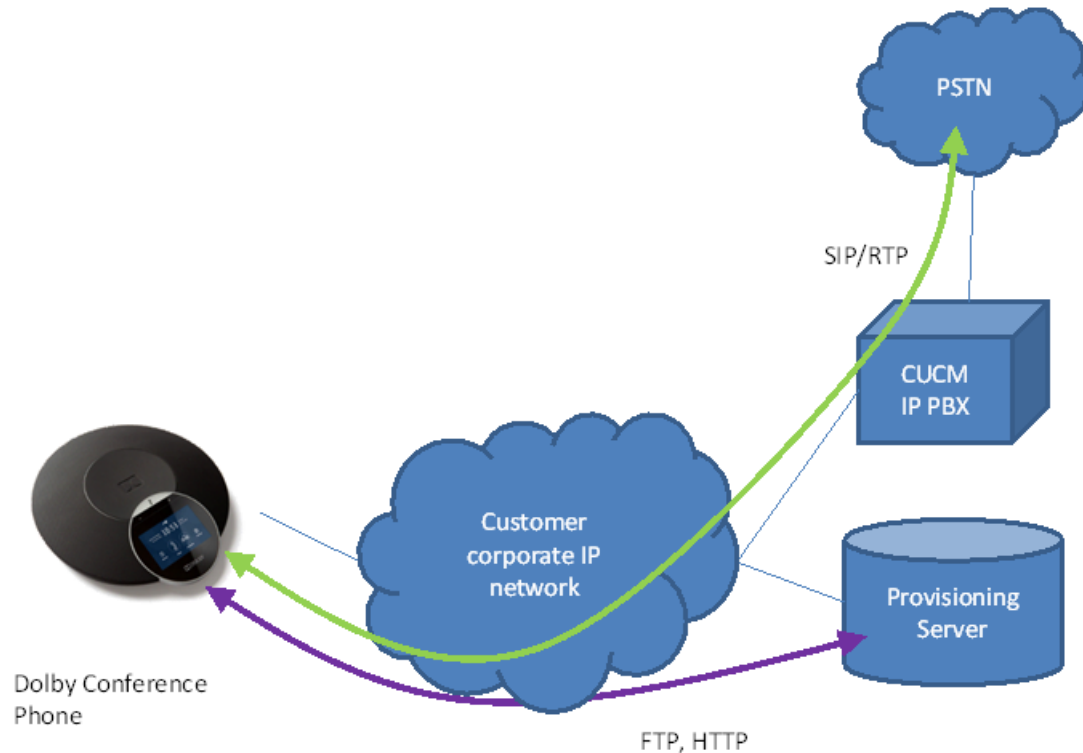
- *dvcp-rel-2.0.1.116.tar.gz* is the name of the phone software image the phones should run.
- *rootCA.cer* and *otherRootCA.cer* are the names of certificate files the phones should load (for example, SSL certificate authority root certificates).

10.13.3 Simple IP PBX-only configuration

This is a basic configuration that enables the Dolby Conference Phone to register with and use an IP PBX for telephone calls.

In the common configuration file, you would place any of the common parameters, such as the proxy server details. In the device-specific configuration file, place the device-specific parameters, such as the SIP registration details to be used and the local time zone of the phone.

Figure 7: Simple IP PBX-only configuration



File: **common.json**

In this configuration it is important to configure the operation mode correctly and configure the SIP proxies as they will be shared, and we recommend you configure common logging options.

```
{
  "Features.OperationMode": "IPPBXONLYMODE",
  "Sip.Pbx.OutboundProxy1": "a.b.c.d",
  "Sip.Pbx.OutboundProxy2": "e.f.g.h",
  "Logging.Upload.Enable": true,
  "Provisioning.Server.UploadDir": "logs"
}
```

In this example:

- *a.b.c.d* and *e.f.g.h* are the IP addresses of one or two call servers or outbound proxy servers that the phone should use.
- *logs* is the name of the subdirectory on the provisioning server where log files will be stored.

File: **[MAC].office.json**

This file contains any configuration items specific to the device you are configuring. Within a configuration file value, the string *[MAC]* is replaced by the MAC address of the specific phone you are configuring.

To create a configuration file specific to the phone with MAC address *1a:2b:3c:4d:5e:6f*, you would create a file called *1a2b3c4d5e6f.user.json*.

A simple user configuration file would include:

```
{
  "Preferences.Localization.Timezone": "tzspec",
  "Sip.Account.DisplayName": "displayname",
  "Sip.Account.UserName": "username",
  "Sip.Credential.Name": "credname",
  "Sip.Credential.Password": "credpass"
}
```

In this example:

- *tzspec* is the name of the time zone in which the user resides (for example, Australia/Sydney).
- *displayname* is the name that should be displayed for the user.
- *username* is the extension number or address of the user.
- *credname* is the name used when authenticating the user.
- *credpass* is the password used when authenticating the user.

Glossary

API

Application programming interface. A set of functions that can be used to access the functions of an operating system or other type of software.

Bluetooth LE

Bluetooth low energy. A chip technology that produces radio waves to connect a device to other Bluetooth devices. The low-energy version uses less power than standard Bluetooth, but maintains a similar communication range.

CLI

Command-line interface.

DHCP

Dynamic Host Configuration Protocol.

DNS

Domain Name System. An Internet service that translates Internet domain and host names to IP addresses and conversely. DNS automatically converts between the name entered in a web browser and the IP addresses of the web server hosting the site whose URL is entered in the web browser.

DNS SRV

Domain Name System service record. A service record (SRV record) defines the location (the host name and port number) of a server in a DNS.

DoS

Denial of service. Events that render systems on a computer network temporarily unusable. DoS can happen accidentally as the result of actions taken by network users, but is often malicious.

DSCP

Differentiated Services Code Point. An IP header field that allows a router to provide different levels of service to different types of network traffic.

DSP

Digital signal processor. A specialized microprocessor optimized for digital signal processing.

DTMF

Dual-tone multifrequency. A system that is used by touch-tone telephones to assign a specific frequency (consisting of two separate tones) to each key so that it can easily be identified by a microprocessor.

FTP

File Transfer Protocol. A network-based protocol designed for transferring data using a client-server architecture.

HTTP

Hypertext Transfer Protocol. An application protocol for hypermedia information systems, and the foundation for data communication for the World Wide Web.

HTTPS

Hypertext Transfer Protocol Secure. An application protocol for secure communication over a network and the Internet that provides authentication of websites and keeps user information private.

IEEE

Institute of Electrical and Electronics Engineers.

IP

Internet Protocol.

IP address

Internet Protocol address. A numerical identifier assigned to a device that is a member of a network that uses the IP for communication.

JSON

JavaScript Object Notation. A lightweight data-interchange format that is easy for humans to read and write, and for machines to parse and generate.

kbps

Kilobits per second.

LAN

Local area network.

LDAP

Lightweight Directory Access Protocol. An application protocol for querying or modifying items in corporate directories that allows sharing of information about users, devices, and applications on a network.

LLDP

Link Layer Discovery Protocol . A vendor-neutral link layer protocol for Ethernet network devices such as switches, routers, and wireless local area network (LAN) access points to announce information about themselves to other nodes on the network and store the information they discover, as defined in IEEE 802.1AB.

MAC address

Media access control address. A unique identifier assigned to a network interface for communications on a network. MAC addresses are typically assigned by the network interface manufacturer.

MPEG

Moving Picture Experts Group. An ISO/IEC working group that develops video and audio coding standards.

NAPTR record

Naming Authority Pointer record. A type of resource record in the Domain Name System (DNS) of the Internet used to map servers and user addresses in the Session Initiation Protocol (SIP). Used most frequently for Internet telephony (VoIP) services.

NAT

Network Address Translation. An Internet standard that enables a local area network (LAN) to use one set of IP addresses for internal traffic and a second set of addresses for external traffic.

NTLMv2

A Microsoft security protocol that provides authentication, integrity, and confidentiality to users. NTLMv2 is part of Windows NT LAN Manager, which is a suite of security protocols.

NTP

Network Time Protocol. A network protocol for clock synchronization on computers.

PAC

Proxy automatic configuration. A file that defines how web browsers and other user agents can automatically choose the appropriate proxy server (access method) for fetching a given URL.

PBX

Private branch exchange. A phone system that is delivered as a hosted service.

PEM

Privacy-Enhanced Electronic Mail. A file format for security certificates in email communication.

PKI

Public key infrastructure. A set of roles, policies, and procedures needed to create, manage, distribute, use, store, and revoke digital certificates and manage public-key encryption.

PoE

Power over Ethernet. A solution in which an electrical current is run to networking hardware over Ethernet category 5 or higher data cabling.

PSTN

Public switched telephone network. A circuit-switched telephone network operated by national, regional, or local operators.

RTCP

RTP Control Protocol. A protocol for communicating quality-of-service information about Real-time Transport Protocol (RTP) sessions.

RTP

Real-time Transport Protocol. An IP network audio and video protocol.

SBC

Session border controller. A piece of network equipment or a collection of functions that controls real-time session traffic at the signaling, call control, and packet layers as they cross a notional packet-to-packet network border between networks, or between network segments.

SIP

Session Initiation Protocol. An application-layer communications protocol used for signaling and controlling communications sessions.

SNMP

Simple Network Management Protocol. A protocol for managing IP network devices

SRTP

Secure Real-time Transport Protocol. A Real-time Transport Protocol (RTP) profile that provides encryption, message authentication, and integrity, and replay protection to the RTP data in both unicast and multicast applications.

SSH

Secure Shell protocol. An encrypted network protocol for secure data communication, remote command-line login, remote command execution, and other secure network services between two networked computers.

SSL

Secure Sockets Layer . A security protocol that works at a socket level.

TCP

Transmission Control Protocol. A communications protocol that specifies how data should be formatted, addressed, transmitted, routed, and received at the destination. Part of the Internet protocols communications suite.

TCP/IP

Transmission Control Protocol/Internet Protocol. Communications protocols that specify how data should be formatted, addressed, transmitted, routed, and received at the destination. Part of the Internet protocols communications suite.

TLS

Transport Layer Security. A cryptographic protocol designed to provide communications security over a computer network.

TLV

Type length value. Within data communication protocols, a format for encoding optional information.

UDP

User Datagram Protocol. A communications protocol that uses no handshaking dialogues to establish a connection with the remote host. UDP is a member of the IP suite.

UI

User interface.

URI

Uniform Resource Identifier. A group of characters identifying a resource on a network (typically, the Internet).

URL

Uniform resource locator.

VLAN

Virtual LAN. Any broadcast domain that is partitioned and isolated in a computer network at the data link layer (OSI layer 2).

VoIP

Voice over Internet Protocol. A methodology and group of technologies for the delivery of voice communications and multimedia sessions over Internet Protocol (IP) networks.

WPAD

Web Proxy Automatic Discovery. An Internet protocol that enables a web browser to automatically connect to a cache server (or proxy server) location in a network to retrieve stored web pages more quickly by than leaving the network to request the web page from the site of the originating web server.



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